



OneATS

User manual

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OneATS user manual

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Definitions, symbols, abbreviations

Symbol	Description
CIM	Common Information Model
DMS	Distributed Management System
DCOM	Distributed Component Object Model
FEP	Front End Processing
HMI	Human Machine Interface
I/O	Input/Output
IED	Intelligent Electronic Device
LAN	Local Area Network
OPC UA	OPC Unified Architecture
RTU	Remote Terminal Unit
SCADA	Supervisory Control and Data Acquisition
SSL	Secure Socker Layer
TCP	Transmission Control Protocol
TLS	Transport Layer Security
XML	Extensible Markup Language

Text color:

Comment

Notes

Warnings

Content

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B.	Database
C.	OneATS DataEditor – Project Management
D.	OneATS FEP
E.	OneATS DataServer
F.	OneATS Grid Studio
G.	OneATS SmartHIS
H.	OneMasterTool



OneATS User Manual

- OneATS FEP

28/Jun/2024

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Software Version	Date of Issue	Change Description
V4.5.8	25/Jun/2024	

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D. OneATS FEP

FEP Server is an application that runs as a **service** that is responsible for connecting and exchanging data with (1) IEDs and (2) data centers according to protocols that support IEC61850, IEC101, IEC103, IEC104, Modbus, OPC, etc.

FEP Server does not have an interface, the declaration and configuration of FEP channels are carried out through **the OneATS DataEditor** application.

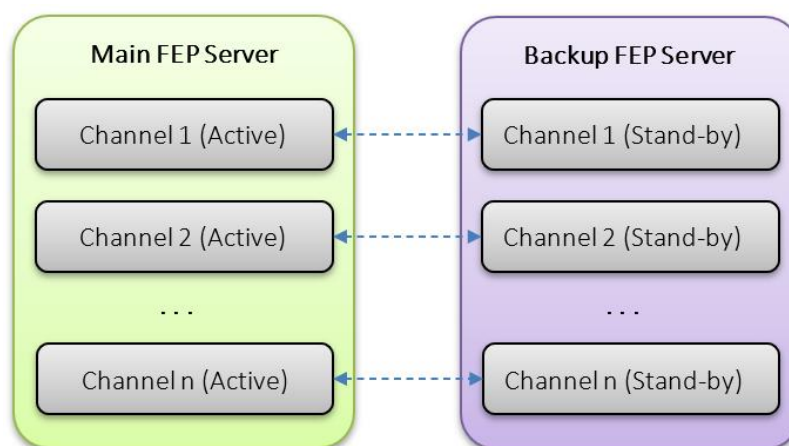
FEP Server supports the following key features:

- Support multiple channels (IEC61850, Modbus, IEC101, IEC103, IEC104, OPC, Sel Fast Message...).
- Ability to manage multiple channels at once.
- Each channel can be configured with one primary connection and multiple redundant connections (up to 4 connections).
- Supports **Force/Unforce** Point values (applicable to some special channels).
- Compatible with **OPC Unified Architecture** (OPC UA) applications.
- The interface on DE is simple and friendly.

D.1. OneATS FEP

D.1.1. OneATS FEP Architecture

OneATS Front-End Processing (FEP) Servers are servers that communicate real-time data with components outside the system. FEP Server exchanges data with these external components, and at the same time exchanges data with internal DataServers in the OneATS system, thereby creating continuous data flows in and out to provide information to the system.



FigureD-1 OneATS Fep redundancy model

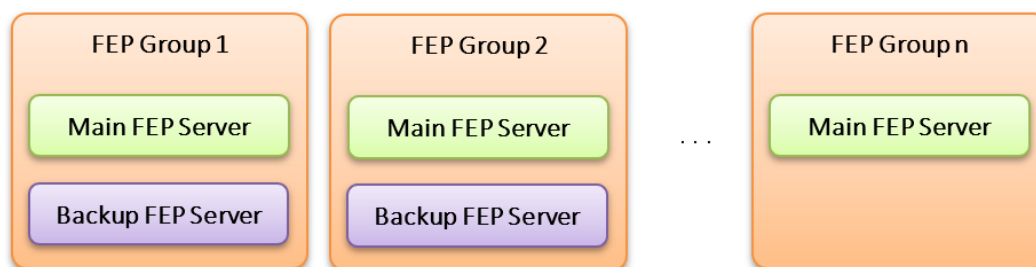
To exchange data with external components, FEPs are designed to consist of smaller components rather than connection channels. These channels are enforced based on data communication standards, which ensure consistency and compatibility when connecting systems of different manufacturers. The connection channels operate independently and can be controlled independently.

To ensure system availability, FEP Server is deployed into pairs of Main FEP Server and Backup FEP Server, the two Main and Backup servers will work together to ensure high system availability. In cases where there are no high availability requirements, it is possible to simply configure the Main FEP Server.

Main FEP Server: the channels on the Main FEP Server will be responsible for permanent data communication.

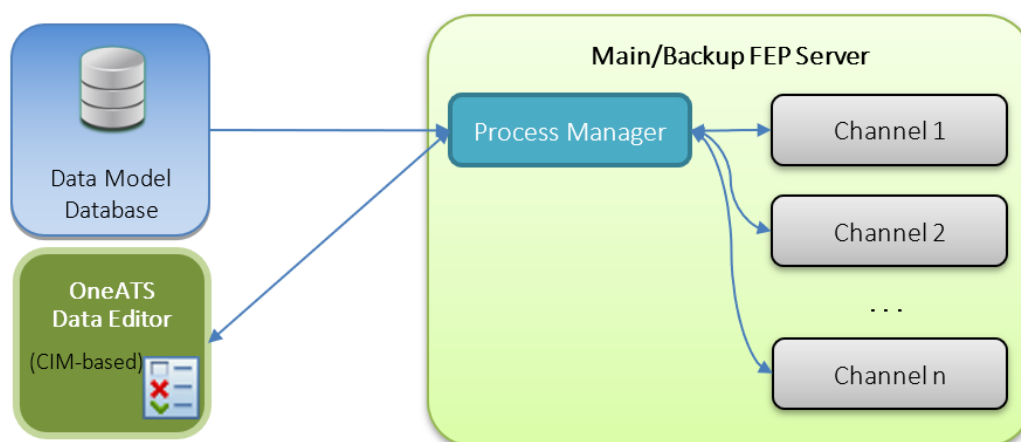
Backup FEP Server: the channels on the Backup FEP Server will be ready to connect, when there is a problem with the channels on the Main FEP Server, the channels on the Backup FEP Server will take over the task.

The channels will be laid out the same on both the Main FEP Server and the Backup FEP Server and the switchover when a problem occurs will be done for each channel and not for the entire server.



FigureD-2 Fep Group Organization

FEP Servers can be subdivided into groups with the Main FEP Servers and Backup FEP Servers of each group. This grouping helps to distribute the load to the physical servers and divide the channel groups according to the specific purpose.



FigureD-3 One ATS Fep module architecture

Process Manager is a service application, which is installed on both the Main FEP Server and the Backup FEP Server. Process Manager performs the following functions:

The Process Manager on each FEP Server will be configured to know if it is running on the Main or Backup machine.

Connect to the configuration database and read the entire configuration for the corresponding FEP Server.

Run all the channels declared to the respective FEP Server, which will be notified as running on the Main or Backup machine.

Monitor the operational status of all channels, if there is a problem with the channel (application is disabled, hanging) then the Process Manager will restart the corresponding channel.

Provide remote channel management services for the DataEditor application.

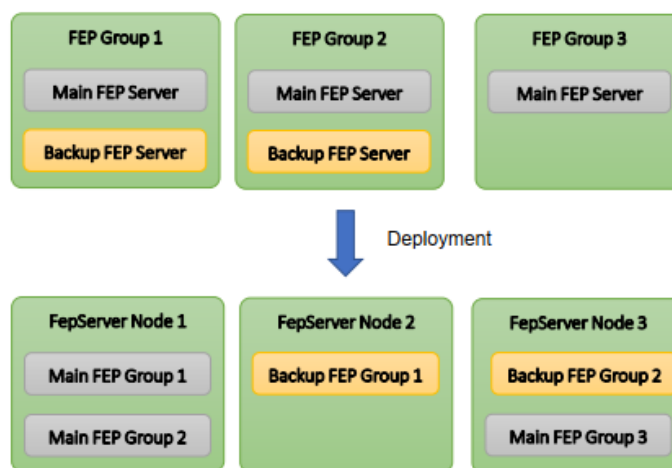
- Track activity status
- Start, Stop Control

- Display logging

D.1.2. Fep Redundancy

D.1.2.1 Implemented at the Group level

The way Fep Group is organized allows for the flexible implementation of redundancy configurations on physical computers. Accordingly, a Fep Server computer can take on multiple groups; correspondingly, each group can play different roles.



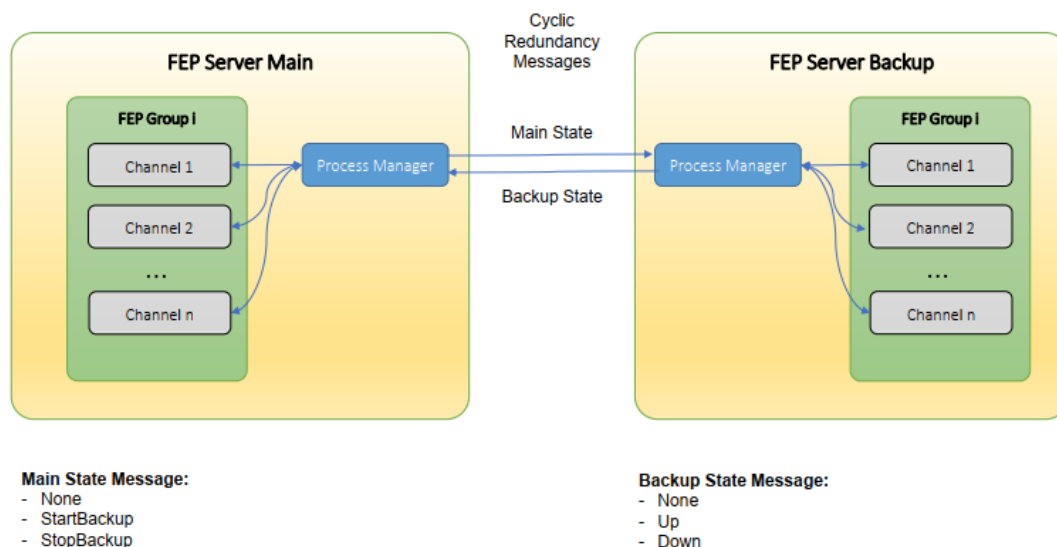
FigureD-4 Example of deploying Fep Server in a system

The above implementation allows for improved distribution, load sharing, flexibility, and scalability of the system.

D.1.2.2 Deploy at the channel level

- **Transmission mechanism between a pair of channels**

The Fep module allows redundancy to be implemented to the per-channel level. Pairs of main-backup channels will constantly send status updates to each other in response to fluctuations.



FigureD-5 Transmission mechanism between main-backup channel pairs

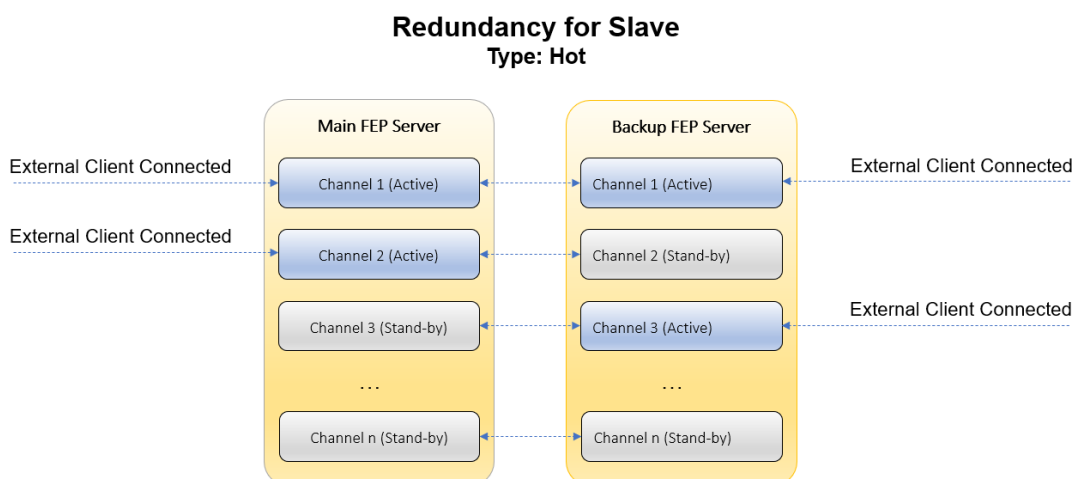
If 2 Fep Server computers lose connection, the backup server will command all channels in the group to run.

■ **Configuring redundancy for slave channels - TCP Server**

This configuration is being deployed for S101, S104, Modbus TCP Slave channels.

Deployment form: hot deployment only.

Mechanism: 2 servers of a pair of main-backup channels open the same port and serve any client that connects to it.



FigureD-6 Hot redundancy configuration for slave channels – TCP Server

■ **Configuring redundancy for master channels - TCP Client**

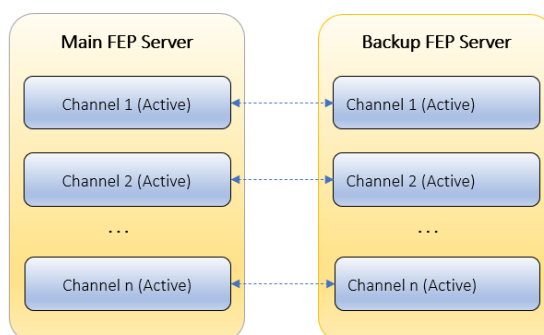
This configuration is being implemented for Ethernet-based channels such as Iec61850Client, M104, SNMP, Ping, OPC, PI Client, Iec62056-21Master.

Implementation form: can choose hot or warm.

○ **Hot Redundancy**

- Mechanism: 2 clients of a pair of main-backup channels connected to the same device/server. DataServer updates data from both channels, prioritizing the main channel.
- Deployment conditions: The connected device/server must be able to serve multiple sessions.

Redundancy for TCPClient Type: Hot

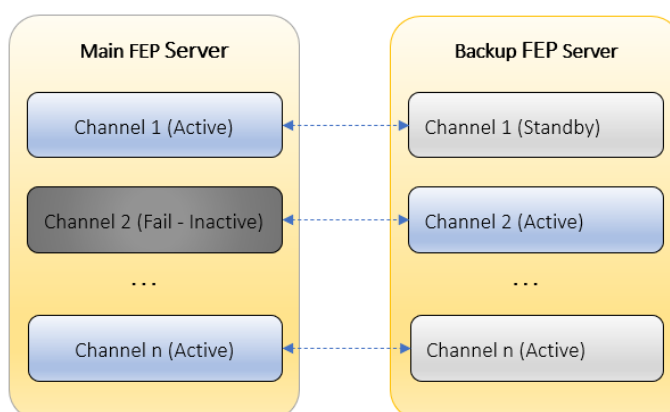


FigureD-7 Hot Redundancy Configuration for Master Channels – TCP Client

○ Warm Redundancy

- Mechanism: one of the two clients of a pair of main-backup channels is connected to the same device/server, prioritizing the main channel connection. If the main channel is not connected, the backup channel will be run; When the main channel reconnects, the backup channel will be disabled.

Redundancy for TCPClient Type: Warm



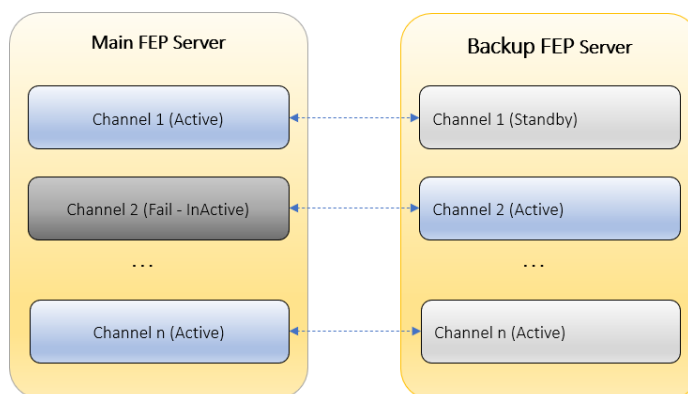
FigureD-8 Configuring warm redundancy for master channels – TCP Client

▪ Configuring redundancy for master – Serial channels

- This configuration is being deployed for serial-based channels such as ModbusMaster, M101, M103, DNP, Sel, HDLC.
- Even if the aforementioned channels use TCPClient connections, the implementation of redundancy is the same as using serial connections.

- Deployment form: only warm deployment.
- Mechanism: similar to the warm redundancy mechanism for the master channel – TCP Client. In addition, channels using the Serial transmission line need a period of time to listen to the transmission test before running, the time to transfer the state between the two pairs of main-backup channels will be longer than the master-TCP Client channel.

Redundancy for Serial Type: Warm



FigureD-9 Configuring warm redundancy for master-Serial channels

D.1.3. List of channels

FEP supports the following types of connection channels:

Channel name	Applicable Standards
IEC60870-5-101 Unbalanced Master	Standard IEC60870-5-101, with channel type for Controlling Station and Unbalanced communication type
IEC60870-5-101 Unbalanced Slave	Standard IEC60870-5-101, with channel type for Controlled Station and Unbalanced communication type
IEC60870-5-104 Master	Standard IEC60870-5-104, with channel type for Controlling Station
IEC60870-5-104 Slave	Standard IEC60870-5-104, with channel type for Controlled Station
IEC60870-5-103 Master	Standard IEC60870-5-103, with channel type for Controlling Station
IEC61850 Client	Standards IEC61850 apply to Client Ed2
IEC62056-21 Master	Standard IEC62056-21
Modbus Master	Modbus standard with Master channel type, including Modbus TCP and Modbus RTU
Modbus Slave	Modbus standard with Slave channel type, including Modbus TCP and Modbus RTU
OPC (Classic) Client	OPC Standard Classic (OPC DA) Client

Ping	ICMP Protocol
SEL	SEL Fast Message and SEL Fast Operate connection standards
SNMP	SNMP Protocol
DNP Master	DNP 3.0 connection standard, with channel type for Controlling Station
DNP OutStation	DNP 3.0 Connection Standard
HDLC	HDLC Connection Standard
Direct Control	Direct Control Links
MelsecNet	Mitsubishi PLC Data Exchange
Mudic Host	Based on PLC Mitsubishi data exchange
PI Client	Standard Connection to OSI PI Database
OPCUA Server	OPC UA Server Standard
OPC UA Client	OPC UA Client Standard

D.2. Install and configure the application

D.2.1. Install the app

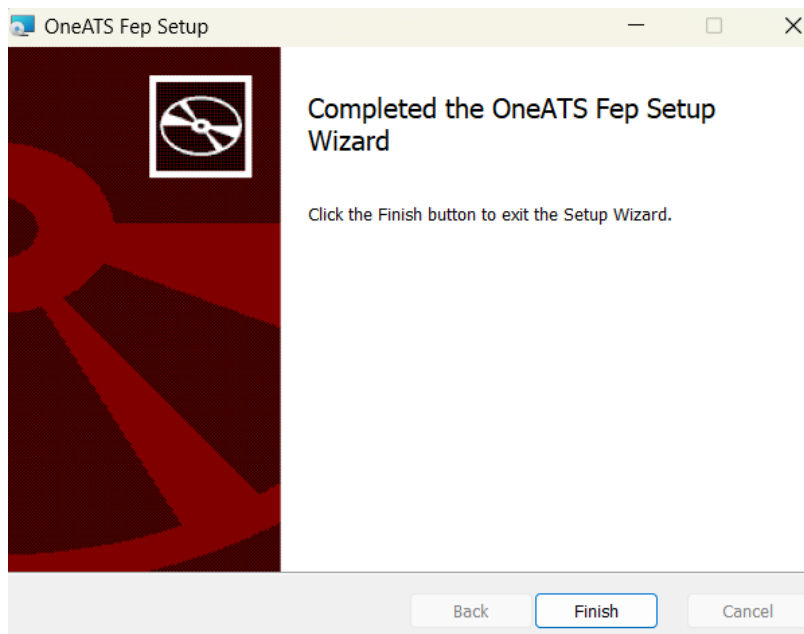
Run the **ATS OneATS FEP** installation file **Setup.msi**.

The installation screen appears, select **I accept the term** in the **License Agreement**, and then select **Install** to continue:



FigureD-10 Start installing OneATS Fep

Select **Finish** to finish the installation process:



FigureD-11. Finish the OneATS FEP installation process

D.2.2. Install and check your license

OneATS FEP Module is licensed for the number of channels/protocols and application runtime.

D.2.2.1 License Installation

Refer to the ChapterH.OneMasterTool documentation

D.2.2.2 Check License

Refer to the ChapterH.OneMasterTool documentation

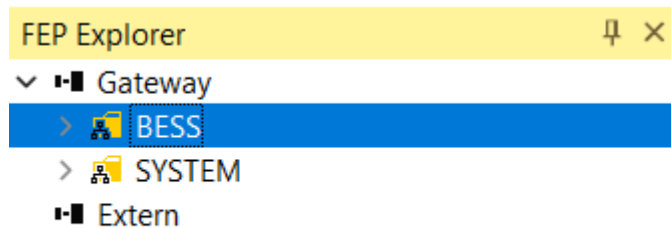
D.2.3. App Configuration

Refer to the ChapterH.OneMasterTool documentation

D.3. Data Configuration

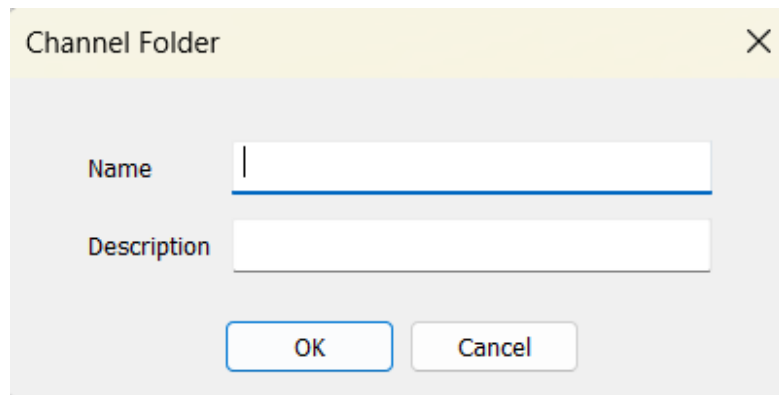
D.3.1. Folders

Folder is a tool to group FEP data channels into separate groups so that they can be viewed, collapsed, and expanded in the form of heading. Data channels can be flexibly grouped into the same folder (for example, the same type of data channel, belonging to the same Substation, etc.).



FigureD-12. FEP Explorer with Folders

In *the FEP Explorer* section, Right Click **Gateway >> New folder**, enter a folder name and a description of the folder, and click **OK**:

The image shows a dialog box titled "Channel Folder" with a close button in the top right corner. It contains two text input fields: "Name" and "Description". Below the fields are two buttons: "OK" and "Cancel".

FigureD-13. Add a new folder

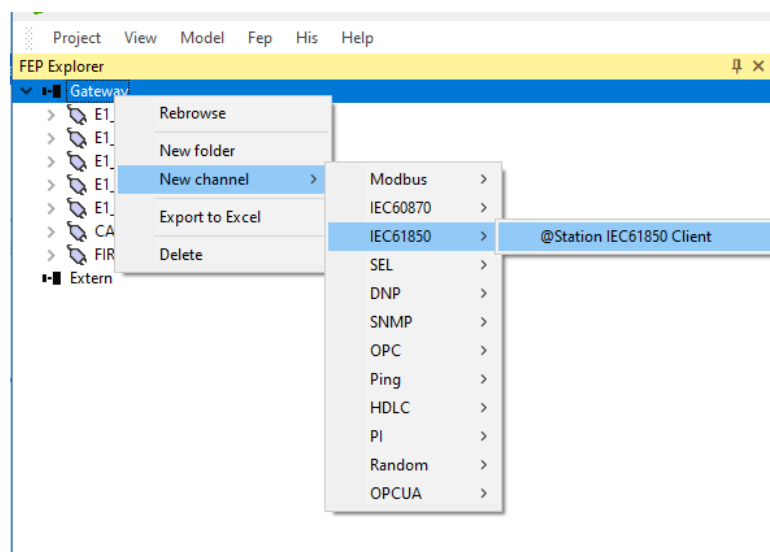
Configuration parameters:

- **Channel Name:** Folder name
- **Description:** Folder description

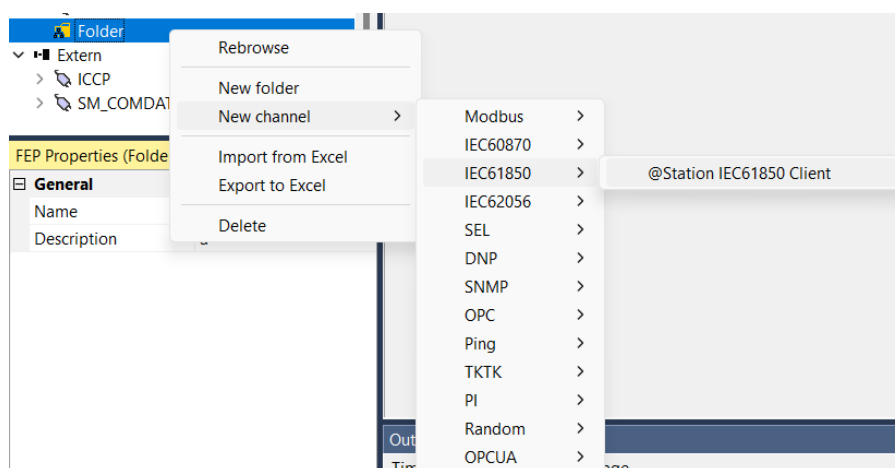
After creating a folder, you can arbitrarily move FEP channels in and out of different folders through dragging and dropping with the left mouse.

D.3.2. IEC61850 Client Channels

D.3.2.1 Add a new channel

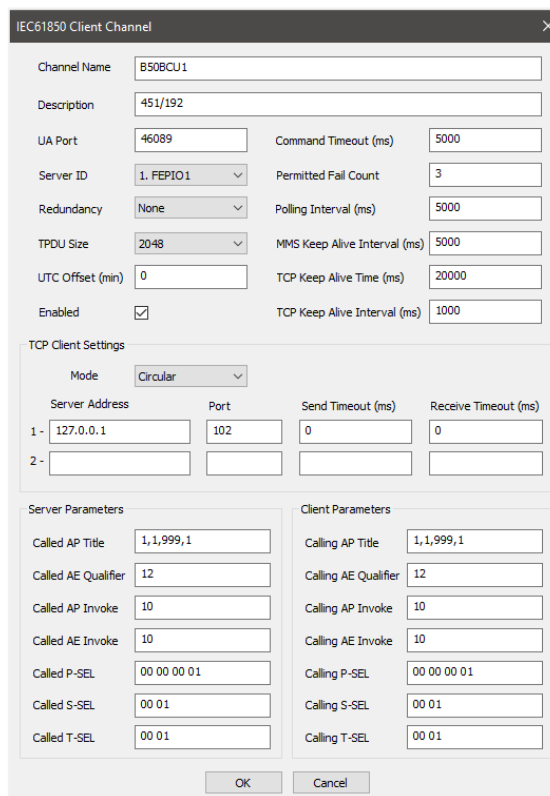


FigureD-14. New IEC 61850 Channel Added from Gateway



FigureD-15. Add a new IEC 61850 channel from the folder

In the **FEP Explorer** section, the Right Click **Gateway** >> **New channel** >> **IEC61850** >> **@Station IEC61850 Client** or Right Click Folder created >> **New channel** >> **IEC61850** >> **@Station IEC61850 Client** to add a new one:



The image shows a screenshot of the 'IEC61850 Client Channel' configuration window. It contains various input fields and dropdown menus for configuring a client channel. The fields are organized into sections: Channel Name, Description, UA Port, Server ID, Redundancy, TPDU Size, UTC Offset, Enabled, TCP Client Settings, Server Parameters, and Client Parameters. The 'Enabled' checkbox is checked. The 'TCP Client Settings' section includes a 'Mode' dropdown set to 'Circular' and a table for 'Server Address' and 'Port' with two rows. The 'Server Parameters' and 'Client Parameters' sections each have a table with fields for Called AP Title, Called AE Qualifier, Called AP Invoke, Called AE Invoke, Called P-SEL, Called S-SEL, and Called T-SEL.

FigureD-16. IEC 61850 Channel Configuration

Fill in the information for the channel, click **OK**.

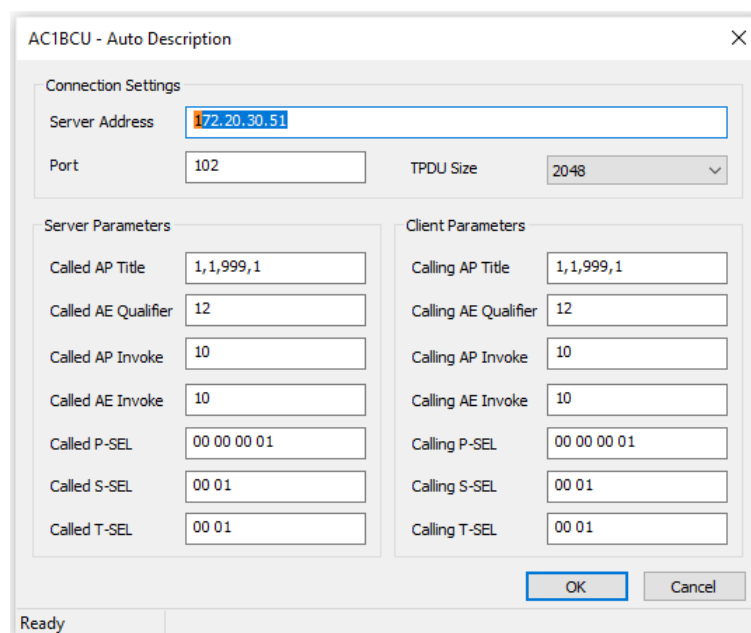
Configuration parameters:

- **Channel Name:** Channel name, valid name using only the letter A... Z; 0....9; Spacing; Underscore and must not start with a number. This rule applies to all channels that FEP supports.
- **Description:** Channel description.
- **UA Port:** The gateway returned to the OPC UA (Software-generated, reconfigurable for ease of management).
- **Server ID:** The ID of the server connecting the channel.
- **Redundancy:**
 - *None:* Do not use.
 - *Warm:* At a time only one FEP IO is running.
 - *Hot:* Both FEP IOs are running at the same time.
- **TCP Client Setting:**
 - *Server Address:* the IP address of the device.
 - *Port:* the protocol port of the device (default is 102).
- **Enabled:** Select/Uncheck to Enable/Disable channel.

D.3.2.2 Create a data list

There are two options for creating a data list: **Auto Description** and **Load from SCL file**.

- **Auto description:** Right click the channel, select **Auto description** from the drop-down list.



The dialog box is titled "AC1BCU - Auto Description". It contains two main sections: "Connection Settings" and "Server Parameters" / "Client Parameters".

Connection Settings:

- Server Address: 172.20.30.51
- Port: 102
- TPDU Size: 2048

Server Parameters:

- Called AP Title: 1,1,999,1
- Called AE Qualifier: 12
- Called AP Invoke: 10
- Called AE Invoke: 10
- Called P-SEL: 00 00 00 01
- Called S-SEL: 00 01
- Called T-SEL: 00 01

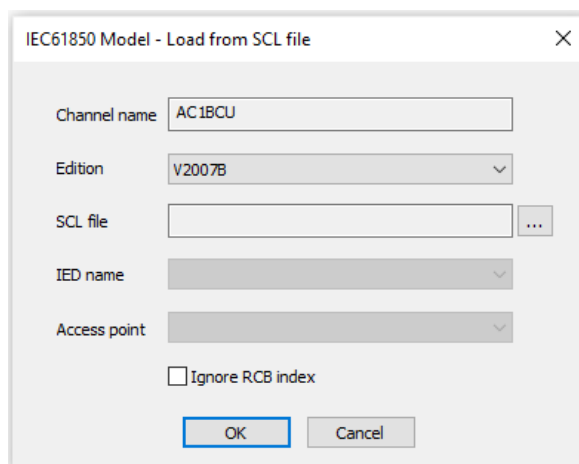
Client Parameters:

- Calling AP Title: 1,1,999,1
- Calling AE Qualifier: 12
- Calling AP Invoke: 10
- Calling AE Invoke: 10
- Calling P-SEL: 00 00 00 01
- Calling S-SEL: 00 01
- Calling T-SEL: 00 01

At the bottom right are "OK" and "Cancel" buttons. At the bottom left is a "Ready" status indicator.

FigureD-17. Create IEC61850Client channel data list using Auto description

- **Load from SCL file:** Right Click the channel, select **Load from SCL file** from the drop-down list.



The dialog box is titled "IEC61850 Model - Load from SCL file". It contains the following fields:

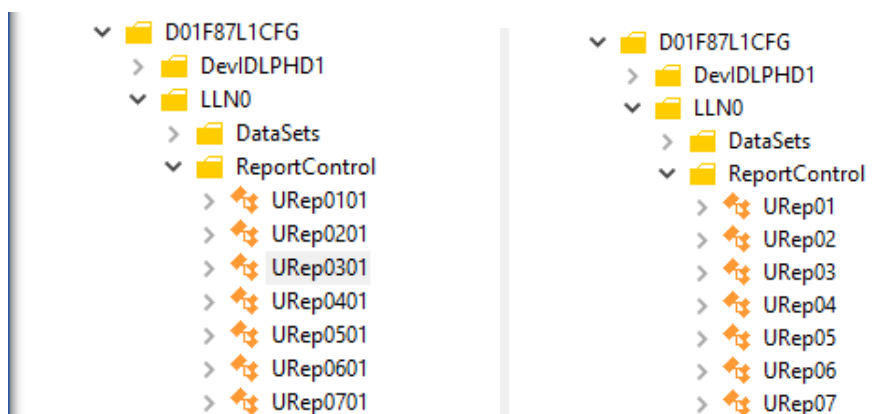
- Channel name: AC1BCU
- Edition: V2007B
- SCL file: (empty field with a browse button "...")
- IED name: (empty dropdown)
- Access point: (empty dropdown)
- ☐ Ignore RCB index

At the bottom are "OK" and "Cancel" buttons.

FigureD-18. Create a list of channel data from an SCL file

Options:

- **Edition:** IEC61850 standard version lookup in .cid file
- **SCL file:** the path to the .cid file
- **Ignore RCB index:** ignore the RCB Index parameter described in the cid file to avoid the names of the report nodes being numbered repeatedly.



Deselect Ignore RCB index

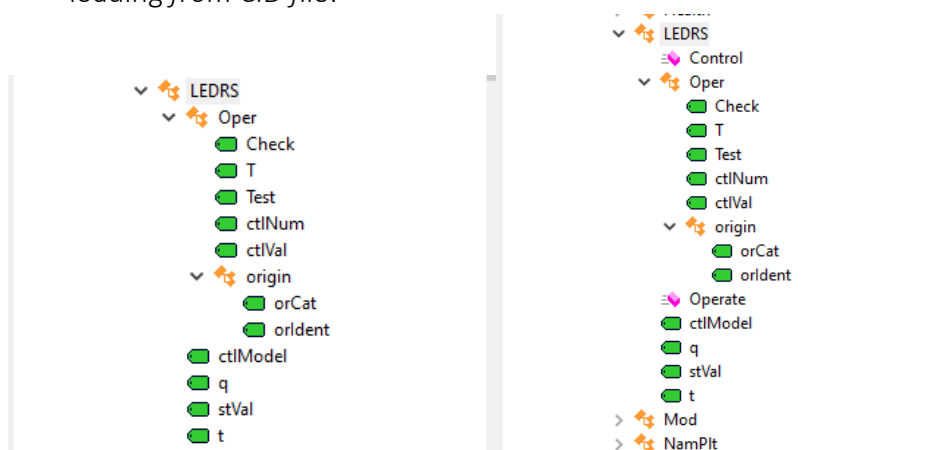
Choose Ignore RCB index

Figure D-19 Skip RCB Index using .cid files

It is not always necessary to select **Ignore RCB index**, it is important that the RCB name on the server (IED) and the client match.

▪ **Note:**

- When performing *Auto description*, the relay information returned to create a data list does not include *DataTypeTemplates* as when *loading from CID file*.
- When performing *Auto description*, the DE module automatically defines the CDC (Common Data Class) of the DOs based on the name of the DO. In some cases where the name of the DO is not common, the CDC cannot be defined, and the corresponding DO may be missing a DA or Method compared to when *loading from CID file*.
- In this situation, it is recommended to create a list of data for the channel by *loading from CID file*.



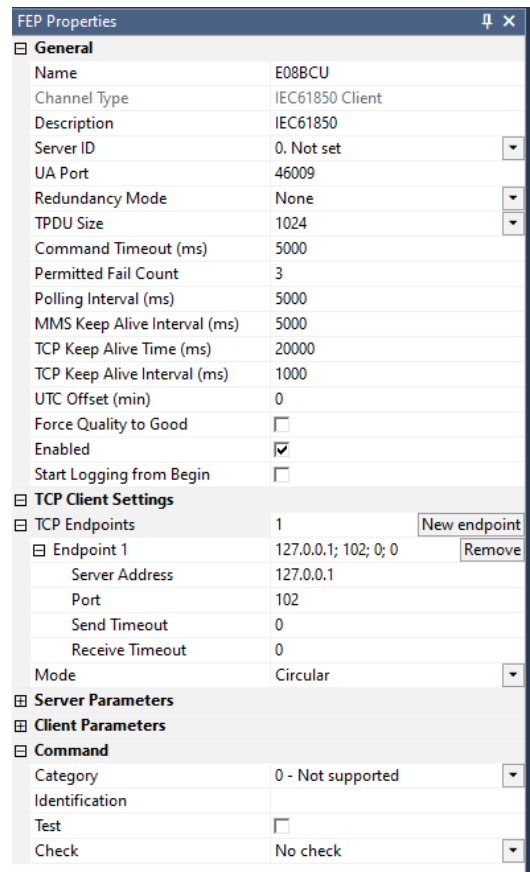
Auto description

Load from .cid file

Figure D-20 CDC unidentified error when auto description

D.3.2.3 Channel Calibration

The FEP Properties **section** allows users to view and change channel information:



The FEP Properties dialog box is shown with the General tab selected. It contains the following fields and values:

Property	Value
Name	E08BCU
Channel Type	IEC61850 Client
Description	IEC61850
Server ID	0. Not set
UA Port	46009
Redundancy Mode	None
TPDU Size	1024
Command Timeout (ms)	5000
Permitted Fail Count	3
Polling Interval (ms)	5000
MMS Keep Alive Interval (ms)	5000
TCP Keep Alive Time (ms)	20000
TCP Keep Alive Interval (ms)	1000
UTC Offset (min)	0
Force Quality to Good	☐
Enabled	☒
Start Logging from Begin	☐

The TCP Client Settings tab is also visible, showing one endpoint:

Endpoint	Server Address	Port	Send Timeout	Receive Timeout	Mode
Endpoint 1	127.0.0.1	102	0	0	Circular

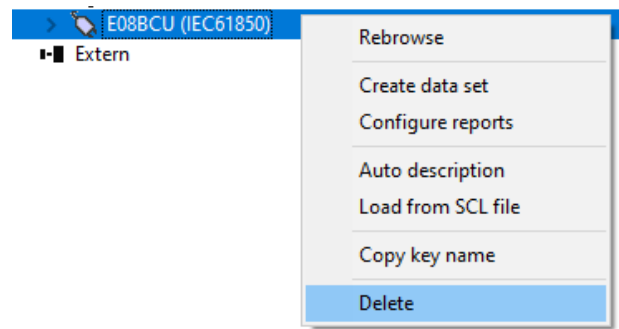
The Server Parameters and Client Parameters tabs are also visible, showing the following fields and values:

Property	Value
Category	0 - Not supported
Identification	
Test	☐
Check	No check

FigureD-21. IEC 61850 Channel Calibration

D.3.2.4 Delete a channel

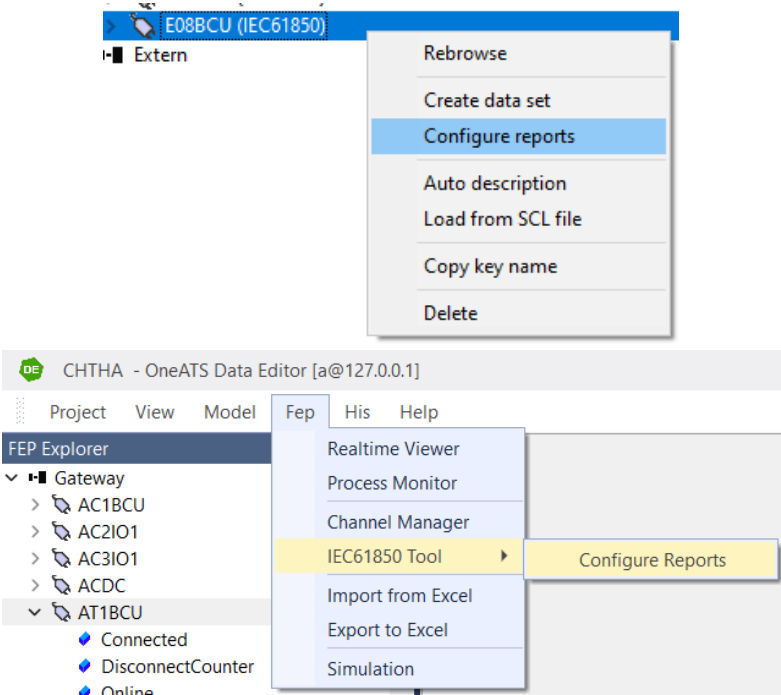
Right-click on the channel you want to delete and select **Delete**.



FigureD-22. IEC 61850 Channel Removal

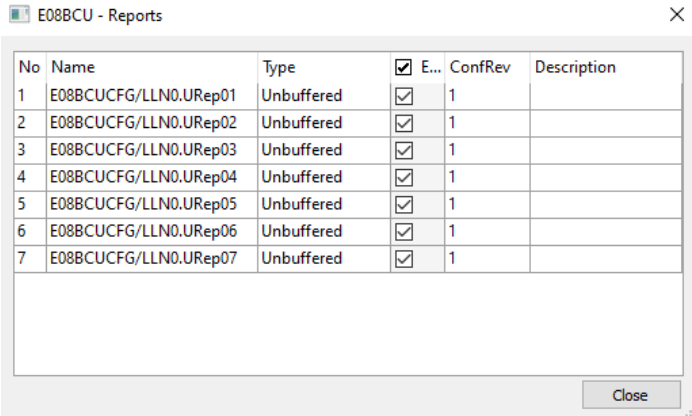
D.3.2.5 Report Configuration

Right-click on the channel and select **Configure report** (shortcut: **Ctrl+R**) or go to the Fep/IEC61850 Tool/Configure Reports tab.



FigureD-23. Channel report configuration menu

A list of Report options appears. Double-click on a specific report to configure it in detail:



The screenshot shows the 'E08BCU - Reports' dialog box. It contains a table with 7 rows of report data. The table has columns: No, Name, Type, ☒ E..., ConfRev, and Description. Below the table is a 'Close' button.

No	Name	Type	☒ E...	ConfRev	Description
1	E08BCUCFG/LLN0.URep01	Unbuffered	☒	1	
2	E08BCUCFG/LLN0.URep02	Unbuffered	☒	1	
3	E08BCUCFG/LLN0.URep03	Unbuffered	☒	1	
4	E08BCUCFG/LLN0.URep04	Unbuffered	☒	1	
5	E08BCUCFG/LLN0.URep05	Unbuffered	☒	1	
6	E08BCUCFG/LLN0.URep06	Unbuffered	☒	1	
7	E08BCUCFG/LLN0.URep07	Unbuffered	☒	1	

No	Channel	Name	Type	Ena...	ConfRev	Description
1	AC1BCU	AC1BCUCFG/LLN0.URep01	Unbuffered	☒	1	
2	AC1BCU	AC1BCUCFG/LLN0.URep02	Unbuffered	☒	1	
3	AC1BCU	AC1BCUCFG/LLN0.URep03	Unbuffered	☒	1	
4	AC1BCU	AC1BCUCFG/LLN0.URep04	Unbuffered	☒	1	
5	AC2IO1	AC2IO1CFG/LLN0.URep01	Unbuffered	☒	1	
6	AC2IO1	AC2IO1CFG/LLN0.URep02	Unbuffered	☒	1	
7	AC2IO1	AC2IO1CFG/LLN0.URep03	Unbuffered	☒	1	
8	AC2IO1	AC2IO1CFG/LLN0.URep04	Unbuffered	☒	1	
9	AC3IO1	AC3IO1CFG/LLN0.URep01	Unbuffered	☒	1	
10	AC3IO1	AC3IO1CFG/LLN0.URep02	Unbuffered	☒	1	
11	AC3IO1	AC3IO1CFG/LLN0.URep03	Unbuffered	☒	1	
12	AC3IO1	AC3IO1CFG/LLN0.URep04	Unbuffered	☒	1	
13	AT1BCU	AT1BCUCFG/LLN0.URep01	Unbuffered	☒	1	
14	AT1BCU	AT1BCUCFG/LLN0.URep02	Unbuffered	☒	1	
15	AT1BCU	AT1BCUCFG/LLN0.URep03	Unbuffered	☒	1	
16	AT1BCU	AT1BCUCFG/LLN0.URep04	Unbuffered	☒	1	
18	AT1F64	AT1F64CFG/LLN0.BRep1	Buffered	☒	1	
19	AT1F64	AT1F64CFG/LLN0.BRep2	Buffered	☒	1	
20	AT1F64	AT1F64CFG/LLN0.BRep3	Buffered	☒	1	
21	AT1F64	AT1F64CFG/LLN0.BRep4	Buffered	☒	1	

FigureD-24. Detailed Configuration Report

E08BCUCFG/LLN0.URep01 - Report (Edit)

Description

Configuration revision

1

Configuration session

Apply for ALL sessions

☒ Enabled report

Report settings

☐ Report ID
☐ Data set
☐ Buffer time
☐ Integrity period

☒ Trigger GI after report is enabled
☐ Resend GI timeout 5000 ms

Trigger options

☐ Data change
☐ Quality change
☐ Data update

☐ Integrity
☐ General interrogation

Optional fields

☐ Sequence number
☐ Report timestamp
☐ Reason for inclusion
☐ Data set name
☐ Data reference

☐ Buffer overflow
☐ Entry ID
☐ Conf revision
☐ Segmentation

OK

Cancel

FigureD-25. The interface supports the selection of report customizations for users

Important Installation Parameters:

- **Enabled report:** select to activate the report.
- **Trigger option:** select the events where the server pushes the report, usually need Data change, quality change, and General interrogation.
- **Trigger GI after report is enabled (ms):** If the GI option in the Trigger option is enabled, this option causes the server to send the report as soon as the report is enabled. This option needs to be enabled so that the report data is fully updated as soon as the channel is connected.

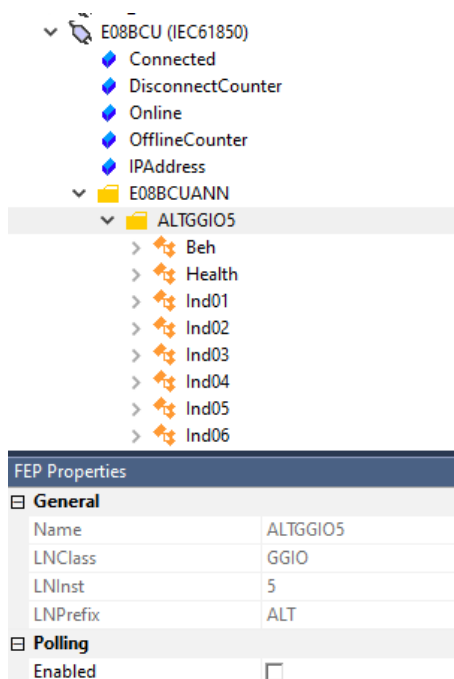
Other parameters:

- **Configuration revision:** a parameter that helps the channel IEC61850 Client manage the configuration versions of the report.
- **Report ID:** The name of the report.
- **Data set:** the reference of the dataset used by the Report.
- **Buffer time (ms):** After each buffer time cycle, the server will collect all the events (data-change, quality-change, data-update) that occurred during that time period to create a single report sent to the client.
- **Integrity period (ms):** If the *Integrity* option in the *Trigger option* is enabled, after each *Integrity period*, the server will send a report containing all the information of the corresponding data-set regardless of whether any event occurs.
- **Purge buffer: (only used for BRCB)** This option causes the server to reset the cache containing the BRCB report.
- **Optional fields:** optional information to be sent by the server in the report newsletter.

D.3.2.6 Polling Configuration

Expand the data tree 61850 → select the FEP **Properties node** → select **Enabled** in the Polling section.

The Polling **data update mechanism** is only complementary and responsive to a finite number of data points and is not recommended as an alternative to **the Report** mechanism in the IEC-61850 protocol

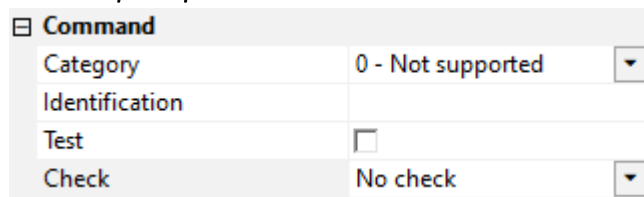


FigureD-26. Polling Configuration

Polling configurations can be applied to LogicalNodes or DataObjects. When polling is enabled for a LogicalNode, all DataObjects within it are considered to be polled.

D.3.2.7 Configure control commands

Open the channel's *Fep Properties* interface and scroll down to the *Command section*



Command	
Category	0 - Not supported
Identification	
Test	☐
Check	No check

FigureD-27 Configure control commands

Configuration parameters:

- **Category:** parameter specifying the control level of the client (0 = Not supported, 1 = Bay control, 2 = Station control, 3 = Remote control, 4 = Automatic Bay, 5 = Automatic station, 6 = Automatic remote, 7 = Maintenance, 8 = Process).

- **Identification:** The integer number of the client being controlled.

Category and **Identification** will be written down to the IED when controlled.

- **Test:** Used during a test run, the IED performs all the test procedures (syncho-check/ interlock-check); but does not act (does not change stVal, the IED does not output control commands to IO).

- **Check:** When selected, the client will ask to check the conditions during the execution of the command.

Syncho-check: Check the electrical parameters of the device (voltage, frequency, phase angle, ...).

Interlock-check: Check the interlocking conditions.

D.3.2.8 Using a channel

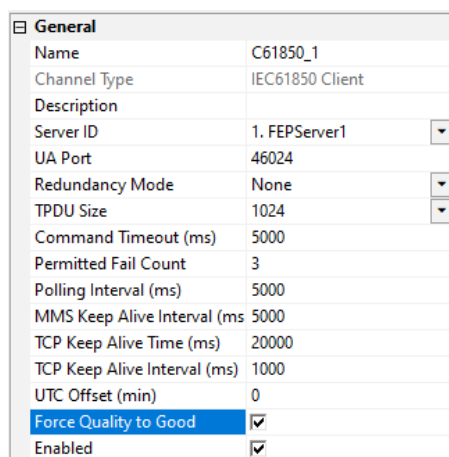
A few notes when using the IEC61850Client channel.

- **Quality of the points**

- Standard IEC61850 specifies that a DataObject consists of 3 distinct points: stVal, q, t. When retrieving data, the value of point q will be used to update the quality of stVal, and the value of point t will be used to update the timestamp for stVal.

- In some cases, point stVal is updated correctly, but point q is not updated correctly (q.value = Reserved, Invalid, or Questionable), the channel will update stVal.quality = Bad by default.

- To fix the need for force quality for stVal to Good by selecting the "Force Quality To Good" option in the channel settings.



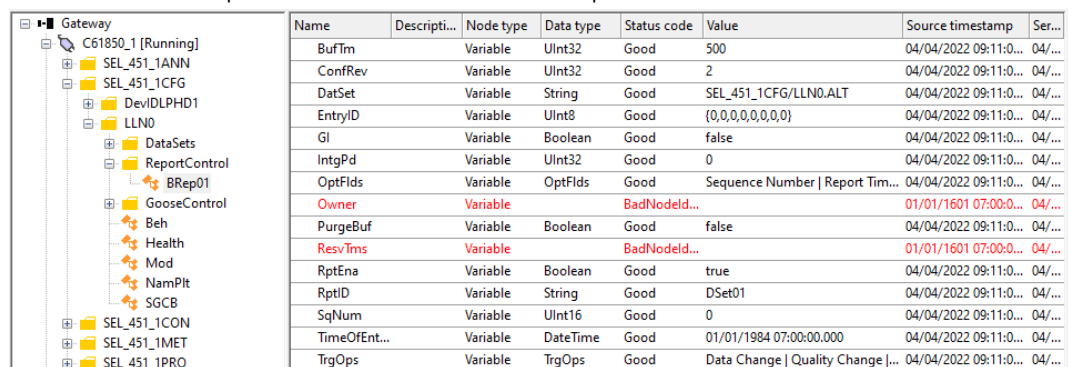
General	
Name	C61850_1
Channel Type	IEC61850 Client
Description	
Server ID	1. FEPServer1
UA Port	46024
Redundancy Mode	None
TPDU Size	1024
Command Timeout (ms)	5000
Permitted Fail Count	3
Polling Interval (ms)	5000
MMS Keep Alive Interval (ms)	5000
TCP Keep Alive Time (ms)	20000
TCP Keep Alive Interval (ms)	1000
UTC Offset (min)	0
Force Quality to Good	☒
Enabled	☒

FigureD-28 Force Quality

Report

When a channel can't get a report, the following checks should be taken:

- Check the Report Control Block via the Fep RealtimeViewer interface.



Name	Descripti...	Node type	Data type	Status code	Value	Source timestamp	Ser...
BufTim		Variable	UInt32	Good	500	04/04/2022 09:11:0...	04/...
ConfRev		Variable	UInt32	Good	2	04/04/2022 09:11:0...	04/...
DatSet		Variable	String	Good	SEL_451_1CFG/LLN0.ALT	04/04/2022 09:11:0...	04/...
EntryID		Variable	UInt8	Good	{0,0,0,0,0,0,0}	04/04/2022 09:11:0...	04/...
GI		Variable	Boolean	Good	false	04/04/2022 09:11:0...	04/...
IntgPd		Variable	UInt32	Good	0	04/04/2022 09:11:0...	04/...
OptFlds		Variable	OptFlds	Good	Sequence Number Report Tim...	04/04/2022 09:11:0...	04/...
Owner		Variable		BadNodeId...	01/01/1601 07:00:0...	04/...	04/...
PurgeBuf		Variable	Boolean	Good	false	04/04/2022 09:11:0...	04/...
ResvTms		Variable		BadNodeId...	01/01/1601 07:00:0...	04/...	04/...
RptEna		Variable	Boolean	Good	true	04/04/2022 09:11:0...	04/...
RptID		Variable	String	Good	DSet01	04/04/2022 09:11:0...	04/...
SqNum		Variable	UInt16	Good	0	04/04/2022 09:11:0...	04/...
TimeOfEnt...		Variable	DateTime	Good	01/01/1984 07:00:00.000	04/04/2022 09:11:0...	04/...
TrgOps		Variable	TrgOps	Good	Data Change Quality Change ...	04/04/2022 09:11:0...	04/...

FigureD-29 Report Control Block Monitoring Interface

- If all the points of the Report Control Block (RCB) are bad, it is necessary to review the process of loading the .cid file, making sure:
 - (1) The IEC61850 version is the same as described in the .cid file
 - (2) The node name of the Client and Server Report Control Block matches, customized by the parameter "Ignore RCB Index".
- If the channel has read about the value of the RCB, make sure:
 - (1) RptEna = true.
 - (2) TrgOps has the desired value, normally TrgOps = Data Change | Quality Change | General Interrogation.
 - (3) The DataSet exists and consists of DOs and DAs that need to retrieve data.
 - (4) The GI for the report has been configured.
- When checking to confirm that the RCB is configured correctly but the channel still has not received the report, it is necessary to continue to consider the following issues:
 - Client-busy

The IEC61850Client channel is designed to accommodate up to 100 commands in the queue.

If there are too many polling points or the polling interval is too small, the IEC61850Client channel will be in the client-busy state and the report configuration command may be canceled, then RptEna = false and the server will not send the report to the client.

- Some IEDs (servers) do not serve multiple clients via an RCB

Then if the server is serving another client, the new client will not get the report. Users may need to reconfigure the server to create multiple RCBs, each serving a report for a client.

- o Other scenarios need to monitor the channel's logs for more analytical information.

- **Control**

When control is not possible through the IEC61850Client channel, the following issues should be checked:

- Control Models

Check the variable value of ctlModel DataObject via Fep RealtimeViewer; if ctlModel = 0 – Status Only, the corresponding DataObject will not receive the control command.

- orCat

Some IEDs only accept commands in certain categories, so it is necessary to declare the Category parameter in the channel's command settings appropriately.

- Client-busy: similar to the case where the report cannot be retrieved.
- Other scenarios need to monitor the channel's logs for more analytical information.

D.3.2.9 Channel Tracking

- **Track channel status**

- Connected: Connected = true after successfully establishing a connection between the client and the server.
- DisconnectedCounter: the number of times the channel has lost connection since launch.
- Online: Online=true after the *Association service* as specified by the IEC61850 protocol is successfully implemented.
- OfflineCounter: the number of times the channel has gone offline since launch.

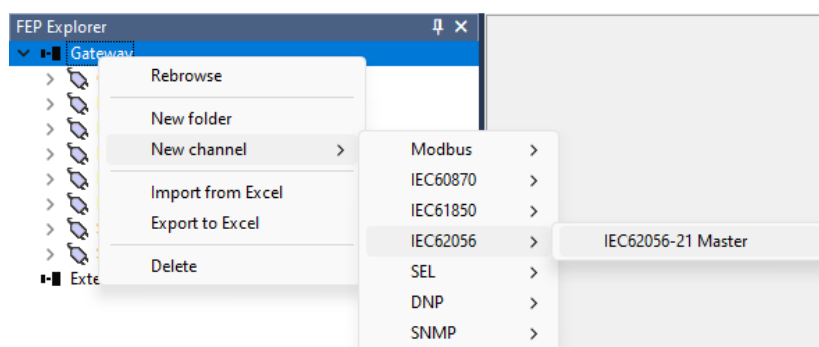
D.3.3. Channel IEC62056-21 Master

D.3.3.1 Add a new channel

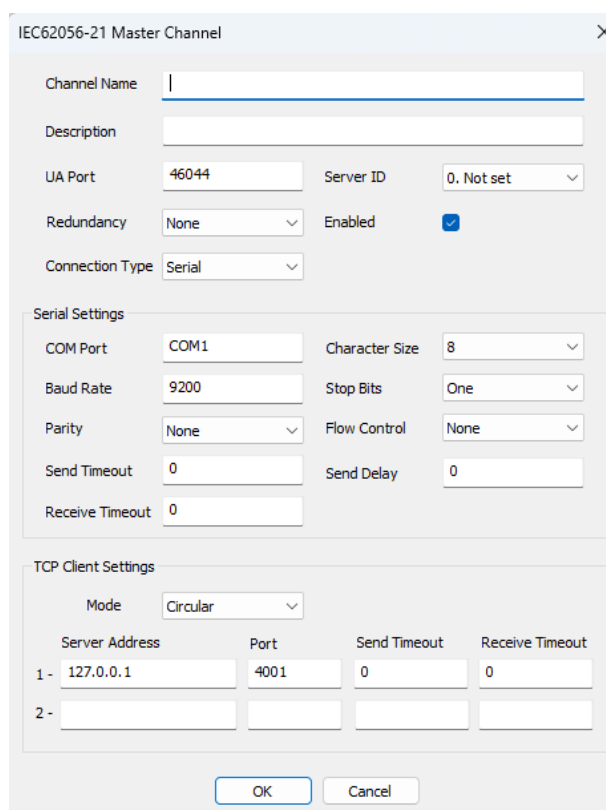
There are two ways to add a new channel: manually and import from a file.

- **Import from excel file:** From the Menu Bar, select *Fep >> Import from Excel*. Select the file path, select the channel, and the Import Mode option. Refer to the IEC101 Master channel using Import from the excel file.
- **Handicraft**
 - **Add a Channel**

In the *FEP Explorer* section, Right Click **Gateway/** Folder has created >> **New channel >> IEC62056 >> IEC62056-21 Master** to add a new one.



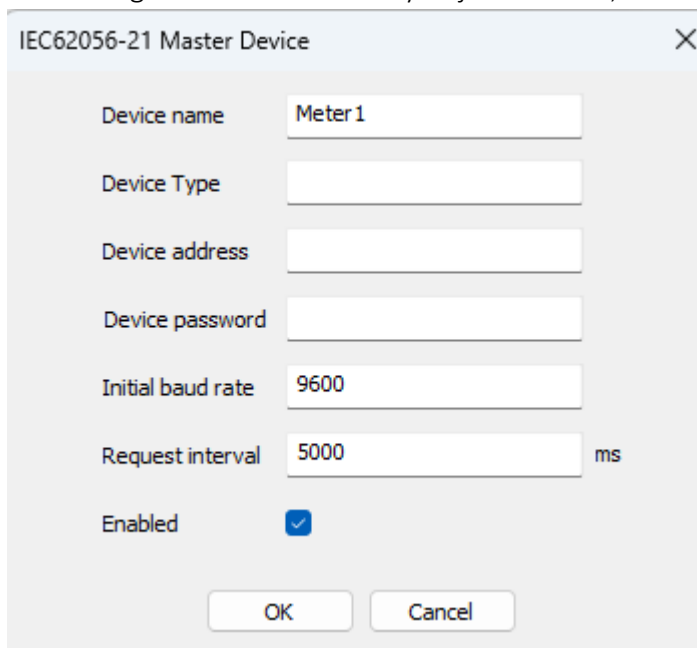
FigureD-30 Added new channel IEC62056-21



FigureD-31 IEC62056-21 Channel Configuration

Configuration parameters:

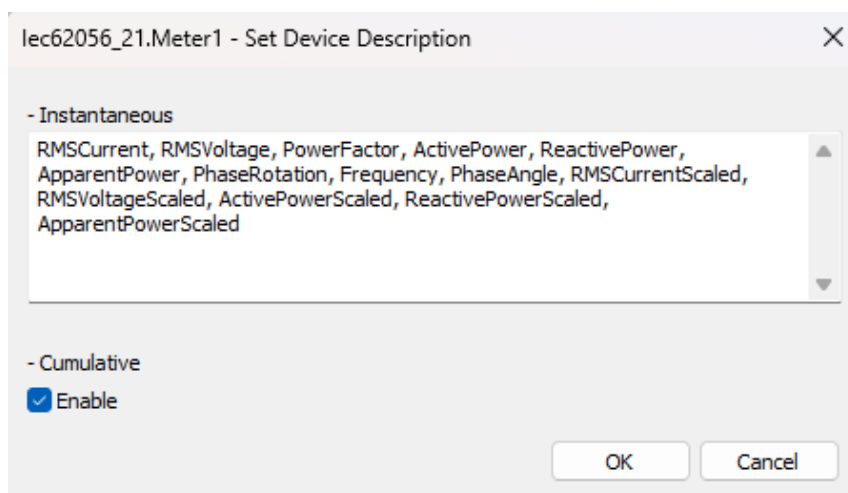
- **Channel Name:** Channel name
 - **Description:** Channel description
 - **UA Port:** Port return for OPC UA
 - **Server ID:** ID of the machine running FEP
 - **Enabled:** Enable/Disable Channel
 - **Redundancy:** Redundancy Mode
- **Add Device:** Right Click the channel you just created, select **Add Device**



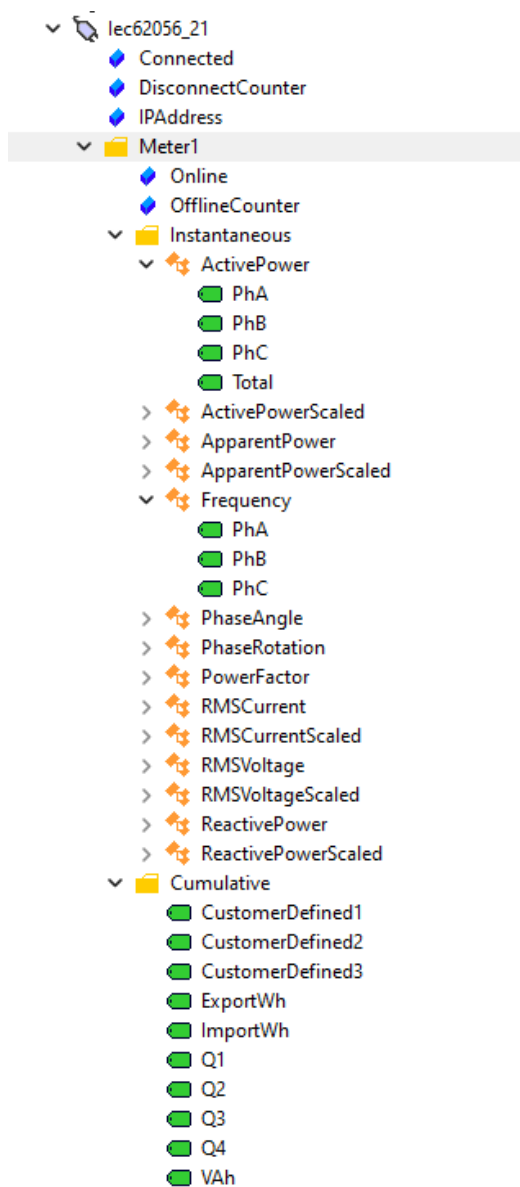
FigureD-32 Configure More Devices

Configuration parameters:

- **Name:** Device name
 - **Device Type:** Device Type
 - **Device Address:** The address of the device
 - **Device Password:** The password of the device
 - **Initial Baud Rate**
 - **Request Interval (ms):** Frequency of data polling
- **Set Device Description:**
 - Right Click the device you just created, select **Set Device Description**.
 - Enter the point name, the list of valid point names is taken from document IEC62056-21.
 - **Enable:** Enable/Disable status
 - Select **OK**.



FigureD-33. Interface Set Device Description



FigureD-34 Example of a list of points

D.3.3.2 Channel Calibration

The FEP Properties **interface** allows users to view and change channel information, devices, and points

D.3.3.3 Delete a channel

Right-click on the channel you want to delete and select **Delete**:

D.3.3.4 Using a channel

- Channel IEC62056-21 collects data from meters in turn.
- At each point, channel IEC62056-21 will log in and exchange data from only one meter. Before collecting data, the data channel needs to log in to Meter.
- The correct Device Number and Device Password are required for the login process.

D.3.3.5 Channel Tracking

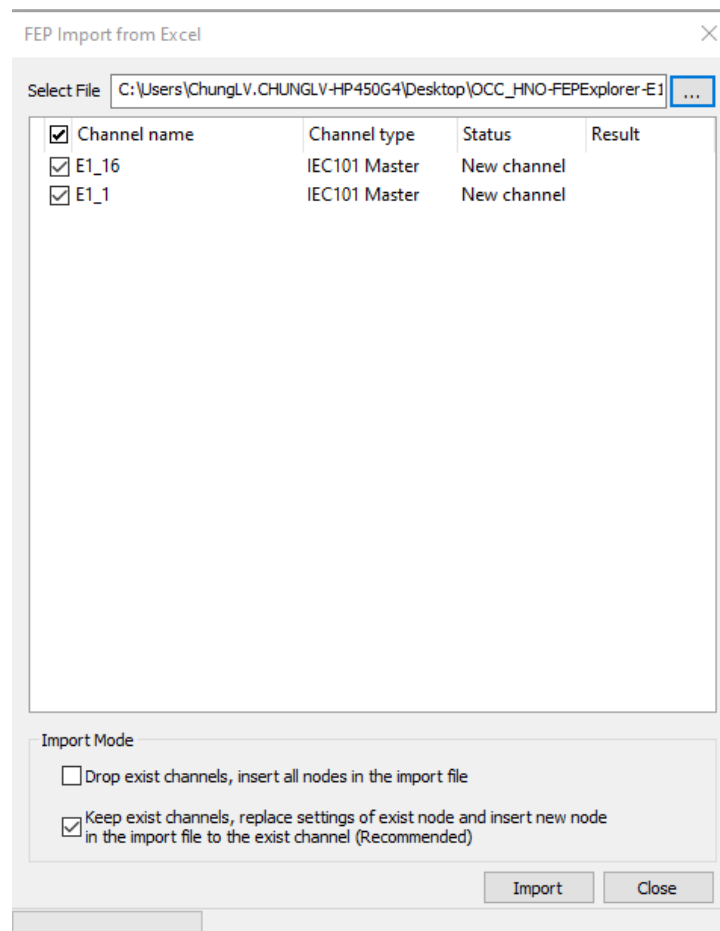
- **Track channel status**
 - Connected: Shows whether the connection was successful or not.
- **Device Status Tracking**
 - Online: Online = true, the device is ready to exchange data with the FEP module.

D.3.4. Channel IEC60870-5-101 Master

D.3.4.1 Add a new channel

There are two ways to add a new channel: manually and import from a file.

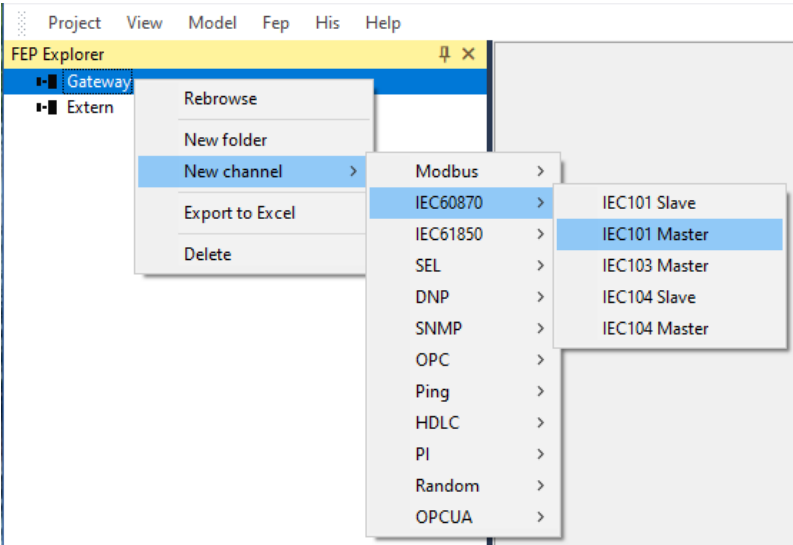
- **Import from excel file:** From the Menu Bar, select **Fep >> Import from Excel**. Select the file path, select the channel, and the Import Mode option.



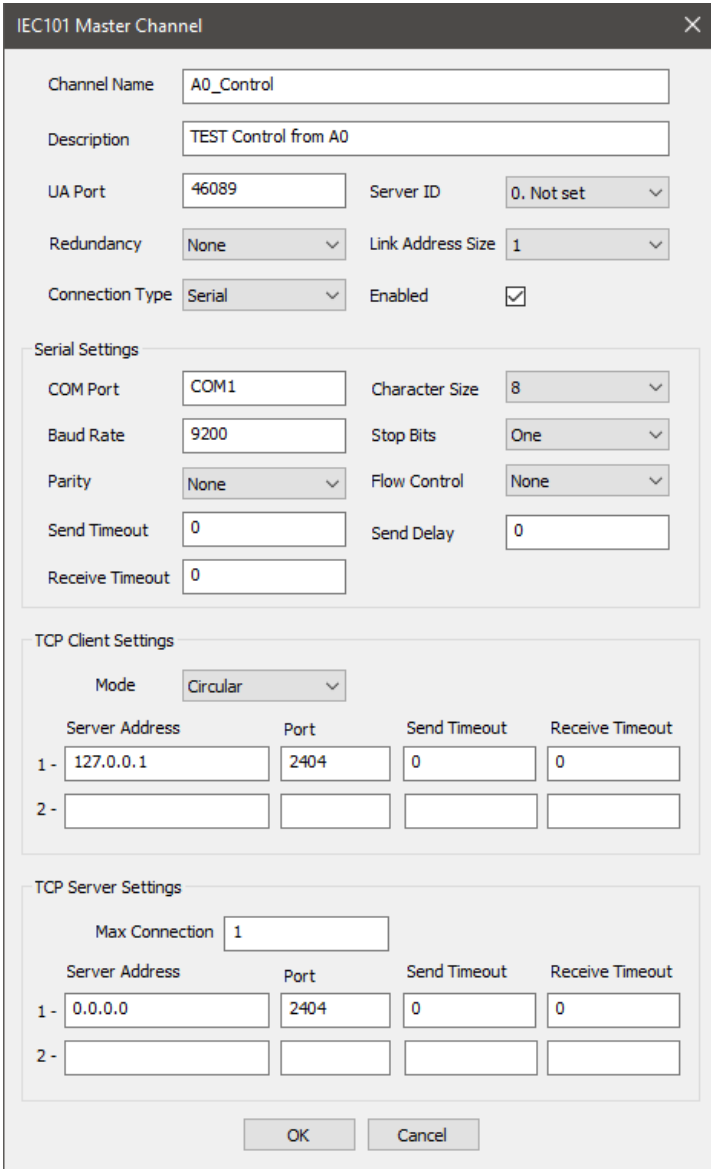
FigureD-35 Add a channel from a file

- **Handicraft**
 - Add a Channel

In *the FEP Explorer* section, Right Click **Gateway**/Folder has created >> **New channel** >> **IEC60870 >> IEC101 Master** to add a new one.



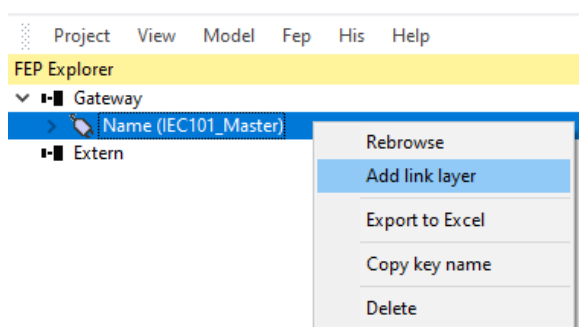
FigureD-36. New Add Menu Channel IEC60870-5-101 Master

A screenshot of the 'IEC101 Master Channel' configuration dialog box. The dialog has a title bar with a close button. It contains several sections for configuration: Channel Name (A0_Control), Description (TEST Control from A0), UA Port (46089), Server ID (0. Not set), Redundancy (None), Link Address Size (1), Connection Type (Serial), and Enabled (checked). Below these are 'Serial Settings' including COM Port (COM1), Character Size (8), Baud Rate (9200), Stop Bits (One), Parity (None), Flow Control (None), Send Timeout (0), Send Delay (0), and Receive Timeout (0). There are also 'TCP Client Settings' and 'TCP Server Settings' sections, each with a table for configuring multiple connections. The 'Mode' is set to 'Circular'. The 'Max Connection' is set to 1. The 'Server Address' and 'Port' are set to 0.0.0.0 and 2404 respectively. The 'Send Timeout' and 'Receive Timeout' are set to 0. The dialog has 'OK' and 'Cancel' buttons at the bottom.

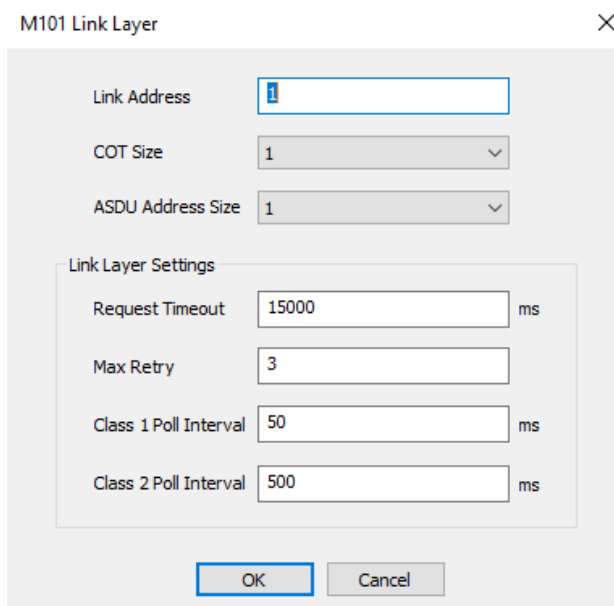
FigureD-37. IEC60870-5-101 Master Channel Configuration

Configuration parameters:

- **Channel Name:** Channel name
- **Description:** Channel description
- **UA Port:** Port return for OPC UA
- **Server ID:** ID of the machine running FEP
- **Link Address Size:** The number of octets used to contain the Link Address in the broadcast frame (1 or 2, default = 1)
- **Enabled:** Enable/Disable Channel
- **Connection Type:** Serial/TcpClient
- **Serial Settings:** the parameter for the Serial connection type
- **Tcp Client Setting:** the parameter for the TcpClient connection type
- **Add Link Layer:** Right Click the newly created channel, select **Add link layer**



FigureD-38. Add a new link layer menu

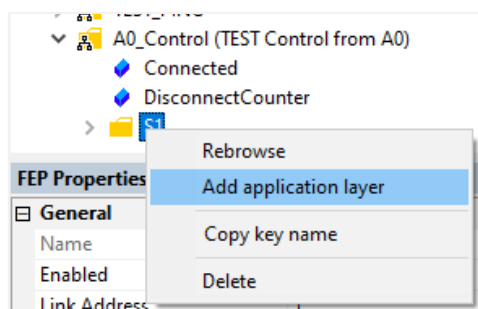


FigureD-39. Configuring a link layer

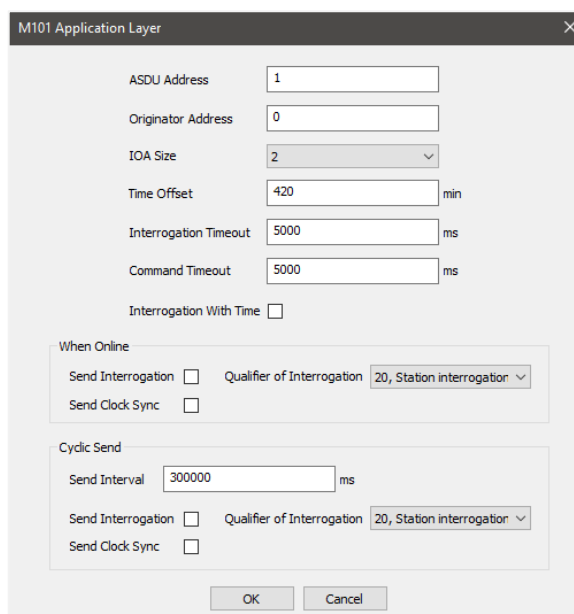
Configuration parameters

- **Link Address:** Link Layer Address
- **COT Size:** The number of octets used to contain the COT (Cause Of Transmission) in the broadcast frame (1 or 2, default = 1)

- **ASDU Address Size:** The number of octets used to contain the ASDU Address in the Broadcast Frame (1 or 2, default = 1)
- **Request Timeout:** Controls the time it takes to execute a request
- **Max Retry:** The maximum number of times a request is executed
- **Class 1 Poll Interval**
- **Class 2 Poll Interval**
- **Add application layer:** Right click the link layer you just created, select **Add application layer**



FigureD-40. Add a new application layer menu

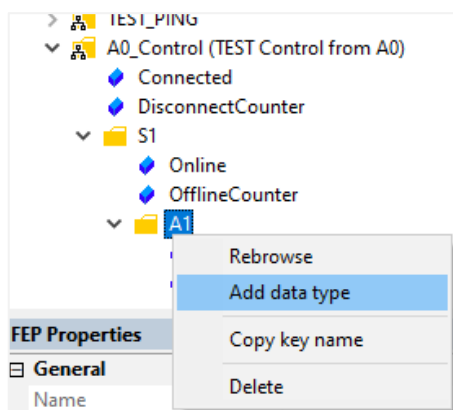


FigureD-41. Configure the application layer

Configuration parameters:

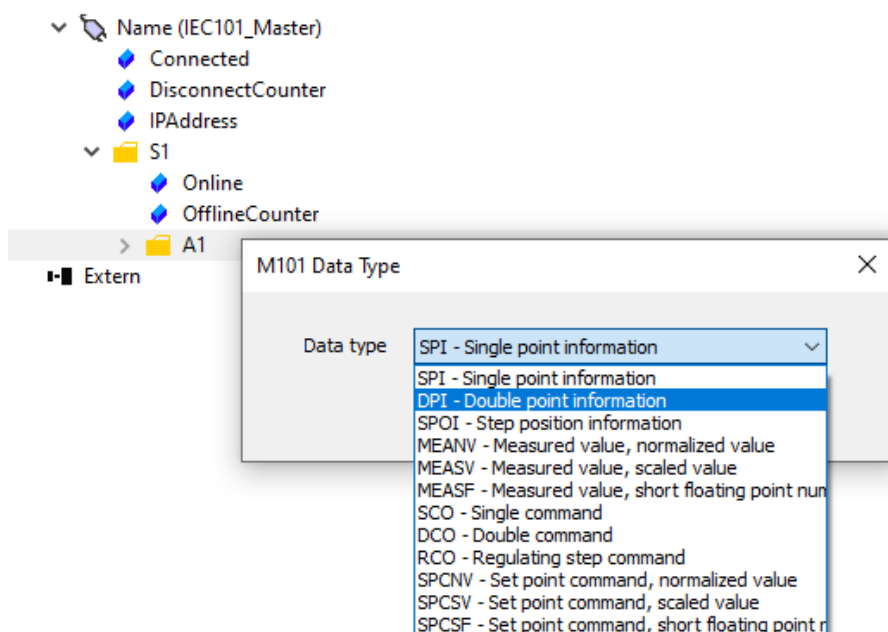
- **ASDU Address:** ASDU address
- **Oginator Address:** in case the system has multiple masters attached to the same slave, this parameter is used to distinguish the masters from each other (default = 0 ~ not used)
- **IOA Size:** The number of octets used to contain the IOA (point address) in the broadcast frame (1, 2, or 3, default = 2)
- **Time Offset (min):** The difference between local time and UTC time in minutes

- **Interrogation Timeout:** Control the time for interrogation commands to be executed
 - **Command Timeout:** Controls the time when a control command is executed
 - **Interrogation With Time:** Assign a timestamp to an interrogation command
 - **When Online:** Commands (GI, ClockSync) are invoked immediately after the application is online
 - **Cyclic Send:** Commands (GI, ClockSync) are invoked cyclically
 - **Qualifier of Interrogation:** The scope of the GI command's data request (default = 20, Station interrogation; 1, Not use)
- **Add DataType:** Right Click the application layer, select **Add data type:**



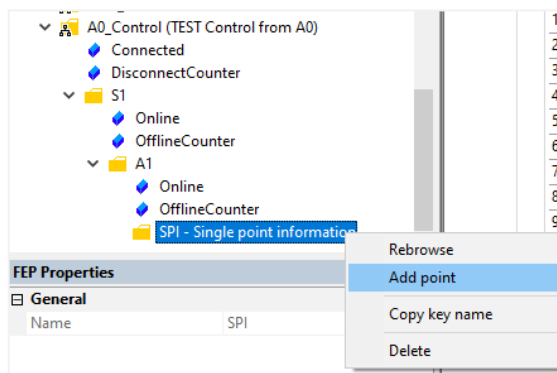
FigureD-42. Add a new data type menu

Select the appropriate data type:



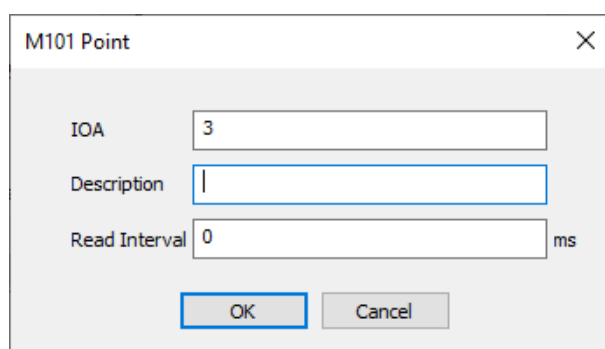
FigureD-43. Select data type

- **Add point:** Right Click Data Type, select **Add point:**



FigureD-44. Menu add new points

○ **Warning and measurement points**

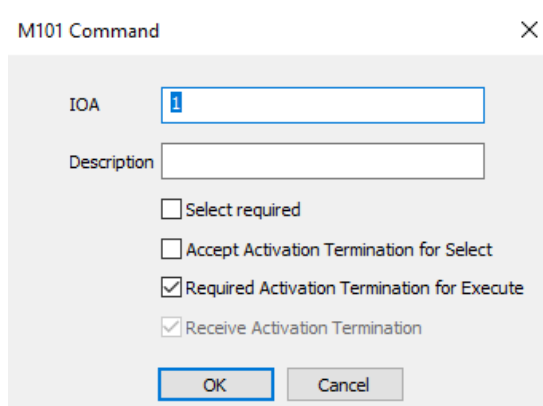


FigureD-45. Configure warning and measurement points

Configuration parameters:

- **IOA:** point address
- **Description:** description point
- **Read Interval:** the cycle of acquiring data using the read request command (default = 0 ~ not used)

○ **Point commands**



FigureD-46 Point Command Configuration

Configuration parameters:

- **IOA:** point address
- **Description:** description point

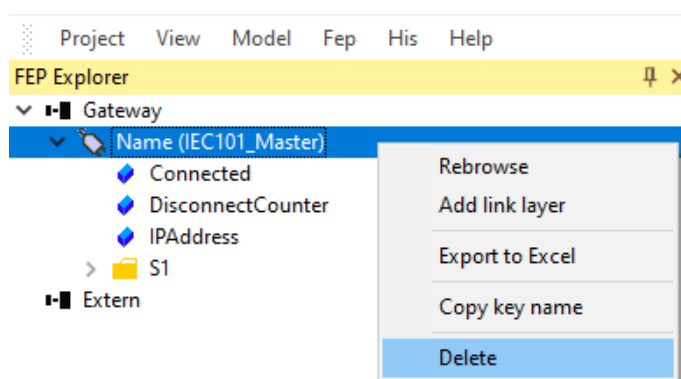
- **Select Required:** Select SBO or Direct control method
- **Accept Activation Termination for Select:** accept ActTerm so slave message returned after executing a select command without canceling the control command (default = false)
- **Required Activation Termination for Execute:** requires the slave to return the ActTerm message after executing the execute command (default = true)

D.3.4.2 Channel Calibration

The FEP Properties *interface* allows users to view and change channel information, layers, and points.

D.3.4.3 Delete a channel

Right-click on the channel you want to delete and select **Delete**:



FigureD-47. Delete Channel IEC60870-5-101 Master

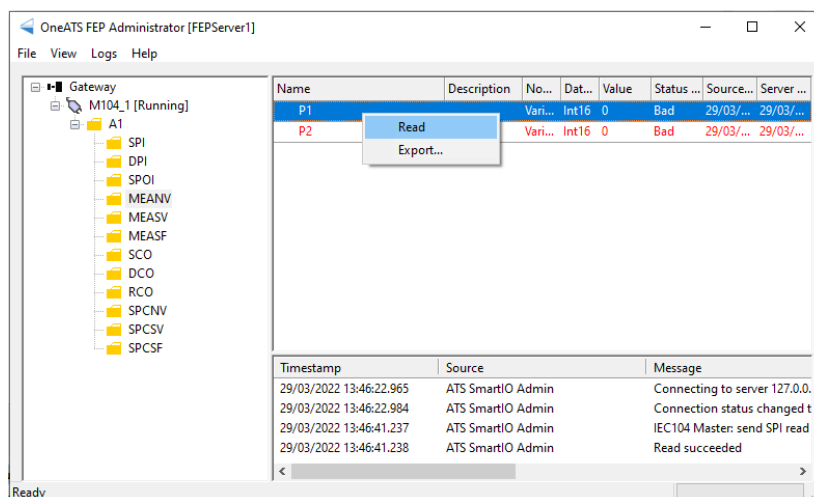
D.3.4.4 Using a channel

- **General Interrogation (GI)**
 - The GI command is used to review all the data of the station, synchronize the master state with the slave
 - When the master sends the GI, the slave sends all the data of the points it has to the master
- **Clock Sync**
 - The Clock Sync command is used to synchronize the slave's time clock with the master
- **Get Data**

Channel M101 is supporting 4 main data acquisition methods:

 - (1) Polling: Slave organizes newsletters or points in class 1 and class 2
 - When the slave notifies the master that the data is in class 1, the M101 channel prioritizes polling the class 1 data first according to the set interval
 - When the slave no longer has data in class 1, the M101 channel polls the data in class 2 according to the set interval

- (2) Event transmission: Slave proactively sends data update newsletters when there is a data/quality change
- (3) Cyclic transmission: The slave actively sends data update bulletins according to the time cycle and the slave-side deadband configuration
- (4) Read Request: The master actively requests data for a specific point of the slave according to the installation cycle. This is a less common method and not all slaves support it. Users can test Read commands through the Fep Realtime Viewer interface.



FigureD-48 Execute a Read Request command on the Fep Realtime Viewer interface

■ Control

In some cases, it may be necessary to adjust the settings for the control points to avoid errors when controlling the SBO.

According to the standard:

- After the Master sends the Select command, the Slave sends back the Activation Confirmation (ActCon) message.
- After the Master sends the Execte command, the Slave sends back 2 Activation Confirmation (ActCon) and Activation Termination (ActTerm) messages, respectively.

However, some companies don't implement slaves like this, which makes the command not work as expected and needs to be customized:

- If the Slave sends both ActCon and ActTerm messages after Select, the **Accept Activation Termination for Select option should be selected**.
- If the slave only sends each ActCon message after Execute, the **Required Activation Termination for Execute option should be unchecked**.

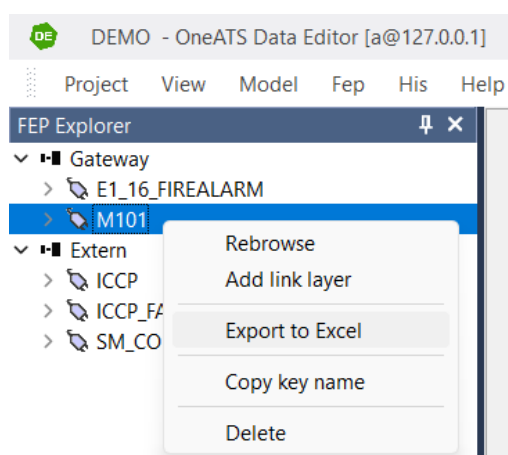
D.3.4.5 Channel Tracking

- Track channel status

- Connected: Connected = true after successfully establishing a connection between master and slave.
- DisconnectedCounter: the number of times the channel has lost connection since launch.
- **Track the status of layer links**
 - Online: Online = true after the link layer in the master and slave shake hands successfully.
 - OfflineCounter: the number of times the linklayer has been offline since launch.
- **Monitor application layer status**
 - Online: Online = true after the application layer in the master and slave shake hands successfully.
 - OfflineCounter: the number of times the application layer has been offline since launch.

D.3.4.6 Export data channels

In *the FEP Explorer*, Right-Click on the >> **Export to Excel data channel** >> select the location where the file is saved >> **Save**.



D.3.5. Channel IEC60870-5-101 Slave

D.3.5.1 Add a new channel

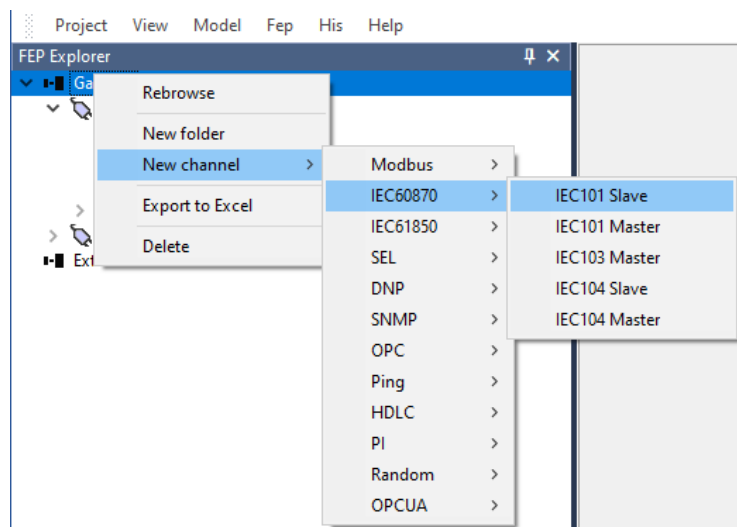
There are two ways to add a new channel: manually and import from a file.

➤ Import from excel file: Similar to channel M101

➤ Craft :

- Add a channel

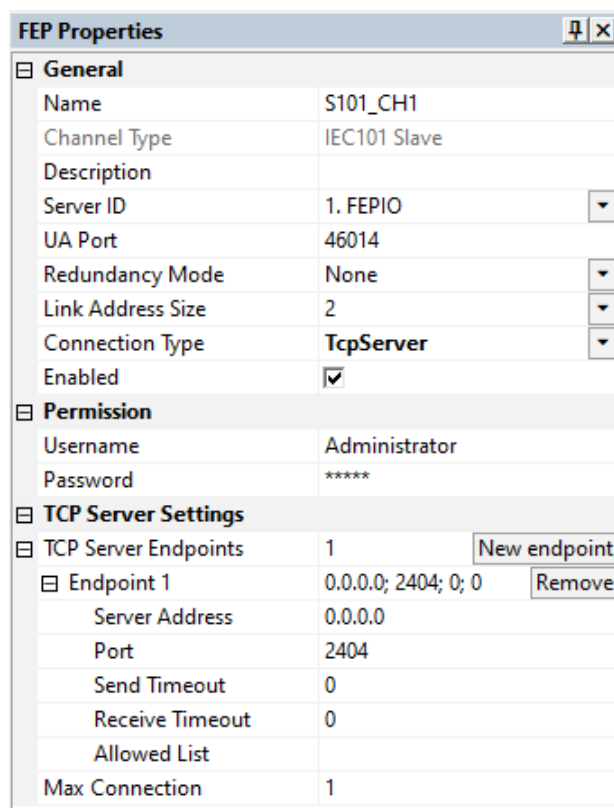
In the *FEP Explorer* section, Right Click *Gateway*/Folder has created >> **New channel >> IEC60870 >> IEC101 Slave**



FigureD-49. New Add Menu Channel IEC60870-5-101 Slave

FEP Properties	
General	
Name	S101_CH1
Channel Type	IEC101 Slave
Description	
Server ID	1. FEPIO
UA Port	46014
Redundancy Mode	None
Link Address Size	2
Connection Type	Serial
Enabled	☒
Permission	
Username	Administrator
Password	*****
Serial Settings	
COM Port	COM1
Baud Rate	9600
Character Size	8
Flow Control	None
Parity	None
Stop Bits	One
Send Timeout	0
Receive Timeout	0
Send Delay	0

FigureD-50. Configuring the IEC60870-5-101 Slave Channel with Serial Connection

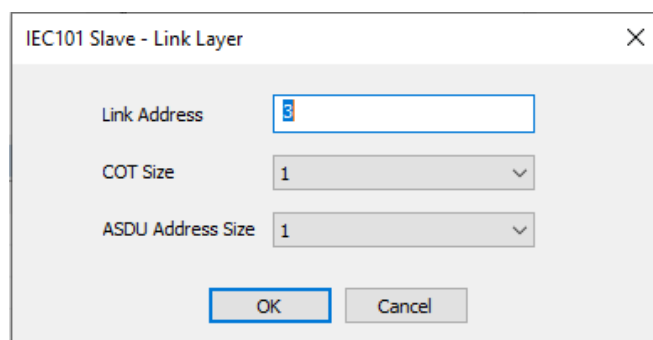


FEP Properties	
General	
Name	S101_CH1
Channel Type	IEC101 Slave
Description	
Server ID	1. FEPIO
UA Port	46014
Redundancy Mode	None
Link Address Size	2
Connection Type	TcpServer
Enabled	☒
Permission	
Username	Administrator
Password	*****
TCP Server Settings	
TCP Server Endpoints	1 New endpoint
Endpoint 1	0.0.0.0; 2404; 0; 0 Remove
Server Address	0.0.0.0
Port	2404
Send Timeout	0
Receive Timeout	0
Allowed List	
Max Connection	1

FigureD-51 Configuring the IEC60870-5-101 Slave Channel with TcpServer Connection

Configuration parameters:

- **Channel Name:** Channel name
 - **Description:** Channel description
 - **UA Port:** Port return for OPC UA
 - **Server ID:** ID of the machine running FEP
 - **Enabled:** Enable/Disable Channel
 - **Connection Type:** TcpServer
 - **Username/Password:** Account to log in to DataServer
 - **Tcp Server Endpoint:**
 - *Max connection:* The maximum number of clients connected.
 - *Server Address:* 0.0.0.0: The client can connect through all network cards (both physical and Wifi). If a specific IP address is typed, the Client can only connect through this Network Card.
 - *Port:* 2404 (Default).
- **Add Link Layer:**
- Right click on the channel you just created and select Add link layer:



The dialog box titled "IEC101 Slave - Link Layer" contains three input fields: "Link Address" with a text box containing the value "8", "COT Size" with a dropdown menu set to "1", and "ASDU Address Size" with a dropdown menu set to "1". At the bottom are "OK" and "Cancel" buttons.

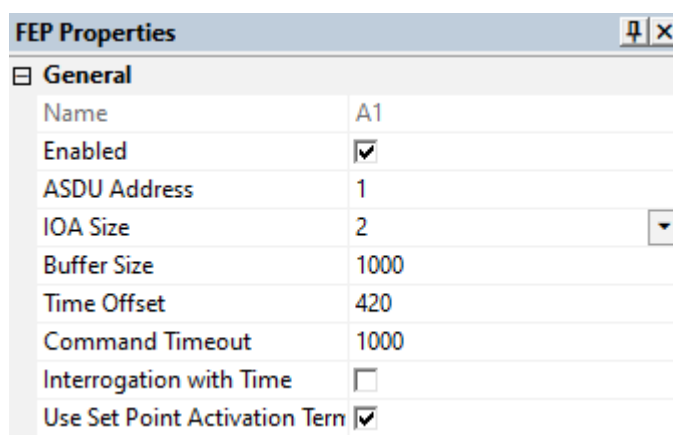
FigureD-52 Add link layer menu

Configuration parameters

- **Link Address:** The address of the link layer.
- **COT Size:** The number of octets used to contain the COT (Cause Of Transmission) in the broadcast frame (1 or 2, default = 1).
- **ASDU Address Size:** The number of octets used to contain the ASDU Address in the broadcast frame (1 or 2, default = 1).

▪ **Adding an Application Layer:**

Right click on the newly created channel and select Add application layer:



The "FEP Properties" dialog box shows the "General" tab. It contains a table of configuration parameters:

Name	A1
Enabled	☒
ASDU Address	1
IOA Size	2
Buffer Size	1000
Time Offset	420
Command Timeout	1000
Interrogation with Time	☐
Use Set Point Activation Term	☒

FigureD-53. Configure the application layer

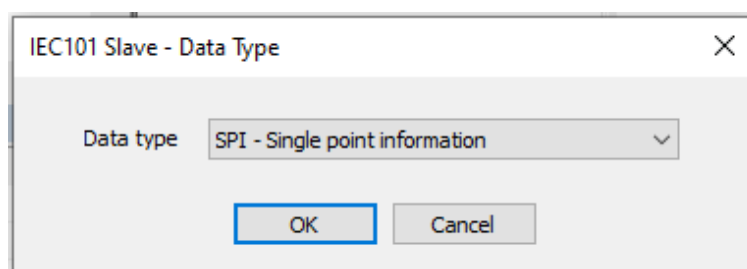
Configuration parameters

- **ASDU Address:** Application Service Data Unit (ASDU) address
- **IOA Size:** The number of octets used to contain the IOA (point address) in the broadcast frame (1, 2, or 3, default = 2)
- **Buffer Size:** The size of the cache used to queue messages
- **Time Offset:** The difference between the computer time and the UTC time in minutes
- **Command Timeout:** Controls the time when a control command is executed
- **Use Set Point Activation Termination (default = true):** For each SetPoint command sent down from the Master, after receiving the command:

- **True:** Slave replies to Master with 2 newsletters (ActCon and ActTerm).
- **False:** The Slave replies to the Master with a newsletter (ActCon).

By default, the **default value = true**. Consider dropping this option only when the Master does not conform to the standard.

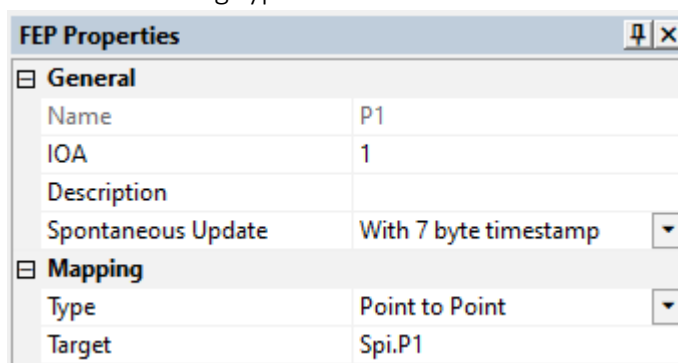
▪ **Add Data type:**



FigureD-54. Add a data type

▪ **Add Point:** Right Click **Data type >> Add point**

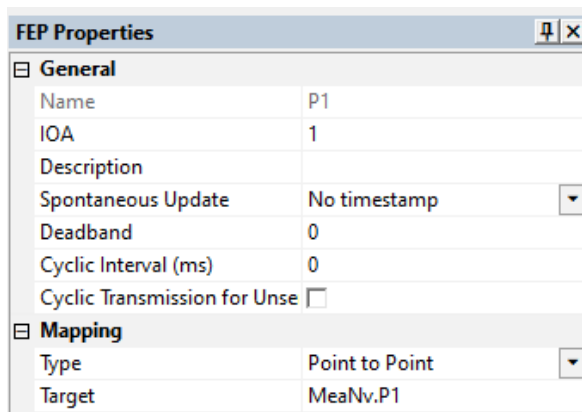
- Points for warning type:



FigureD-55. Configuring Alert Points

Configuration parameters:

- **IOA:** point address
- **Spontaneous Update:** timestamp Slave sends up to Master
 - *No timestamp:* Slave does not send timestamp to Master
 - *With 3 byte timestamp:* timestamp format h:m:s
 - *With 7 byte timestamp:* timestamp format d/m/y h:m:s
- **Mapping:** mapping S101 channel data with DataModel
 - *Type:* Mapping type (point to point, Expression, Method)
 - *Tagget:* KeyName of the mapped node on the DataModel
- Points for measurement type:



FEP Properties	
General	
Name	P1
IOA	1
Description	
Spontaneous Update	No timestamp
Deadband	0
Cyclic Interval (ms)	0
Cyclic Transmission for Unse	☐
Mapping	
Type	Point to Point
Target	MeaNv.P1

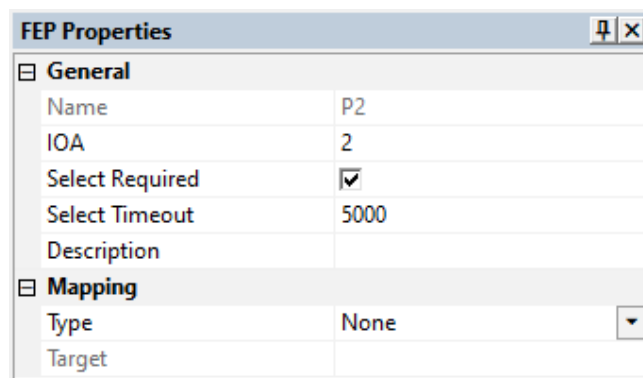
FigureD-56 Measurement point configuration

Configuration parameters:

- **Spontaneous Update:** Similar warning points
- **Deadband and Cyclic Interval and Cyclic Transmission For Unsent Event Only:** settings when using the cyclic transmission method, see section [D.3.4.4 Channel Usage for details](#)

By default, Deadband = 0 and Cyclic Interval = 0 correspond to not using cyclic transmission, points are updated with event transmission

- Control Points:



FEP Properties	
General	
Name	P2
IOA	2
Select Required	☒
Select Timeout	5000
Description	
Mapping	
Type	None
Target	

FigureD-57. Control Point Configuration

Configuration parameters

- **IOA:** point address
- **Description:** description point
- **Select Required:** select direct control method or SBO
- **Select Timeout:** the time that allows the slave to wait for the control command after selecting
- **Mapping:** Similar measurement points

D.3.5.2 Channel Calibration

The FEP Properties *interface* allows users to view and change channel information, layers, and points.

D.3.5.3 Delete a channel

Right-click on the channel you want to delete and select **Delete**.

D.3.5.4 Using a channel

▪ Role in OneATS system

In the OneATS system, the S101 channel interacts with DataServer with both roles: UAServer and UAClient; where the UAClient role is the primary function.

• UAClient

- As a UAClient, the S101 channel can retrieve data from the DataServer for updates to point information, or call DataServer methods through mapping.
- Channel S101 needs to declare an account (user/password) to have full access to the DataServer.
- Channel S101 supports data simulation with DataServer's FEP Simulation module.

• UAServer

- As a UAServer, the S101 channel supports Force/Unforce methods for simple data simulation.

▪ Configuring cyclic transmission for measurement points

- Cyclic transmission is a data transmission method of the IEC60870-5-101 standard, applicable to measurement points (MEANV, MEASV, MEASF).
- Compared to event transmission, cyclic transmission helps to save transmission bandwidth and optimize channel operation.
- Set the cyclic transmission parameters for the measurement points.

- **Deadband and Cyclic Interval:**

The slave checks the datachange after each cyclic interval cycle; Datachange is calculated as the difference $|\text{new value} - \text{old value}|$ of point.

If the Datachange > Deadband then the Slave updates the data to the Master immediately.

If the Datachange < Deadband then the Slave updates the data to the Master cyclic interval.

- **Cyclic Transmission For Unsent Event Only (default = false):** Used to minimize data updates to the Master (but still ensure that the data is updated if there is a datachange). When this option is selected, the slave only sends data to the master when the datachange > deadband.

- If the value of the point is changed using the **Force()** or **Unforce() methods**, the Slave always updates the data to the Master immediately without regard to the Deadband.

▪ Buffer Size

- Buffer size is the maximum number of messages used to send data in the queue of the S101 channel.

- When the number of messages in the queue is too large, the S101 channel will automatically delete some event transmission messages, which may lead to data synchronization between master and slave.

- When this phenomenon occurs, it is necessary to increase the buffer size or use the cyclic transmission method for the measurement points to reduce the number of event transmissions.

D.3.5.5 Channel Tracking

- **Track channel status**
 - **Connected:** Connected = true after the S101 channel opens the TCP port or successfully accesses the serial port.
 - **DisconnectedCounter:** the number of times the channel has lost connection since launch.
 - **ConnectedCount:** the number of clients connecting to channel S101 in the case of using a TCP connection.
 - **ConnectionIdentity:** the IP address of the clients connecting to the S101 channel in the case of using a TCP connection.

D.3.5.6 Export data channels

Similar to the M101 channel.

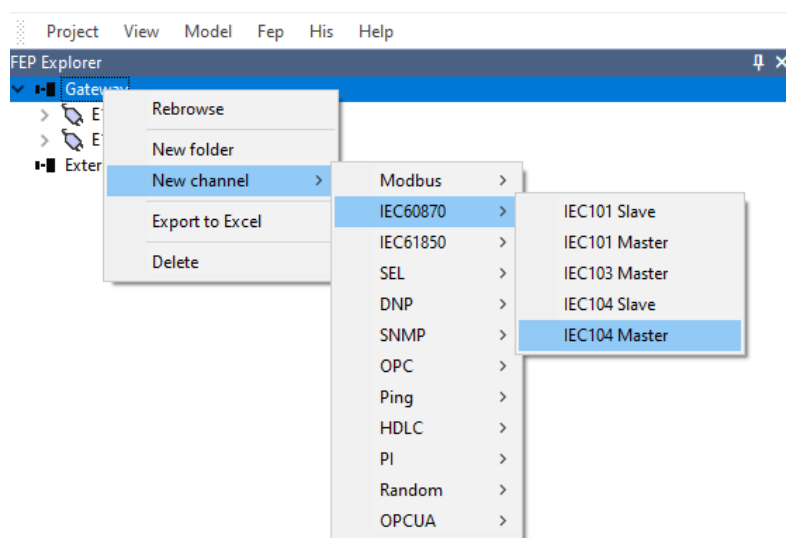
D.3.6. Channel IEC60870-5-104 Master

D.3.6.1 Add a new channel

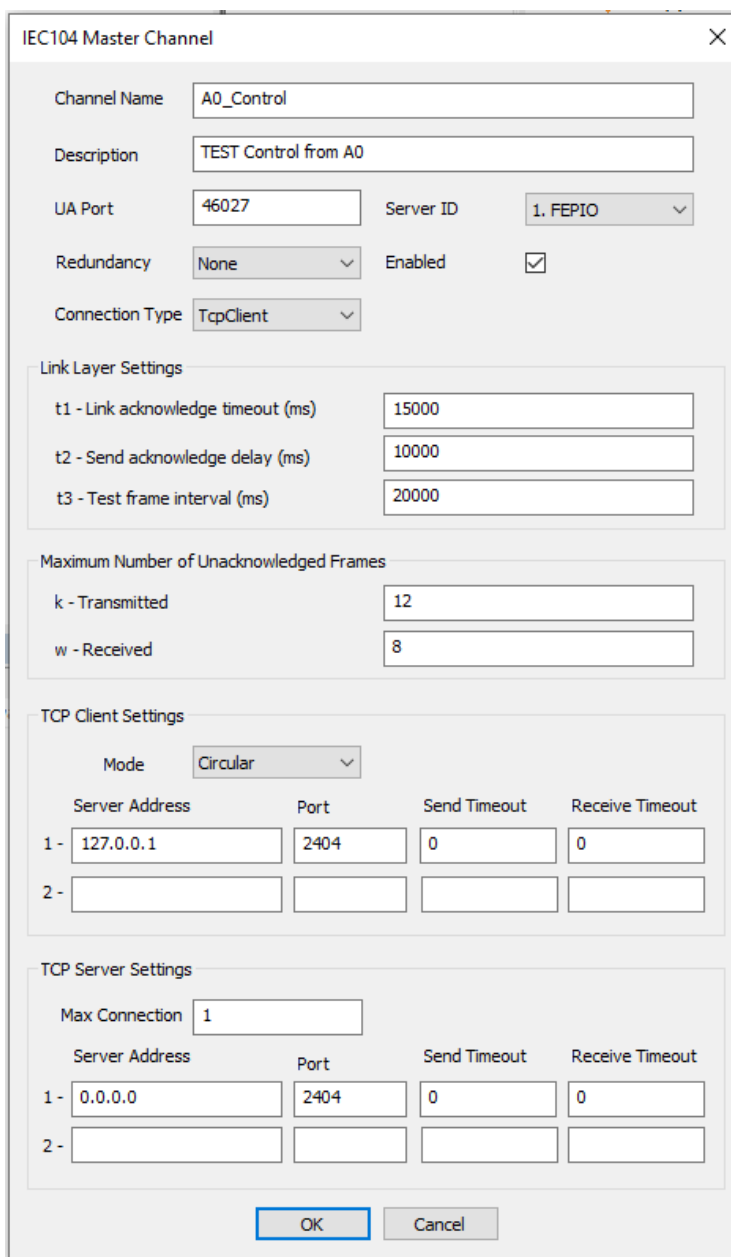
There are 2 ways to add a new channel: manually and import from an excel file.

- Import from excel file: Similar to channel M101.
- Handicraft:

In *the FEP Explorer* section, Right Click **Gateway**/Folder has created >> **New channel** >> **IEC60870** >> **IEC104 Master** to add a new one.



FigureD-58. New Add Menu Channel IEC60870-5-104 Master



The dialog box is titled "IEC104 Master Channel". It contains the following fields and sections:

- Channel Name:** A0_Control
- Description:** TEST Control from A0
- UA Port:** 46027
- Server ID:** 1. FEPIO (dropdown)
- Redundancy:** None (dropdown)
- Enabled:** ☒
- Connection Type:** TcpClient (dropdown)
- Link Layer Settings:**
 - t1 - Link acknowledge timeout (ms): 15000
 - t2 - Send acknowledge delay (ms): 10000
 - t3 - Test frame interval (ms): 20000
- Maximum Number of Unacknowledged Frames:**
 - k - Transmitted: 12
 - w - Received: 8
- TCP Client Settings:**
 - Mode:** Circular (dropdown)
 - Table with 4 columns: Index, Server Address, Port, Send Timeout, Receive Timeout.

	Server Address	Port	Send Timeout	Receive Timeout
1 -	127.0.0.1	2404	0	0
2 -				
- TCP Server Settings:**
 - Max Connection:** 1
 - Table with 4 columns: Index, Server Address, Port, Send Timeout, Receive Timeout.

	Server Address	Port	Send Timeout	Receive Timeout
1 -	0.0.0.0	2404	0	0
2 -				

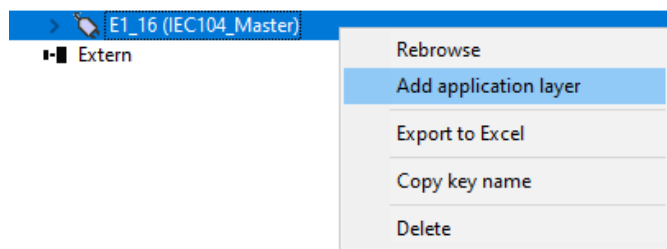
Buttons: OK, Cancel

FigureD-59. IEC60870-5-104 Master Channel Configuration

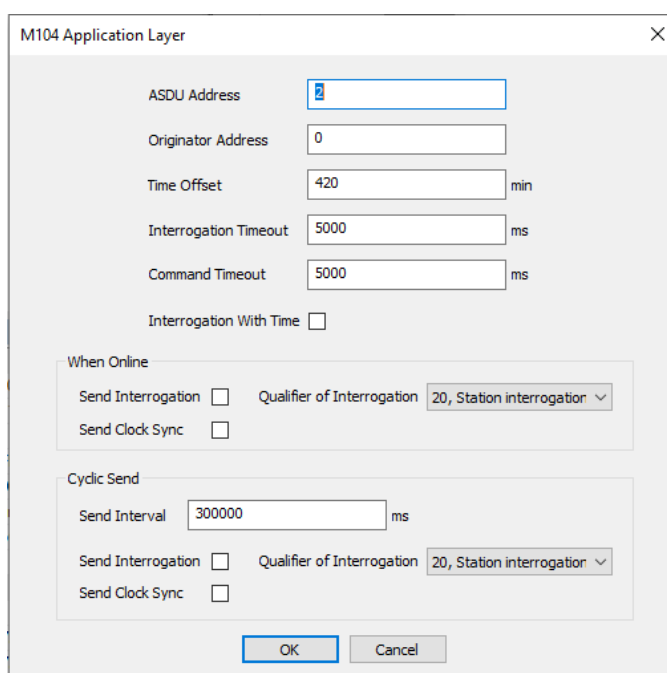
Configuration parameters:

- **Channel Name:** Channel name
- **Description:** Channel description
- **UA Port:** Port return for OPC UA
- **Server ID:** ID of the machine running FEP
- **Enabled:** Enable/Disable Channel
- **Connection Type:** TcpClient
- **Tcp Client Setting:** the parameter for the TcpClient connection type. Multiple TCPClient Endpoints can be declared in this section when the M104 channel needs to be connected to a redundant server

- Add **application layer**: Right click the link layer you just created, select **Add application layer**



FigureD-60. Add a new application layer menu

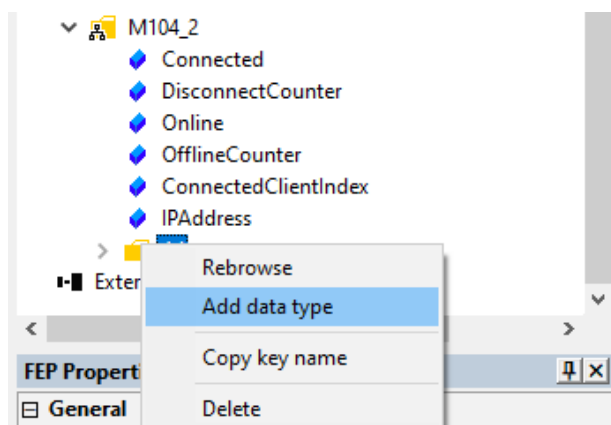


FigureD-61. Configure the application layer

Configuration parameters:

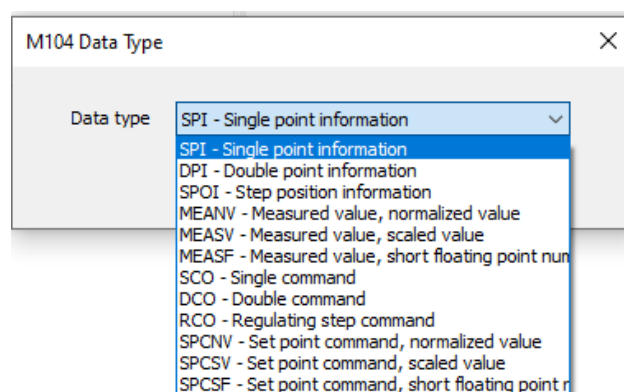
- **ASDU Address**: ASDU address
- **Originator Address**: in case the system has multiple masters attached to the same slave, this parameter is used to distinguish the masters from each other (default = 0 ~ not used)
- **Time Offset**: The difference between local time and UTC time in minutes
- **Interrogation Timeout**: Control the time for interrogation commands to be executed
- **Command Timeout**: Controls the time when a control command is executed
- **Interrogation With Time**: Assign a timestamp to an interrogation command
- **When Online**: Commands (GI, ClockSync) are invoked immediately after the application is online
- **Cyclic Send**: Commands (GI, ClockSync) are invoked cyclically

- **Qualifier of Interrogation:** The scope of the GI command's data request (default = 20, Station interrogation; 1, Not use)
- **Add DataType:** Right Click the application layer, select **Add data type**:



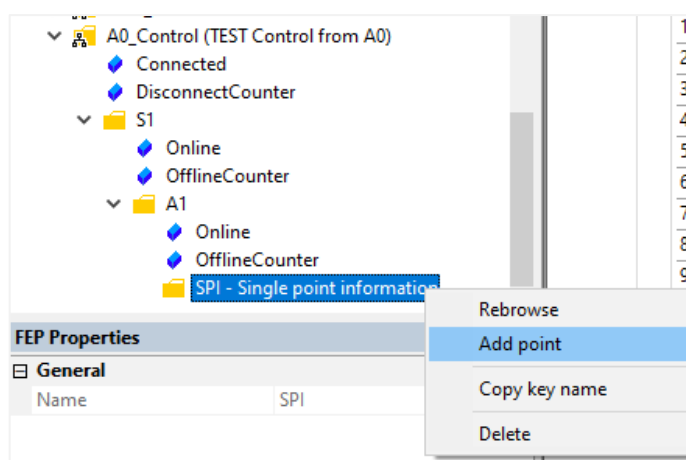
FigureD-62. Add a new data type menu

Select the appropriate data type:



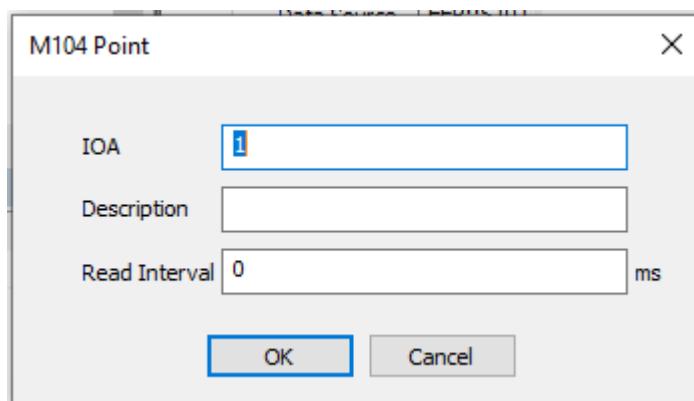
FigureD-63. Select data type

- **Add point:** Right Click Data Type, select **Add point**:



FigureD-64. Menu add new points

- **Warning and measurement points**

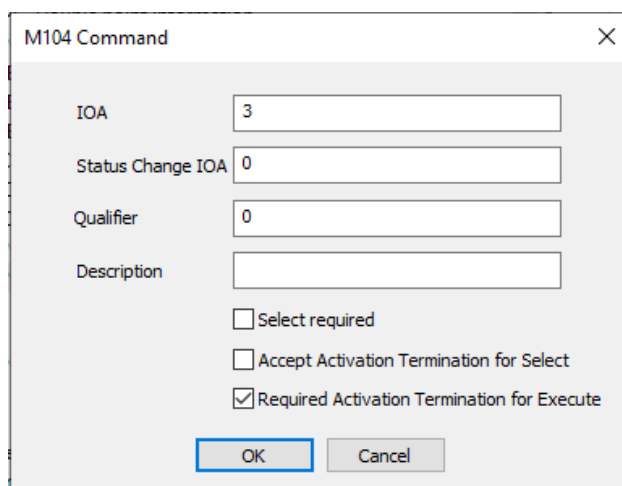


The M104 Point dialog box contains three input fields: IOA with the value '1', Description (empty), and Read Interval with the value '0' and a unit 'ms'. At the bottom are 'OK' and 'Cancel' buttons.

FigureD-65. Configure warning and measurement points

Configuration parameters:

- **IOA:** point address
- **Description:** description point
- **Read Interval:** the cycle of acquiring data using the read request command (default = 0 ~ not used)
- o **Point commands**



The M104 Command dialog box contains four input fields: IOA with the value '3', Status Change IOA with the value '0', Qualifier with the value '0', and Description (empty). Below these are three checkboxes: 'Select required' (unchecked), 'Accept Activation Termination for Select' (unchecked), and 'Required Activation Termination for Execute' (checked). At the bottom are 'OK' and 'Cancel' buttons.

FigureD-66 Point Command Configuration

Configuration parameters

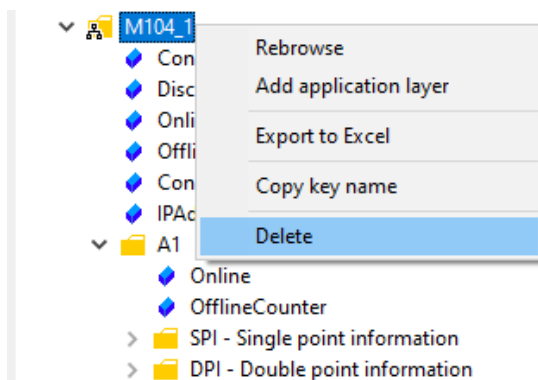
- **IOA:** point address
- **Description:** description point
- **Select Required:** Select SBO or Direct control method
- **Accept Activation Termination for Select:** accept ActTerm so slave message returned after executing a select command without canceling the control command (default = false)
- **Required Activation Termination for Execute:** requires the slave to return the ActTerm message after executing the execute command (default = true)

D.3.6.2 Channel Calibration

The FEP Properties *interface* allows users to view and change channel information, application layers, and points.

D.3.6.3 Delete a channel

Right-click on the channel you want to delete and select **Delete**:



FigureD-67. Delete Channel IEC60870-5-104 Master

D.3.6.4 Using a channel

- **General Interrogation (GI)**
 - The GI command is used to review all the data of the station, synchronize the master state with the slave.
 - When the master sends the GI, the slave sends all the data of the points it has to the master.
- **Clock Sync**
 - The Clock Sync command is used to synchronize the slave's time clock with the master.
- **Get Data**

The M104 channel supports 3 main data acquisition methods:

 - (1) Event transmission: Slave actively sends data update newsletters when there is a data/quality change.
 - (2) Cyclic transmission: The slave actively sends data update bulletins according to the time cycle and the slave-side deadband configuration.
 - (3) Read Request: The master actively requests data for a specific point of the slave according to the installation cycle. Users can test the Read command via the Fep Realtime Viewer interface (similar to the M101 channel).

Note:

 - Event transmission is the default data retrieval method of the M104 channel. The use of other methods depends on whether the slave supports it or not.
- **Control**

In some cases, it may be necessary to adjust the settings for the control points to avoid errors when controlling the SBO.

According to the standard:

- After the Master sends the Select command, the Slave sends back the Activation Confirmation (ActCon) message.

- After the Master sends the Execute command, the Slave sends back 2 Activation Confirmation (ActCon) and Activation Termination (ActTerm) messages, respectively.

However, some companies don't implement slaves like this, which makes the command not work as expected and needs to be customized:

- If the Slave sends both ActCon and ActTerm messages after Select, the **Accept Activation Termination for Select option should be selected**.

- If the slave only sends each ActCon message after Execute, the **Required Activation Termination for Execute option should be unchecked**.

D.3.6.5 Channel Tracking

- **Track channel status**

- **Connected**: Connected = true after successfully establishing a TCP connection between master and slave.

- **DisconnectedCounter**: the number of times the channel has lost connection since launch.

- **Online**: Online=true The station initialization procedure is completed.

- **OfflineCounter**: The number of times the channel has gone offline since launch.

- **Monitor application layer status**

- **Online**: Online = true after the application layer in the master and slave shake hands successfully.

- **OfflineCounter**: the number of times the channel has lost connection to the adjacent application layer since launch.

D.3.6.6 Export data channels

Similar to the M101 channel.

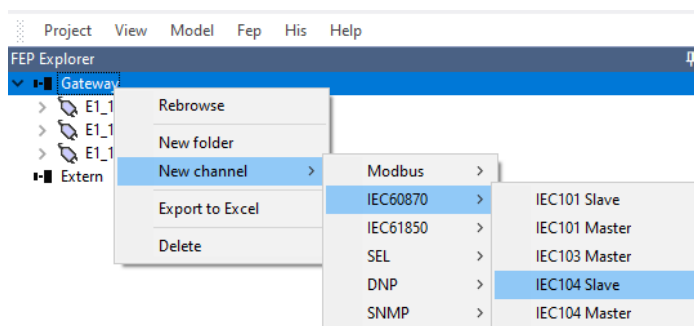
D.3.7. Channel IEC60870-5-104 Slave

D.3.7.1 Add a new channel

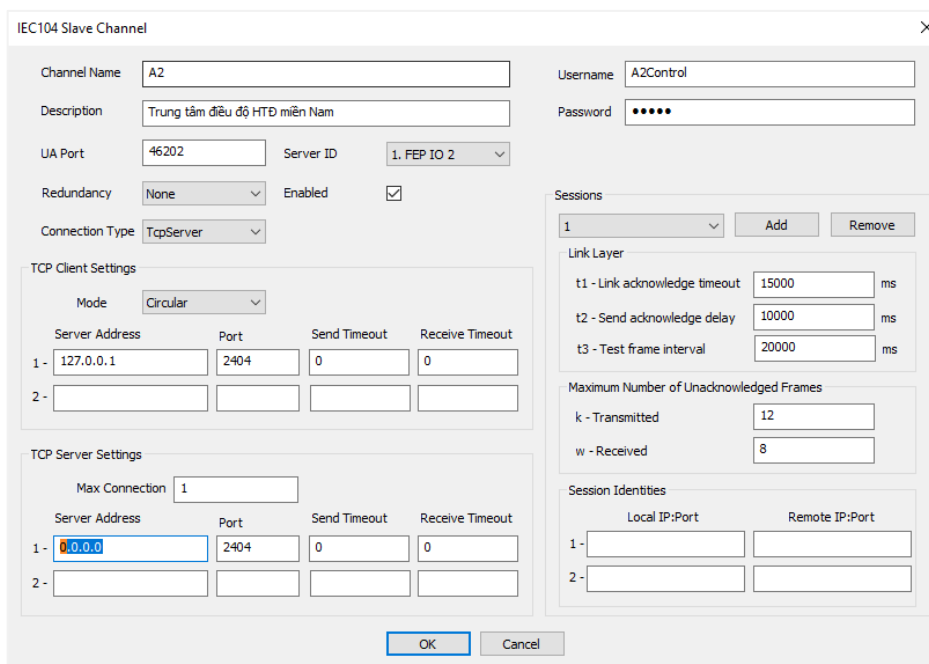
There are 2 ways to add a new channel: manually and import from an excel file.

- Import from excel file: Similar to channel IEC60870-5-101.
- Handicraft:
 - Add a Channel

In the *FEP Explorer section*, Right Click **Gateway**/Folder has created >> **New channel** >> **IEC60870** >> **IEC104 Slave**.



FigureD-68. New Add Menu Channel IEC60870-5-104 Slave



FigureD-69. IEC60870-5-104 Slave Channel Configuration

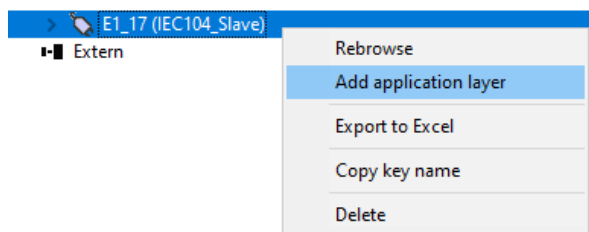
Configuration parameters:

- **Channel Name**: Channel name
- **Description**: Channel description
- **UA Port**: Port return for OPC UA
- **Server ID**: ID of the machine running FEP

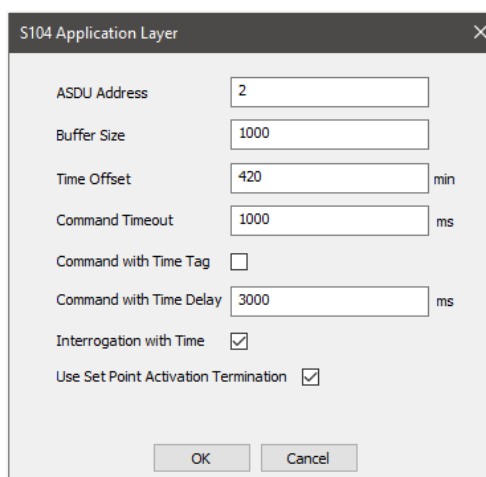
- **Enabled:** Enable/Disable Channel
- **Connection Type:** TcpServer
- **Username/Password:** The username/password for the channel to log in to the DataServer
- **Tcp Server Setting:**
 - **Max connection:** The maximum number of clients connected to
 - **Server Address:** 0.0.0.0: The client can connect through all network cards (both physical and Wifi). If a specific IP address is typed, the Client can only connect through this Network Card.
 - **Port:** 2404 (Default)
- **Sessions:**
 - Number of incoming Active Clients, can add or subtract the number of Sessions
 - **LinkLayer Setting:** includes 5 parameters t1, t2, t3, k, w; related to the transmission control mechanism and the order of the transmitted messages of the standard, usually set by default as the recommended standard
 - **Session Identities:** includes 4 parameters of local IP, local port, remote IP, remote port; used to distinguish sessions from each other when the S104 channel needs to serve many connected masters.

➤ **Adding an Application Layer:**

Right click on the newly created channel and select Add application layer:



FigureD-70. Add application layer menu



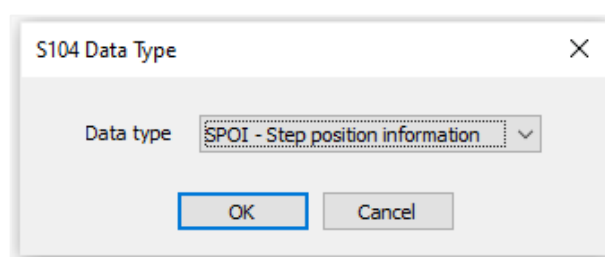
FigureD-71. Configure the application layer

Configuration parameters:

- **ASDU Address:** Application Service Data Unit (ASDU) address.
- **Command with Time Tag:** assigns a timestamp to the control command messages.
- **Command with Time Delay:** The delay allowed from the time the master sends the command until the slave confirms it.
- **Use Set Point Activation Termination (default = true):** For each SetPoint command sent down from the Master, after receiving the command:
 - **True:** Slave replies to Master with 2 newsletters (ActCon and ActTerm).
 - **False:** The Slave replies to the Master with a newsletter (ActCon).

By default, the default value = true. Consider dropping this option only when the Master does not conform to the standard.

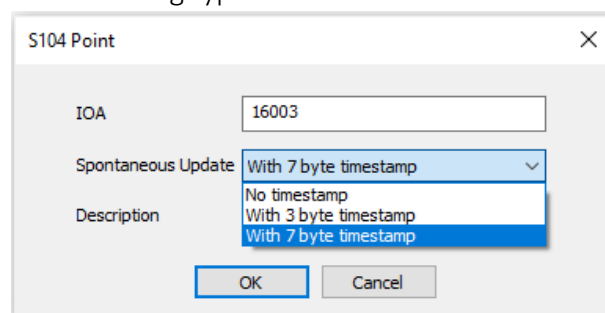
➤ Add Data type:



FigureD-72. Add a data type

➤ Add Point: Right Click **Data type**>> **Add point**

- Points for warning type:

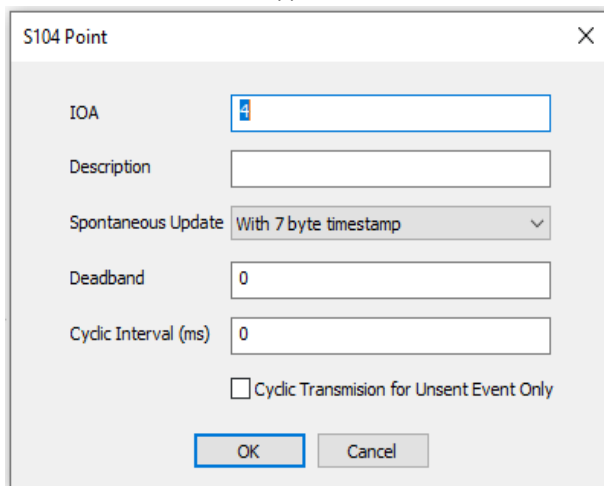


FigureD-73. More warning and measurement points

Configuration parameters:

- **IOA:** point address
- **Description:** description point
- **Spontaneous Update:** timestamp Slave sends up to Master
 - **No timestamp:** Slave does not send timestamp to Master

- *With 3 byte timestamp*: timestamp format h:m:s
- *With 7 byte timestamp*: timestamp format d/m/y h:m:s
- **Mapping**: mapping S104 channel data with DataModel
 - *Type*: Mapping type (point to point, Expression, Method)
 - *Tagget*: KeyName of the mapped node on the DataModel
- Points for measurement type:



The S104 Point dialog box contains the following fields and controls:

- IOA**: Text input field with the value '4'.
- Description**: Empty text input field.
- Spontaneous Update**: Dropdown menu set to 'With 7 byte timestamp'.
- Deadband**: Text input field with the value '0'.
- Cyclic Interval (ms)**: Text input field with the value '0'.
- Cyclic Transmission for Unsent Event Only**: Unchecked checkbox.
- Buttons**: 'OK' and 'Cancel' buttons at the bottom.

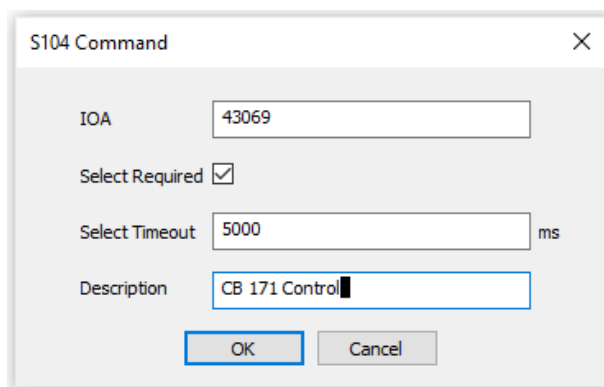
FigureD-74 Measurement point configuration

Configuration parameters:

- **IOA**: point address
- **Description**: description point
- **Spontaneous Update**: Similar warning points
- **Deadband and Cyclic Interval and Cyclic Transmission For Unsent Event Only**: settings when using the cyclic transmission method, see section [D.3.6.4 Channel Usage](#)

By default, Deadband = 0 and Cyclic Interval = 0 correspond to not using cyclic transmission, points are updated with events

- Control Points:



The S104 Command dialog box contains the following fields and controls:

- IOA**: Text input field with the value '43069'.
- Select Required**: Checked checkbox.
- Select Timeout**: Text input field with the value '5000' and a 'ms' unit label.
- Description**: Text input field with the value 'CB 171 Control'.
- Buttons**: 'OK' and 'Cancel' buttons at the bottom.

FigureD-75. Add control points

Configuration parameters:

- **IOA**: point address

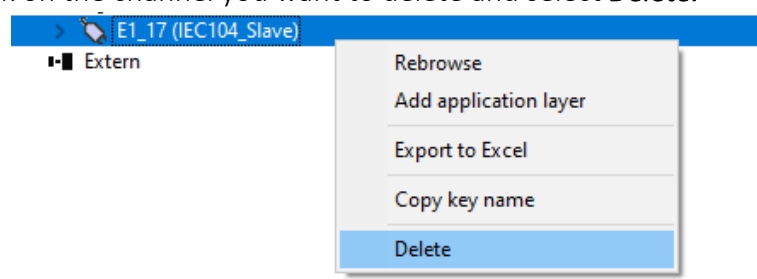
- **Description:** description point
- **Select Required:** select direct control method or SBO
- **Select Timeout:** the time that allows the slave to wait for the control command after selecting
- **Mapping:** Similar measurement points

D.3.7.2 Channel Calibration

The FEP Properties **interface** allows users to view and change channel information, application layers, and points.

D.3.7.3 Delete a channel

Right-click on the channel you want to delete and select **Delete**.



FigureD-76. Delete Channel IEC60870-5-104 Slave

D.3.7.4 Using a channel

▪ Role in OneATS system

In the OneATS system, the S104 channel interacts with DataServer with both roles: UAServer and UAClient; where the UAClient role is the primary function.

• UAClient

- As a UAClient, the S104 channel can retrieve data from the DataServer for updates to point information, or call DataServer methods through mapping.
- Channel S104 needs to declare an account (user/password) to have enough access to the DataServer.
- Channel S104 is supported with data simulation with DataServer's FEP Simulation module.

• UAServer

- As a UAServer, the S104 channel supports Force/Unforce methods for simple data simulation.

▪ Configuring cyclic transmission for measurement points

- Cyclic transmission is a data transmission method of the IEC60870-5-101 standard, applicable to measurement points (MEANV, MEASV, MEASF).
- Compared to event transmission, cyclic transmission helps to save transmission bandwidth and optimize channel operation.
- Set the cyclic transmission parameters for the measurement points.
 - **Deadband and Cyclic Interval:**

The slave checks the datachange after each cyclic interval cycle; Datachange is calculated as the difference $|\text{new value} - \text{old value}|$ of point.

If the Datachange $>$ Deadband then the Slave updates the data to the Master immediately.

If the Datachange $<$ Deadband then the Slave updates the data to the Master cyclic interval.

- ***Cyclic Transmission For Unsent Event Only (default = false):***

Used to minimize the update of data to the Master (but still ensure that the data is updated if there is a datachange). When this option is selected, the slave only sends data to the master when the datachange $>$ deadband.

- If the value of the point is changed using the Force() or Unforce() methods, the Slave always updates the data to the Master immediately without regard to the Deadband.

▪ **Buffer Size**

- Buffer size is the maximum number of messages used to send data in the queue of the S104 channel.
- When the number of messages in the queue is too large, the S104 channel will automatically delete some of the transmission event messages, which may lead to data synchronization between master and slave.
- When this phenomenon occurs, it is necessary to increase the buffer size or use the cyclic transmission method for the measurement points to reduce the number of event transmissions.

▪ **Multi-session**

- Due to the IEC60870-5 standard's message ordering control mechanism, in order for an S104 channel to serve multiple incoming masters, each connection must be managed by a mutually distinct session.
- The information of the performance session is as follows:
 $\langle \text{localIP} \rangle : \langle \text{localPort} \rangle - \langle \text{remoteIP} \rangle : \langle \text{remotePort} \rangle$
 In which:
 - Local IP is the network card address used by the S104 channel
 - Local port is the port used by the S104 channel to open the TCP Server
 - Remote IP is the IP address of the Master channel
 - Remote port is the port used by the master channel for TCP Client; This parameter is randomly assigned by the client computer, so it is not used to declare a session and is defaulted by a *
- Sessions must differ in at least 1 of the 4 parameters of local IP, local port, remote IP, remote port (usually only the first 3 parameters are used). When declaring that it is necessary to specify the difference between sessions, the remaining parameters can be defaulted with a *
- Some examples
 - The S104 channel serves 2 masters located at 2 different IP addresses

How to declare:

- Session 1: *.* - <RemoteIP1>.*
- Session 2: *.* - <RemoteIP2>.*
- Endpoint: only need to declare 1 endpoint

- o Channel S104 serves 2 masters located at the same IP address

How to declare:

- Session 1: *.*:<LocalPort1> - *.*
- Session 2: *.*:<LocalPort2> - *.*
- Endpoint: here the S104 channel uses 2 different ports, so it is necessary to declare 2 endpoints with 2 corresponding local ports
- On the master side, the 2 master channels must use 2 different ports to connect

D.3.7.5 Channel Tracking

- **Track channel status**
 - **Connected**: Connected = true after channel S104 successfully opens TCP port
 - **DisconnectedCounter**: the number of times the channel has lost connection since launch
 - **ConnectedCount**: the number of clients connecting to the channel
 - **StartedConnectedCount**: the number of clients that are connecting and exchanging data with the channel
 - **ConnectionIdentity**: the IP address of the clients connecting to the channel
- **ApplicationLayer Status Monitoring**
 - **OnlineSessionCounter**: The number of sessions that are connecting and exchanging data to the Application Layer

D.3.7.6 Export data channels

Similar to the M101 channel.

D.3.8. Channel IEC60870-5-103 Master

D.3.8.1 Add a new channel

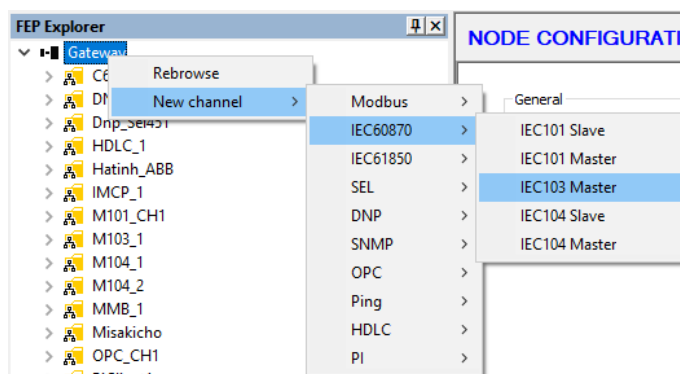
There are two ways to add a new channel: manually and import from a file.

➤ Import from excel file: Similar to channel M101

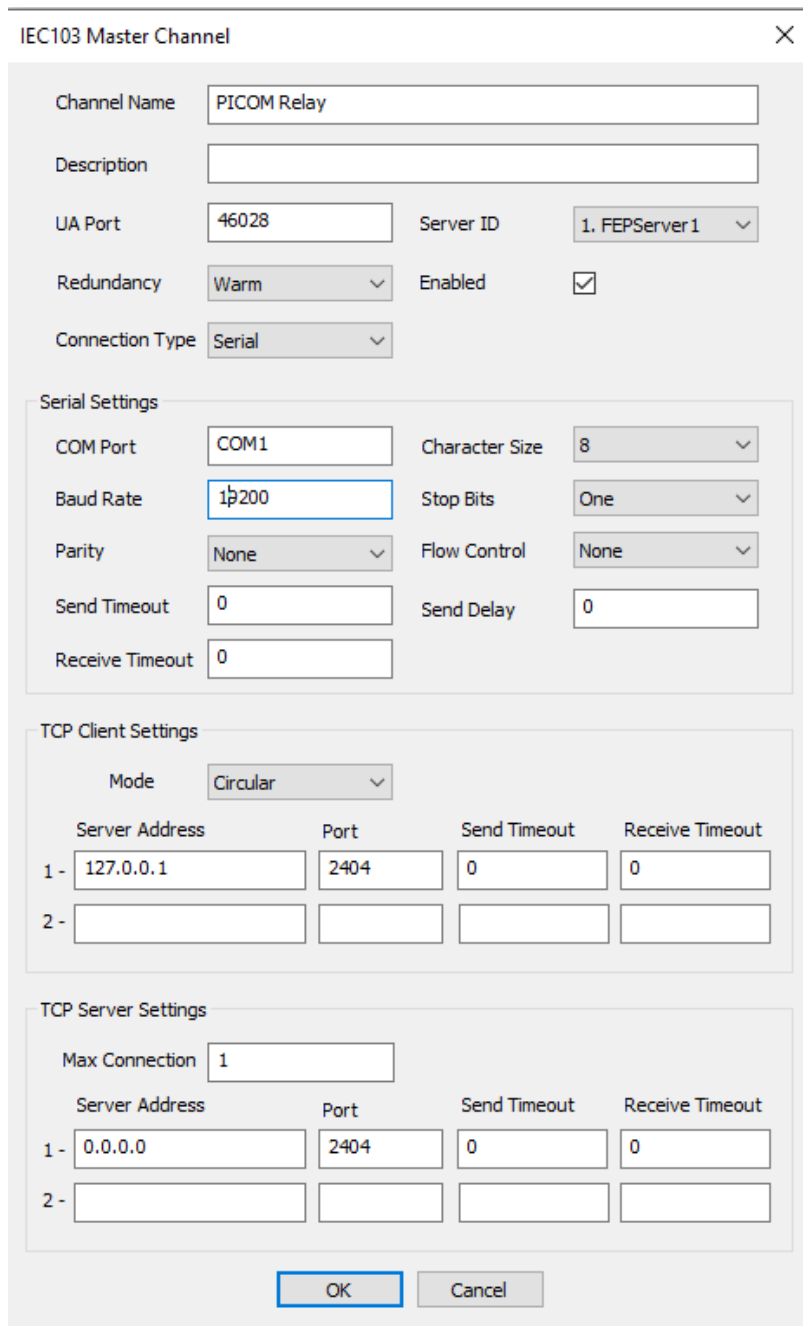
➤ Handicraft

▪ Add a Channel

In *the FEP Explorer* section, Right Click **Gateway**/Folder has created >> **New channel**
>> **IEC60870** >> **IEC103 Master** to add a new one.



FigureD-77. New Add Menu Channel IEC60870-5-103 Master



The dialog box is titled "IEC103 Master Channel" and contains the following sections:

- Main Configuration:**
 - Channel Name: PICOM Relay
 - Description: (empty)
 - UA Port: 46028
 - Server ID: 1. FEPServer1
 - Redundancy: Warm
 - Enabled: ☒
 - Connection Type: Serial
- Serial Settings:**
 - COM Port: COM1
 - Character Size: 8
 - Baud Rate: 19200
 - Stop Bits: One
 - Parity: None
 - Flow Control: None
 - Send Timeout: 0
 - Send Delay: 0
 - Receive Timeout: 0
- TCP Client Settings:**
 - Mode: Circular
 - Table with 4 columns: Index, Server Address, Port, Send Timeout, Receive Timeout.

	Server Address	Port	Send Timeout	Receive Timeout
1 -	127.0.0.1	2404	0	0
2 -				
- TCP Server Settings:**
 - Max Connection: 1
 - Table with 4 columns: Index, Server Address, Port, Send Timeout, Receive Timeout.

	Server Address	Port	Send Timeout	Receive Timeout
1 -	0.0.0.0	2404	0	0
2 -				

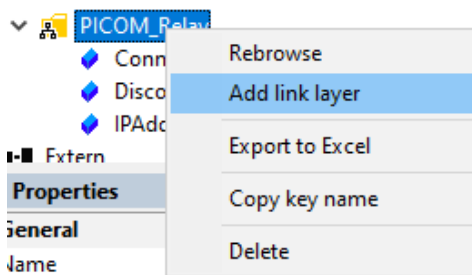
Buttons: OK, Cancel

FigureD-78. IEC60870-5-103 Master Channel Configuration

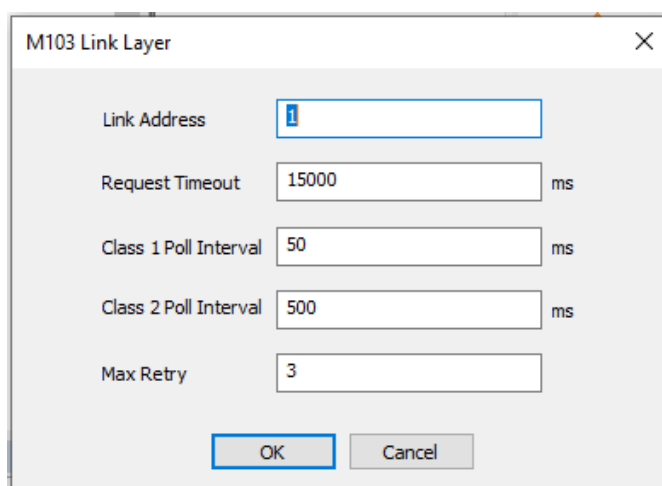
Main configuration parameters:

- **Channel Name:** Channel name
- **Description:** Channel description
- **UA Port:** Port return for OPC UA
- **Server ID:** ID of the machine running FEP
- **Enabled:** Enable/Disable Channel
- **Connection Type:** Serial/TcpClient
- **Serial Settings:** the parameter for the Serial connection type
- **Tcp Client Setting:** the parameter for the TcpClient connection type

- **Add Link Layer:** Right Click the newly created channel, select **Add link layer**



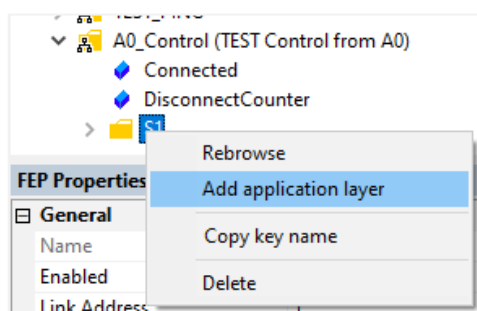
FigureD-79. Add a new link layer menu



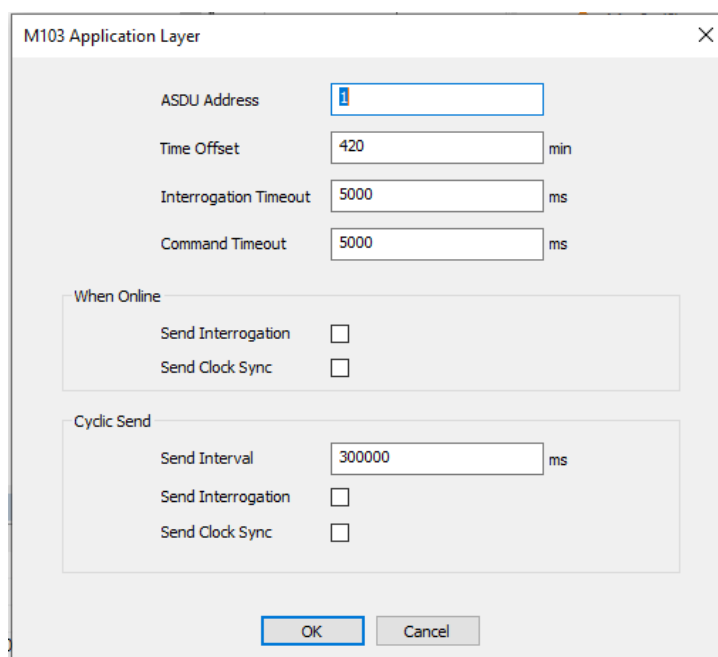
FigureD-80. Configuring a link layer

Configuration parameters:

- **Link Address:** Link Layer Address
 - **Request Timeout:** Controls the time it takes to execute a request
 - **Max Retry:** The maximum number of times a request is executed
 - **Class 1 Poll Interval**
 - **Class 2 Poll Interval**
- **Add application layer:** Right click the link layer you just created, select **Add application layer**



FigureD-81. Add a new application layer menu



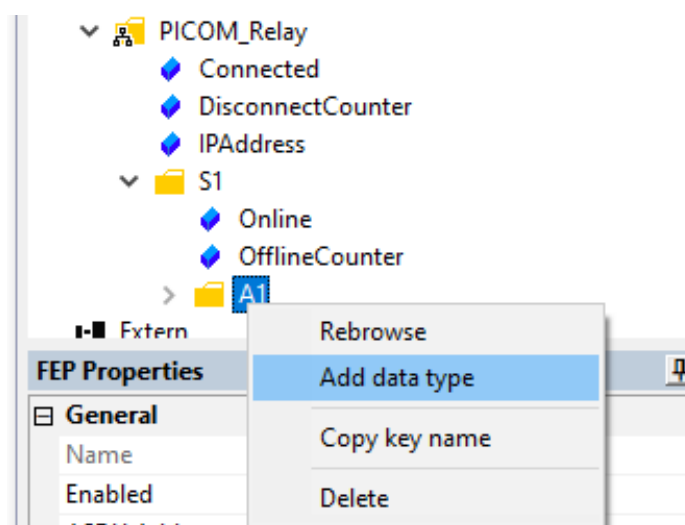
The dialog box is titled "M103 Application Layer". It contains the following fields and options:

- ASDU Address:** A text box containing the value "1".
- Time Offset:** A text box containing "420" followed by a "min" unit label.
- Interrogation Timeout:** A text box containing "5000" followed by a "ms" unit label.
- Command Timeout:** A text box containing "5000" followed by a "ms" unit label.
- When Online:** A section containing two checkboxes: "Send Interrogation" and "Send Clock Sync", both of which are currently unchecked.
- Cyclic Send:** A section containing a "Send Interval" text box with the value "300000" and a "ms" unit label, and two checkboxes: "Send Interrogation" and "Send Clock Sync", both of which are currently unchecked.
- At the bottom are "OK" and "Cancel" buttons.

FigureD-82. Configure the application layer

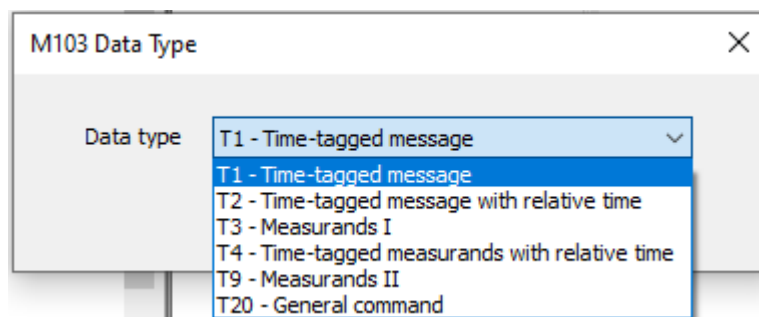
Configuration parameters:

- **ASDU Address:** ASDU address
- **Time Offset:** The difference between local time and UTC time in minutes
- **Interrogation Timeout:** Control the time for interrogation commands to be executed
- **Command Timeout:** Controls the time when a control command is executed
- **Interrogation With Time:** Assign a timestamp to an interrogation command
- **When Online:** Commands (GI, ClockSync) are invoked immediately after the application is online
- **Cyclic Send:** Commands (GI, ClockSync) are invoked cyclically
- **Add DataType:** Right Click the application layer, select **Add data type**:



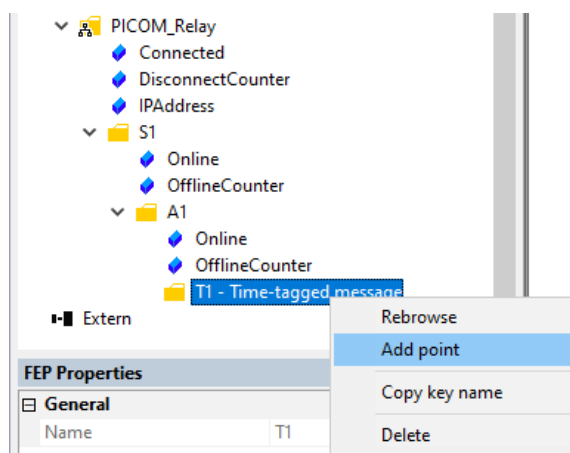
FigureD-83. Add a new data type menu

Select the appropriate data type:



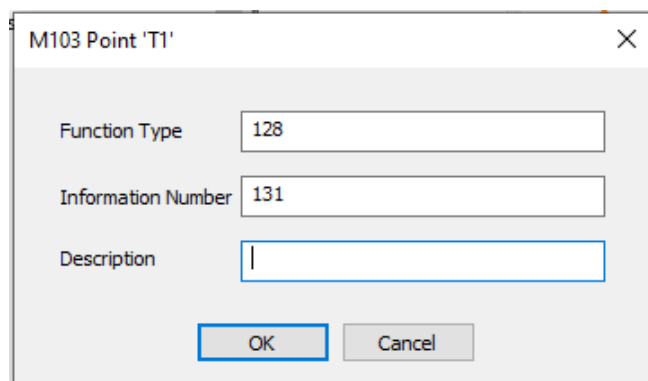
FigureD-84. Select data type

- **Add point:** Right Click Data Type, select **Add point**:



FigureD-85. Menu add new points

- **Warning points**

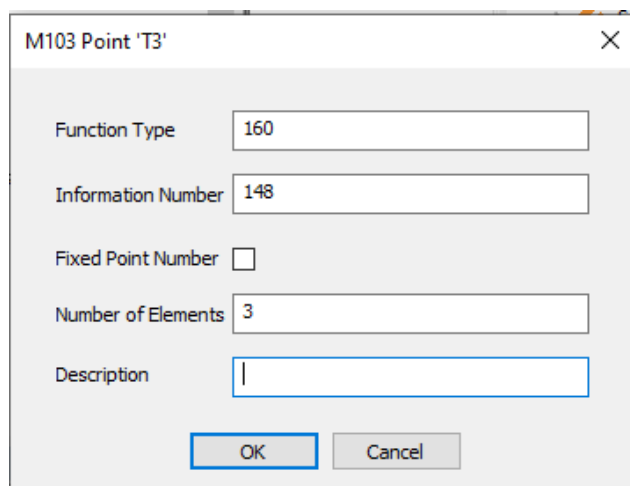


FigureD-86. Configuring Alert Points

Parameters:

- **Function Type:** The parameter that specifies the device function described by the slave
- **Information Number:** address point
- **Description:** description point

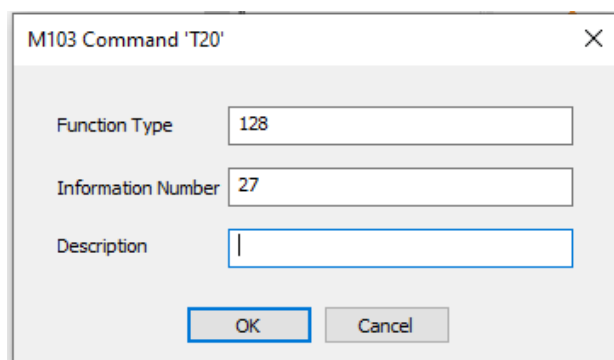
- **Measurement Points**



FigureD-87 Measurement point configuration

Parameters:

- **Function Type:** The parameter that specifies the device function described by the slave
 - **Information Number:** address point
 - **Number of Element:** The number of component variables of the point
 - **Fixed Point Number:** when enabled, the data will be updated as a float number; when disabled, the data will be updated as an integer number.
float number value = integer number value / 32768
 - **Description:** description point
- **Control points**



FigureD-88 Control Point Configuration

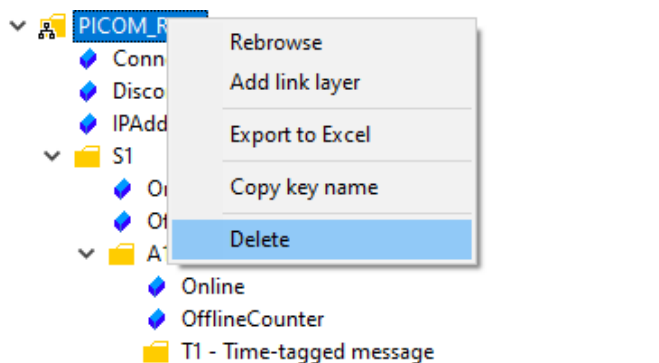
Parameters: Similar warning point

D.3.8.2 Channel Calibration

The FEP Properties *interface* allows users to view and change channel information, layers, and points.

D.3.8.3 Delete a channel

Right-click on the channel you want to delete and select **Delete**:



FigureD-89. Delete Channel IEC60870-5-103 Master

D.3.8.4 Using a channel

- **General Interrogation (GI)**
 - The GI command is used to review all the data of the station, synchronize the master state with the slave.
 - When the master sends the GI, the slave sends all the data of the points it has to the master.
- **Clock Sync**
 - The Clock Sync command is used to synchronize the slave's time clock with the master.

D.3.8.5 Channel Tracking

- **Track channel status**
 - Connected: Connected = true after successfully establishing a connection between master and slave.
 - DisconnectedCounter: the number of times the channel has lost connection since launch.
- **Track the status of layer links**
 - Online: Online = true after the link layer in the master and slave shake hands successfully.
 - OfflineCounter: the number of times the channel has lost its connection to the adjacent linklayer since launch.
- **Monitor application layer status**
 - Online: Online = true after the application layer in the master and slave shake hands successfully.
 - OfflineCounter: the number of times the channel has lost connection to the adjacent application layer since launch.

D.3.8.6 Export data channels

Similar to the M101 channel.

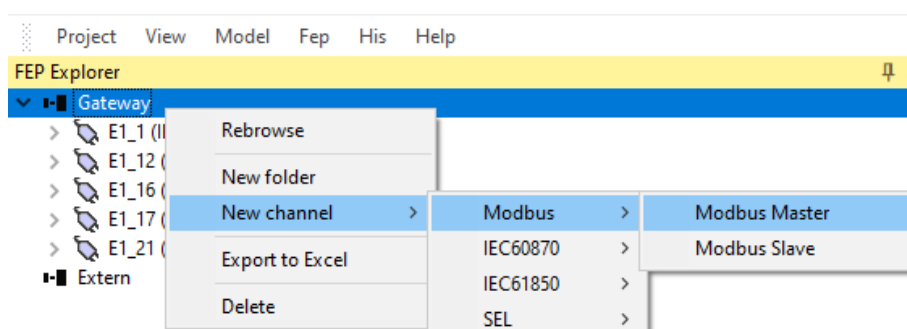
D.3.9. Modbus Master Channel

D.3.9.1 Add a new channel

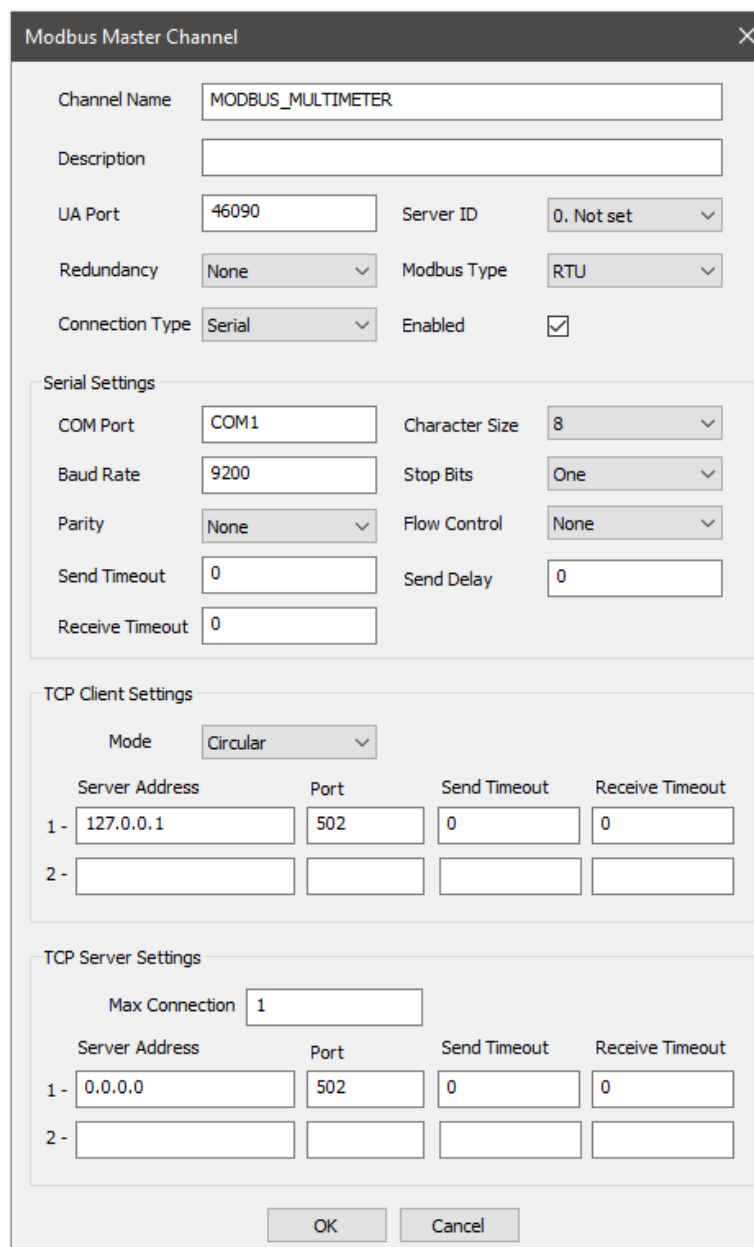
There are 2 ways to add a new channel: Manual and Import from file.

- Import from excel file: Similar to channel M101
- Handicraft:

In the *FEP Explorer* section, Right Click **Gateway**/Folder created >> **New channel** >> **Modbus** >> **Modbus Master** to add new:



FigureD-90. Menu add new modbus master channel



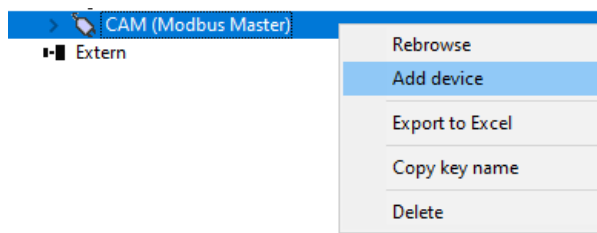
The image shows a 'Modbus Master Channel' configuration window. It contains several sections for setting up a Modbus channel. The 'Channel Name' is 'MODBUS_MULTIMETER'. The 'Description' is empty. The 'UA Port' is '46090'. The 'Server ID' is '0. Not set'. The 'Redundancy' is 'None'. The 'Modbus Type' is 'RTU'. The 'Connection Type' is 'Serial'. The 'Enabled' checkbox is checked. The 'Serial Settings' section includes 'COM Port' (COM1), 'Character Size' (8), 'Baud Rate' (9200), 'Stop Bits' (One), 'Parity' (None), 'Flow Control' (None), 'Send Timeout' (0), 'Send Delay' (0), and 'Receive Timeout' (0). The 'TCP Client Settings' section includes a 'Mode' dropdown set to 'Circular', and two rows of settings for 'Server Address', 'Port', 'Send Timeout', and 'Receive Timeout'. The first row is filled with '127.0.0.1', '502', '0', and '0'. The second row is empty. The 'TCP Server Settings' section includes a 'Max Connection' dropdown set to '1', and two rows of settings for 'Server Address', 'Port', 'Send Timeout', and 'Receive Timeout'. The first row is filled with '0.0.0.0', '502', '0', and '0'. The second row is empty. At the bottom are 'OK' and 'Cancel' buttons.

FigureD-91. Modbus master channel configuration

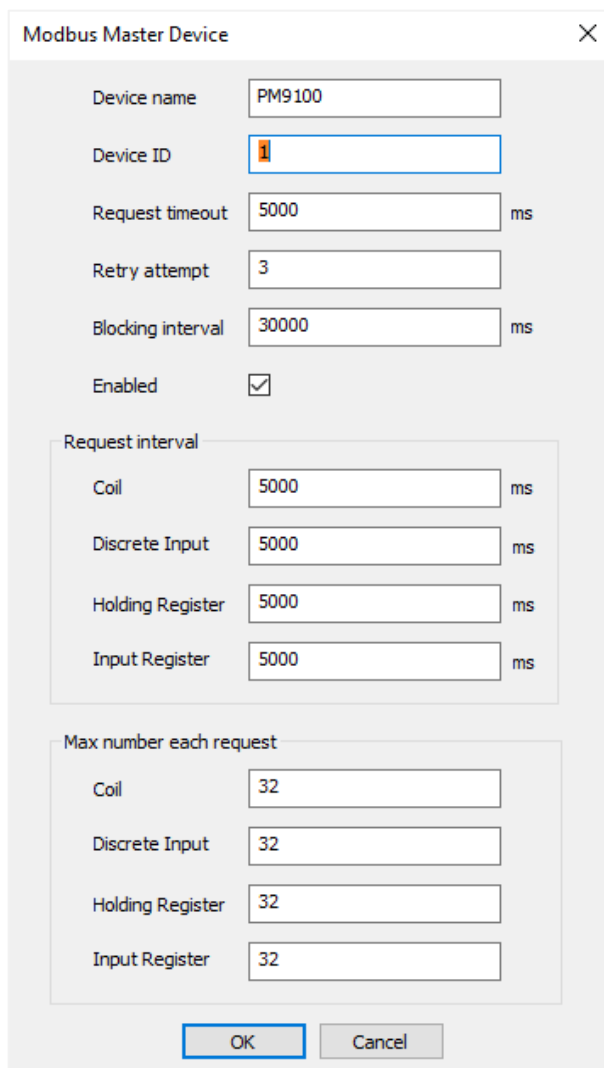
Main parameters:

- **Channel Name:** Channel name
- **Description:** Channel description
- **UA Port:** Port return for OPC UA
- **Server ID:** ID of the machine running FEP
- **Modbus Type:** select RTU or TCP
- **Enabled:** Enable/Disable Channel
- **Connection Type:** Serial/TcpClient
- **Serial Settings:** the parameter for the Serial connection type
- **Tcp Client Setting:** the parameter for the TcpClient connection type

- **Add Device:** Right Click the channel, select **Add device:**



FigureD-92. Add device menu

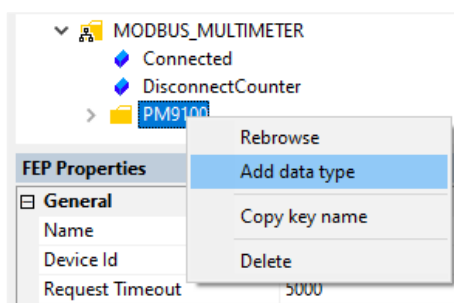
A screenshot of a 'Modbus Master Device' configuration dialog box. The dialog has a title bar with a close button. It contains several input fields and checkboxes. The 'Device name' field is set to 'PM9100'. The 'Device ID' field is set to '1'. The 'Request timeout' field is set to '5000' ms. The 'Retry attempt' field is set to '3'. The 'Blocking interval' field is set to '30000' ms. There is a checked 'Enabled' checkbox. Below these are three sections: 'Request interval' with fields for 'Coil' (5000 ms), 'Discrete Input' (5000 ms), 'Holding Register' (5000 ms), and 'Input Register' (5000 ms); and 'Max number each request' with fields for 'Coil' (32), 'Discrete Input' (32), 'Holding Register' (32), and 'Input Register' (32). At the bottom are 'OK' and 'Cancel' buttons.

FigureD-93. Modbus master device configuration

Configuration parameters:

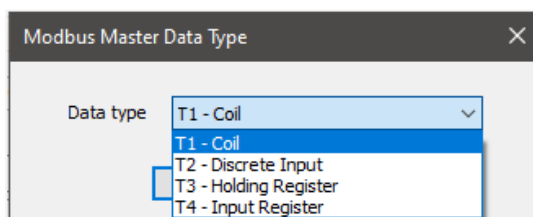
- **Device ID:** Device address
- **Request Timeout:** Controls the time it takes to execute a read/write command
- **Retry Attempt:** The maximum number of times an order is re-executed

- **Blocking interval:** The cycle of arranging the device's points into read commands
- **Enabled:** Enable/Disable device
- **Request Interval:** The polling period corresponding to each data type
- **Max number each request:** The maximum number of points that can be placed in a read command
- **Add DataType:** Right Click device, select **Add data type:**



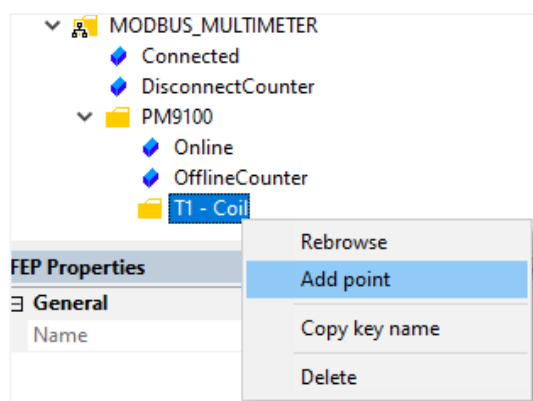
FigureD-94. Add data type menu

Select the appropriate data type:



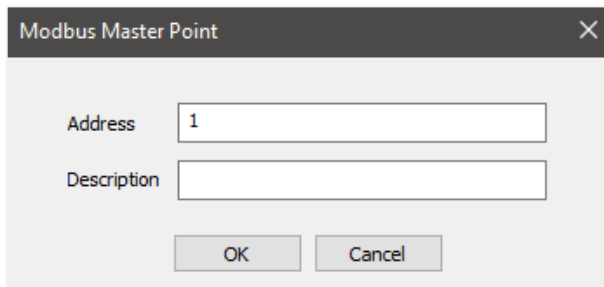
FigureD-95. Choose the modbus master data type

- **Add point:** Right Click Data Type, select **Add point:**



FigureD-96. Menu to add points

- Add digital point (coil, discrete input)

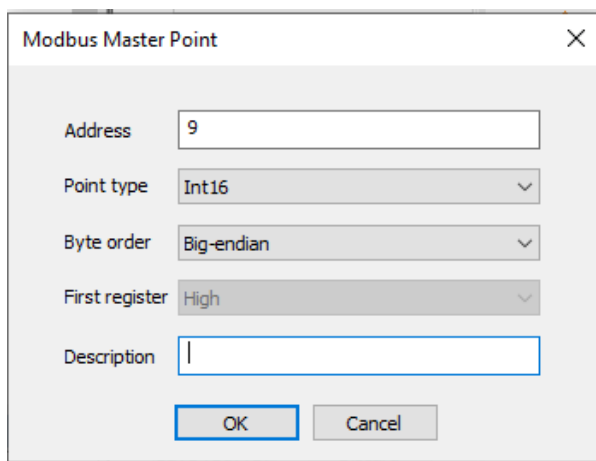


The dialog box titled "Modbus Master Point" has a close button (X) in the top right corner. It contains two text input fields: "Address" with the value "1" and "Description" which is empty. At the bottom are "OK" and "Cancel" buttons.

FigureD-97. Digital point configuration

Configuration parameters:

- **Address:** Point address
- **Description:** Point description
- Add analog points (input register, holding register)



The dialog box titled "Modbus Master Point" has a close button (X) in the top right corner. It contains several fields: "Address" with the value "9", "Point type" with a dropdown menu showing "Int16", "Byte order" with a dropdown menu showing "Big-endian", "First register" with a dropdown menu showing "High", and "Description" which is empty. At the bottom are "OK" and "Cancel" buttons.

FigureD-98 Analog Point Configuration

Configuration parameters:

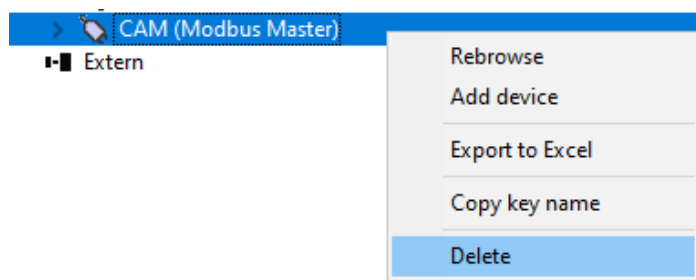
- **Address:** The address of the first register of the point that the point occupies
- **Description:** Point description
- **Point type:** The data type of the point
The data type of the point defines the registers that the point occupies, each of which is 2 bytes in size
- **Byte order:** The order of the bits in a register (Big-edian/ Little-edian), default Big-edian
- **First register:** Specifies the scorer with the largest weight when converting between the point value and the value of the registers occupied by the point

D.3.9.2 Channel Calibration

The FEP Properties **interface** allows users to view and change channel, device, and point information.

D.3.9.3 Delete a channel

Right-click on the channel you want to delete and select **Delete**:



FigureD-99. Modbus channel deletion menu

D.3.9.4 Using a channel

- **Polling**
 - To optimize bandwidth, the MMB channel arranges points with continuous addresses in the same read request when polling. The table below is an example of request sorting.

Settings	
Max number each request	
Coil	3
HoldingRegister	4

Points		Blocking Requests
Coil	Coil Address	
1	1	Read Coils - Request 1
2	2	
3	3	
4	4	Read Coils - Request 2
5	5	
7	7	Read Coils - Request 3
8	8	
10	10	Read Coils - Request 4

Points			Blocking Requests
Holding Register	DataType	Register Address	
1	Int16	1	Read Holding Register - Request 1
2	Int32	2	
		3	
4	UInt64	4	Read Holding Register - Request 3
		5	
		6	
		7	

8	Int16	8	Read Holding Register - Request 3
10	Int32	10	Read Holding Register - Request 4
		11	

- Note: if a read request fails (for example, because one of the points does not exist in the slave), all points in that read request will not be updated. To clarify this phenomenon, it is necessary to set the *Max number each request* parameter for the corresponding datatype to 1 for further analysis.

▪ **Values of analog points**

- On the transmission line, the newsletter contains information about the value of the registers.
- After receiving the newsletters, the MMB channel combines the values of the registers that each point occupies to calculate the point value. The *Byte order* and *First Register parameters* specify the combination method.
- If the slave and master do not agree on the method of combining the values of the registers, the analog point value in the MMB channel will be wrong.

▪ **Data address offset**

- The MMB channel regulates the numbering of point addresses according to data addresses (zero-based). In some cases, the slave device has a different way of numbering the data address (e.g., one-based), then it is necessary to declare and translate the address of all the points so that the master and slave are different.

D.3.9.5 Channel Tracking

▪ **Track channel status**

- Connected: Connected = true after successfully establishing a connection between master and slave.
- DisconnectedCounter: the number of times the channel has lost connection since launch.

▪ **Monitor device status**

- Online: The device's online variable is updated according to the channel's connected variable value (all devices are online as soon as the channel connects).
- OfflineCounter: the number of times the channel has lost connection since launch.

D.3.9.6 Export data channels

Similar to the M101 channel.

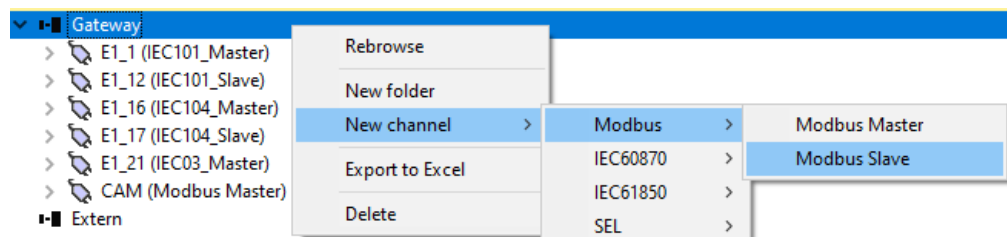
D.3.10. Modbus Slave Channel

D.3.10.1 Add a new channel

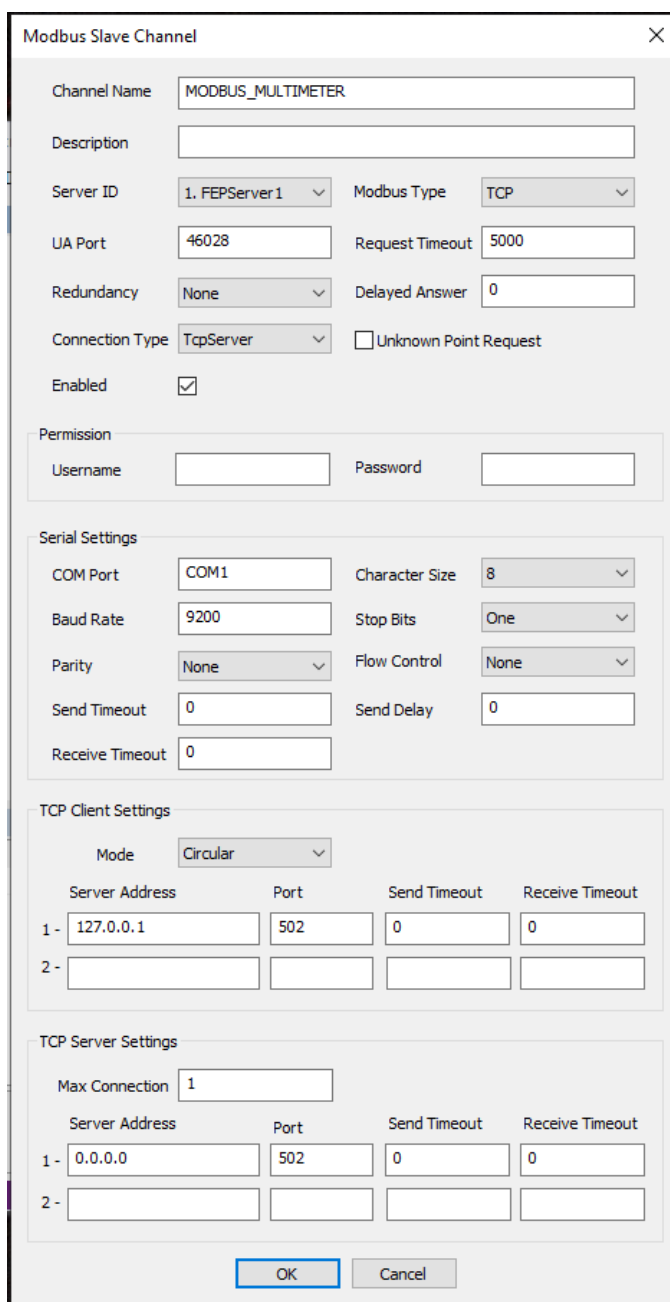
There are 2 ways to add a new channel: Manual and Import from file.

- Import from excel file: Similar to channel M101
- Handicraft:

In *the FEP Explorer* section, Right Click **Gateway**/Folder created >> **New channel** >> **Modbus** >> **Modbus Slave** to add new:



FigureD-100. Menu adds new modbus slave channel



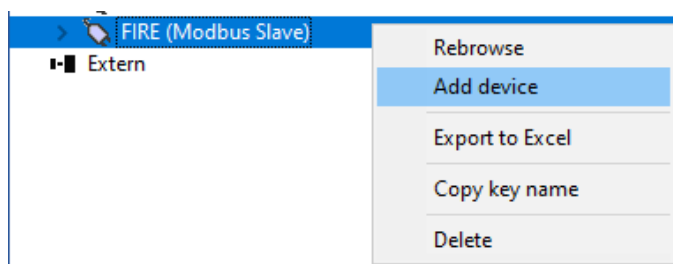
The image shows a 'Modbus Slave Channel' configuration window. It contains several sections: Channel Name (MODBUS_MULTIMETER), Description (empty), Server ID (1. FEPServer1), Modbus Type (TCP), UA Port (46028), Request Timeout (5000), Redundancy (None), Delayed Answer (0), Connection Type (TcpServer), and an 'Unknown Point Request' checkbox. There is an 'Enabled' checkbox which is checked. Below this is a 'Permission' section with 'Username' and 'Password' fields. The 'Serial Settings' section includes COM Port (COM1), Character Size (8), Baud Rate (9200), Stop Bits (One), Parity (None), Flow Control (None), Send Timeout (0), Send Delay (0), and Receive Timeout (0). The 'TCP Client Settings' section has a 'Mode' dropdown set to 'Circular' and a table for two server addresses (127.0.0.1 and an empty field) with ports (502) and timeouts (0). The 'TCP Server Settings' section has a 'Max Connection' field set to 1 and a similar table for two server addresses (0.0.0.0 and an empty field) with ports (502) and timeouts (0). At the bottom are 'OK' and 'Cancel' buttons.

FigureD-101. Modbus slave channel configuration

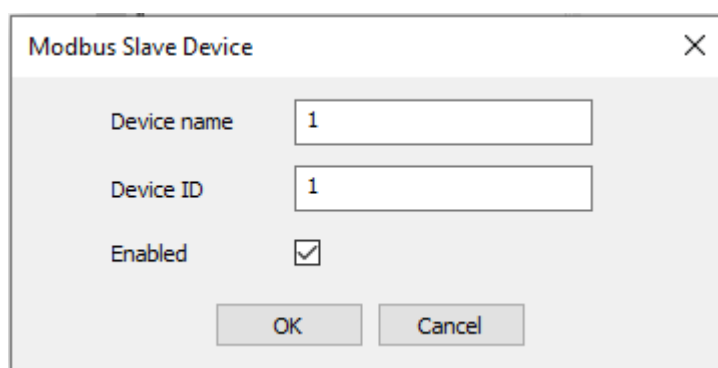
Configuration parameters:

- **Channel Name:** Channel name
- **Description:** Channel description
- **UA Port:** Port return for OPC UA
- **Server ID:** ID of the machine running FEP
- **Enabled:** Enable/Disable Channel
- **Connection Type:** TcpServer / Serial
- **Modbus Type:** the version of the Modbus protocol used (TCP/RTU)
- **Username/Password:** The username/password for the channel to log in to the DataServer

- **Tcp Server Setting:**
 - **Max connection:** The maximum number of clients connected to
 - **Server Address:** 0.0.0.0: The client can connect through all network cards (both physical and Wifi). If a specific IP address is typed, the Client can only connect through this Network Card.
 - **Port:** 502 (Default)
 - **Serial Setting**
- **Add Device:** Right Click the channel, select **Add device:**



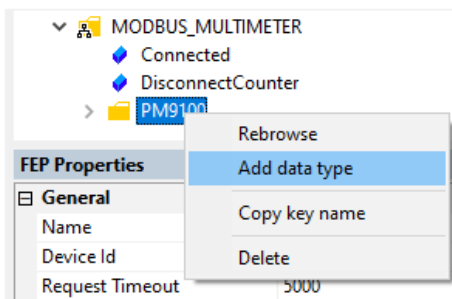
FigureD-102. Add device menu



FigureD-103. Modbus slave device configuration

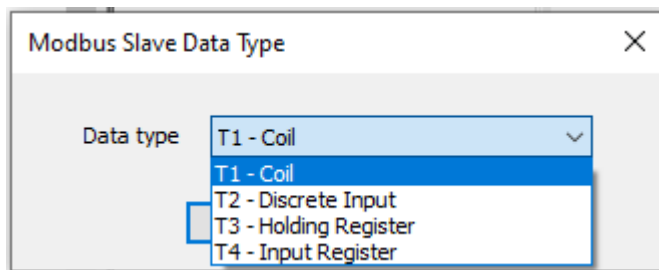
Configuration parameters:

- **Device ID:** Device address
- **Add DataType:** Right Click device, select **Add data type:**



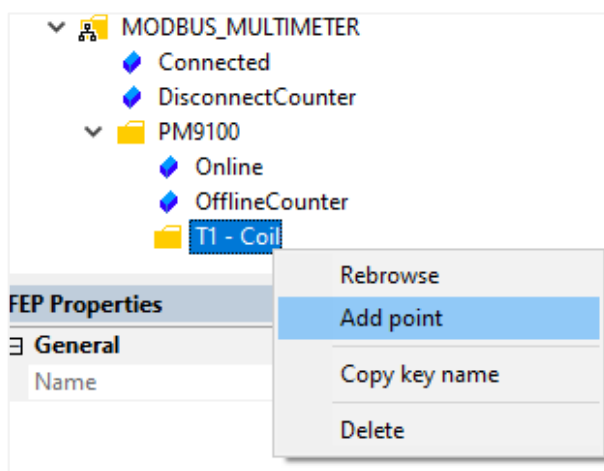
FigureD-104. Add data type menu

Select the appropriate data type:



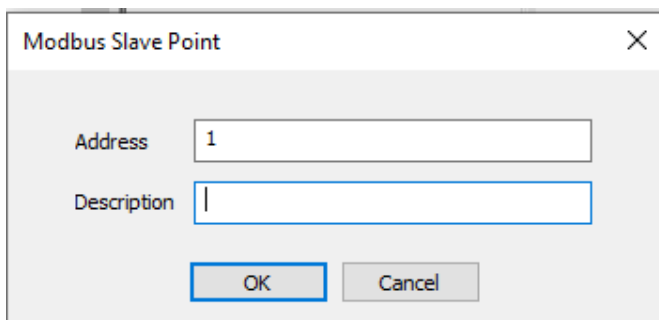
FigureD-105. Choosing a modbus slave data type

- **Add point:** Right Click Data Type, select **Add point:**



FigureD-106. Menu to add points

- Add digital point (coil, discrete input)

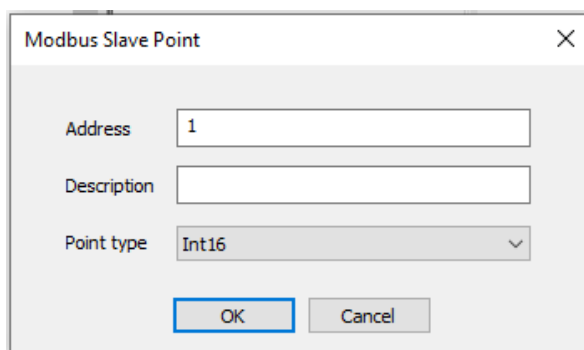


FigureD-107. Digital point configuration

Configuration parameters:

- **Address:** Point address
- **Description:** Point description
- **Mapping:** mapping S101 channel data with DataModel
 - **Type:** Mapping type (point to point, Expression, Method)
 - **Taget:** KeyName of the mapped node on the DataModel

- Add analog points (input register, holding register)



FigureD-108 Analog Point Configuration

Configuration parameters:

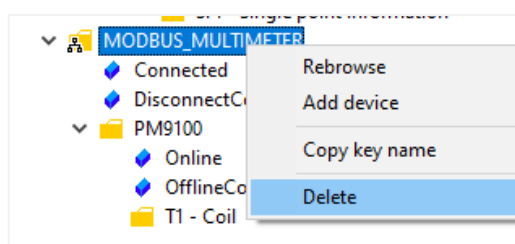
- **Address:** The address of the first register of the point that the point occupies
- **Description:** Point description
- **Point type:** The data type of the point
The data type of the point defines the registers that the point occupies, each of which is 2 bytes in size
- **Mapping:** mapping S101 channel data with DataModel
 - **Type:** Mapping type (point to point, Expression, Method)
 - **Tagget:** KeyName of the mapped node on the DataModel

D.3.10.2 Channel Calibration

The FEP Properties **interface** allows users to view and change channel, device, and point information.

D.3.10.3 Delete a channel

Right-click on the channel you want to delete and select **Delete**:



FigureD-109. Modbus slave channel deletion menu

D.3.10.4 Using a channel

▪ Role in OneATS system

In the OneATS system, the SMB channel interacts with the DataServer with both roles: UAServer and UAClient; where the UAClient role is the primary function.

• UAClient

- As a UAClient, the SMB channel can retrieve data from the DataServer for updates to point information, or call DataServer methods through mapping.

- The SMB channel needs to declare an account (user/password) to have sufficient access to the DataServer.
- SMB channels are supported to simulate data with DataServer's FEP Simulation module.
- **UAServer**
 - As a UAServer, the SMB channel supports Force/Unforce methods for simple data simulation.

D.3.10.5 Channel Tracking

- **Track channel status**
 - Connected: Connected = true after the SMB channel successfully opens the TCP port or serial port.
 - DisconnectedCounter: the number of times the channel has lost connection since launch.
 - ConnectedCount: the number of clients connecting to the SMB channel in the case of using a TCP connection.

D.3.10.6 Export data channels

Similar to the M101 channel.

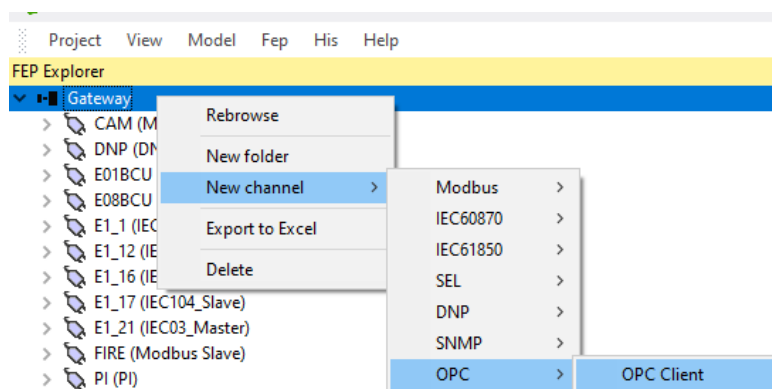
D.3.11. OPC Client Channel

D.3.11.1 Add a new channel

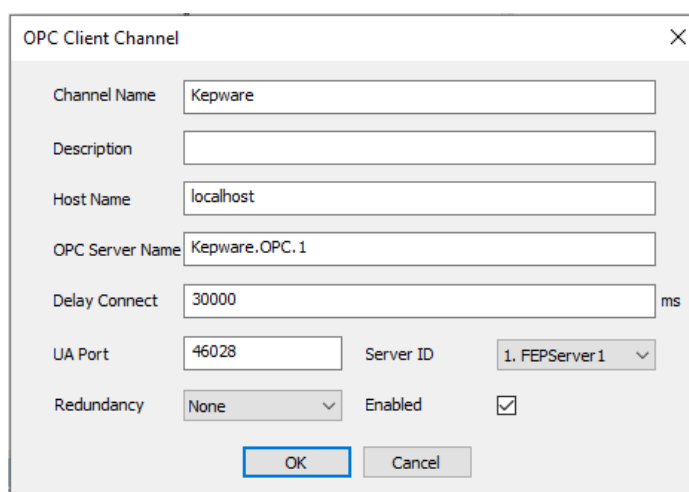
There are 2 ways to add a new channel: Manual and Import from file.

- Import from excel file: Similar to channel M101
- Handicraft:

In *the FEP Explorer* section, Right *Click Gateway*/Folder has created >> *New channel* >> *OPC* >> *OPC Client* to add a new one:



FigureD-110. Add a new OPC Client channel menu

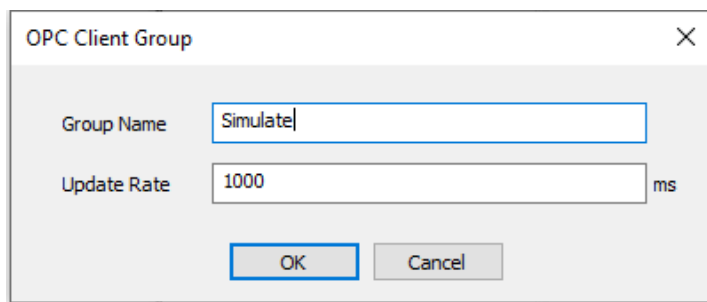


FigureD-111. OPC Client Channel Configuration

Main parameters:

- **Channel Name:** Channel name
- **Description:** Channel description
- **UA Port:** Port return for OPC UA
- **Server ID:** ID of the machine running FEP
- **Enabled:** Enable/Disable Channel
- **Host Name:** the name of the computer running OPC Server
- **OPC Server Name:** The name of the OPC Server
- **Delay Connect:** reserved

- **Add Group:** Right Click the channel, select **Add group:**

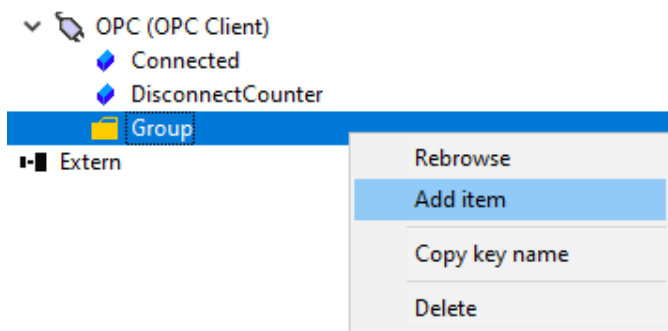


The dialog box is titled "OPC Client Group". It contains two input fields: "Group Name" with the text "Simulate" and "Update Rate" with the value "1000" and a unit "ms". At the bottom, there are "OK" and "Cancel" buttons.

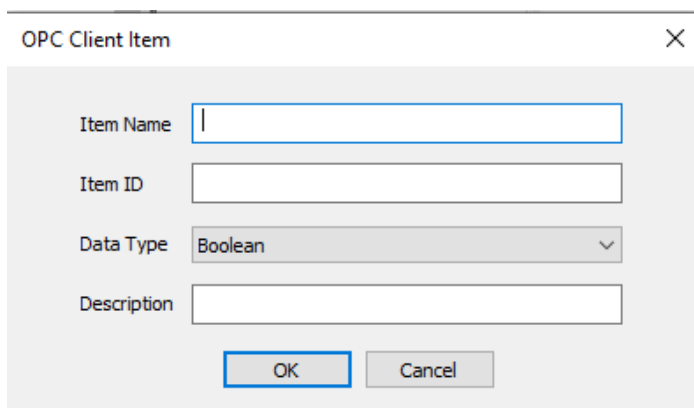
FigureD-112. Add group menu

Configuration parameters:

- **Group Name:** Group name
 - **Update Rate:** The cycle of checking data changes of server-side data, points in the group are updated if there is a data change
- **Add Item (Point):** Right Click group, select **Add Item:**



FigureD-113. Add item menu



The dialog box is titled "OPC Client Item". It contains four input fields: "Item Name" (empty), "Item ID" (empty), "Data Type" (set to "Boolean" with a dropdown arrow), and "Description" (empty). At the bottom, there are "OK" and "Cancel" buttons.

FigureD-114. Configure OPC Client Item

Configuration parameters:

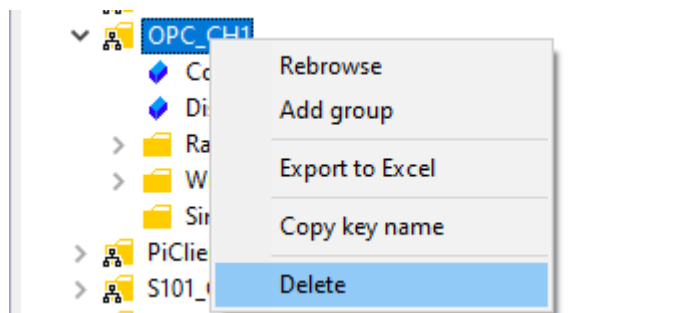
- **Item Name:** The name of the point on the channel's UA Server
- **Item ID:** the name of the point on the OPC Server connected to
- **Datatype:** data type
- **Description:** description point

D.3.11.2 Channel Calibration

The FEP Properties *interface* allows users to view and change channel, group, and item information.

D.3.11.3 Delete a channel

Right-click on the channel you want to delete and select **Delete**:



FigureD-115. OPC Client channel deletion menu

D.3.11.4 Using a channel

- **Configure DCOM for Window**
 - The computer running the FEP OPC Client channel needs to be configured with DCOM for Window in order to be able to connect the OPC successfully.
 - The user running the OAFEP service must be included in the DCOM configuration. The user running the OAFEP service defaults to NETWORK SERVICE.
 - The detailed DCOM configuration procedure is not included in this document.
- **Installing OPC Redistributable**
 - In order for the FEP OPC Client channel to operate stably and fully-featured, the computer needs to have the OPC Redistributable package installed.
 - It is recommended to update and use the latest version of OPC Redistributable. The current stable version is v3.00.108.

D.3.11.5 Channel Tracking

- **Track channel status**
 - Connected: Connected = true after successfully establishing a connection between the client and the server.
 - DisconnectedCounter: the number of times the channel has lost connection since launch.

D.3.11.6 Export data channels

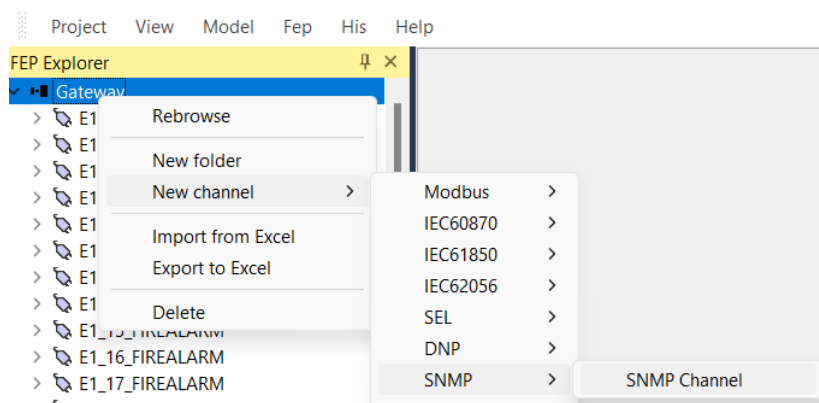
Similar to the M101 channel.

D.3.12. SNMP Channel

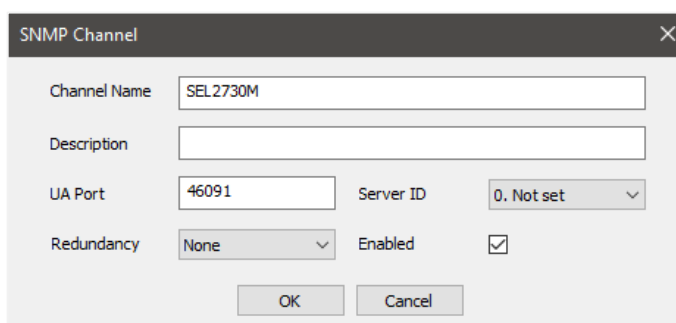
The SNMP channel supports the Simple Network Management Protocol.

D.3.12.1 Add a new channel

In the *FEP Explorer* section, Right Click **Gateway**/Folder has created >> **New channel** >> **SNMP** >> **SNMP Channel** to add a new one:



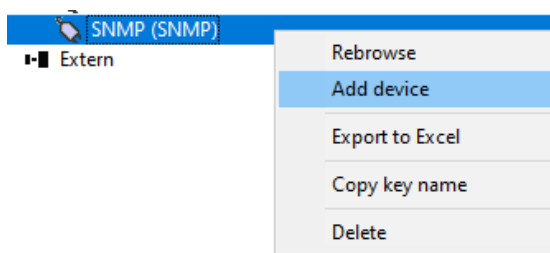
FigureD-116. Add SNMP Channel Menu



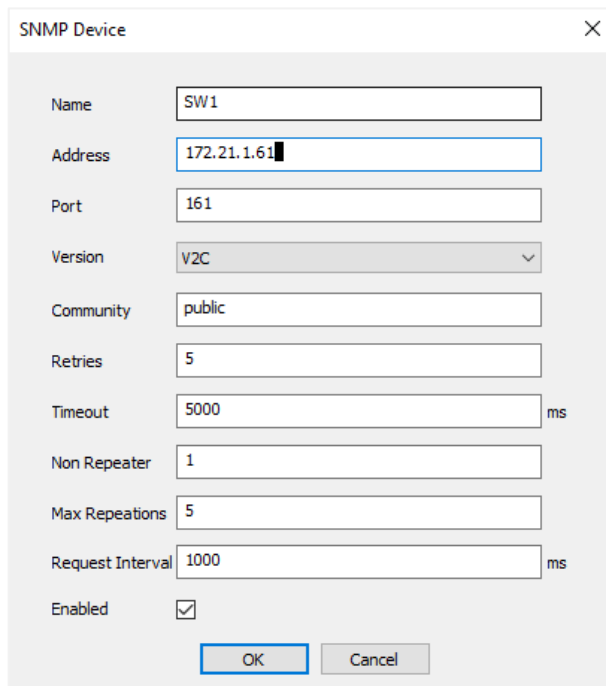
FigureD-117. SNMP Channel Configuration

Configuration parameters:

- **Channel Name:** Channel name
 - **Description:** Channel description
 - **UA Port:** Port return for OPC UA
 - **Server ID:** ID of the machine running FEP
 - **Enabled:** Enable/Disable Channel
- Add **Device:** Right Click the channel, select **Add device**



FigureD-118. Add device menu

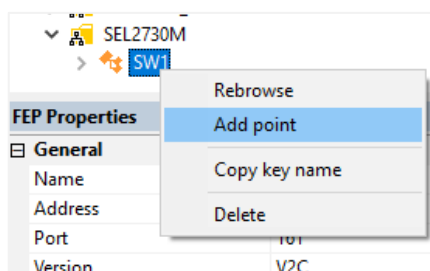


The 'SNMP Device' dialog box contains the following fields and controls:

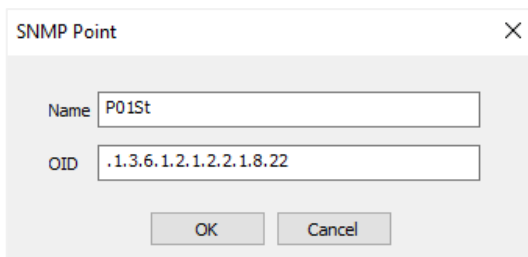
- Name: Text box with 'SW1' entered.
- Address: Text box with '172.21.1.61' entered.
- Port: Text box with '161' entered.
- Version: Dropdown menu with 'V2C' selected.
- Community: Text box with 'public' entered.
- Retries: Text box with '5' entered.
- Timeout: Text box with '5000' entered, followed by 'ms'.
- Non Repeater: Text box with '1' entered.
- Max Repeations: Text box with '5' entered.
- Request Interval: Text box with '1000' entered, followed by 'ms'.
- Enabled: Check box, which is checked.
- Buttons: 'OK' and 'Cancel' at the bottom right.

FigureD-119. SNMP device configuration

- Add point: Right Click device, select **Add point**



FigureD-120. Menu to add points



The 'SNMP Point' dialog box contains the following fields and controls:

- Name: Text box with 'P01St' entered.
- OID: Text box with '.1.3.6.1.2.1.2.2.1.8.22' entered.
- Buttons: 'OK' and 'Cancel' at the bottom right.

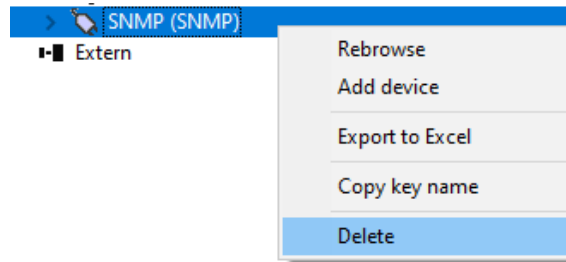
FigureD-121. SNMP point configuration

D.3.12.2 Channel Calibration

Similar to the channel correction section IEC61850 Client.

D.3.12.3 Delete a channel

Right-click on the channel you want to delete and select **Delete**:



FigureD-122. SNMP Channel Deletion Menu

D.3.12.4 Export data channels

Similar to the M101 channel.

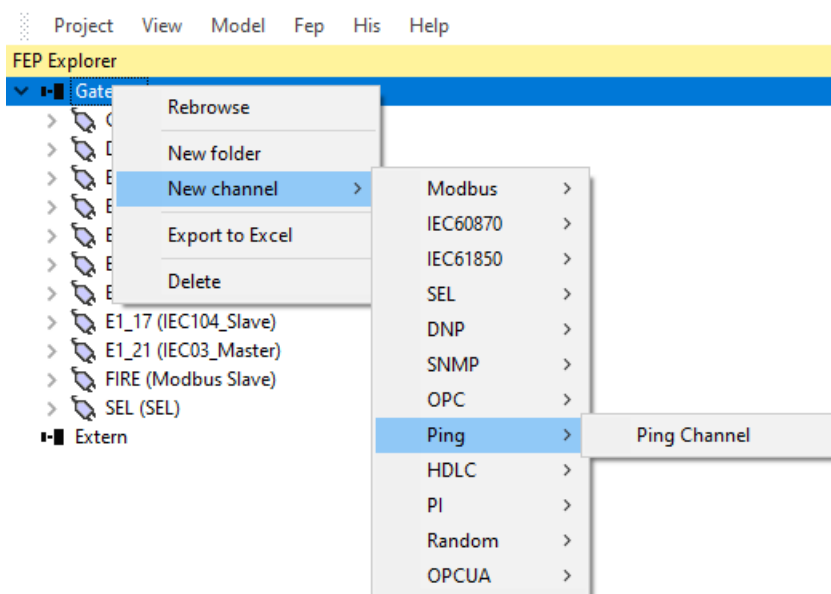
D.3.13. Ping Channel

The Ping channel supports the ICMP protocol for the purpose of line monitoring.

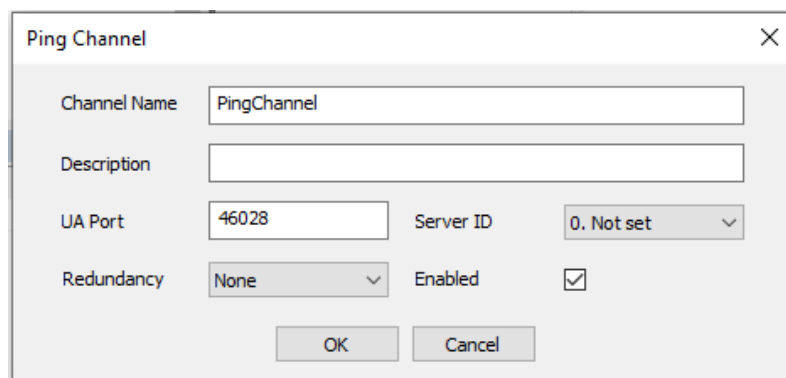
D.3.13.1 Add a new channel

- Add a channel:

In the *FEP Explorer* section, Right Click **Gateway**/Folder created >> **New channel** >> **Ping** >> **Ping Channel** to add new:



FigureD-123. Add Ping Channel Menu

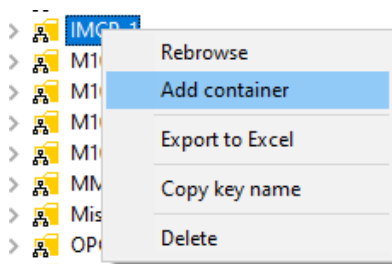
The screenshot shows the 'Ping Channel' configuration dialog box. It contains the following fields and controls: 'Channel Name' with the text 'PingChannel'; 'Description' with an empty text box; 'UA Port' with the value '46028'; 'Server ID' with a dropdown menu showing '0. Not set'; 'Redundancy' with a dropdown menu showing 'None'; and an 'Enabled' checkbox which is checked. At the bottom are 'OK' and 'Cancel' buttons.

FigureD-124. Ping Channel Configuration

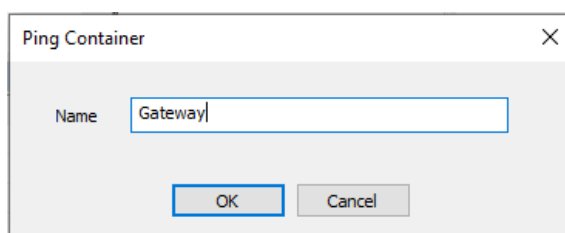
Configuration parameters:

- **Channel Name:** Channel name
- **Description:** Channel description
- **UA Port:** Port return for OPC UA
- **Server ID:** ID of the machine running FEP
- **Enabled:** Enable/Disable Channel

- **Add Container:** Right Click the channel, select **Add container**

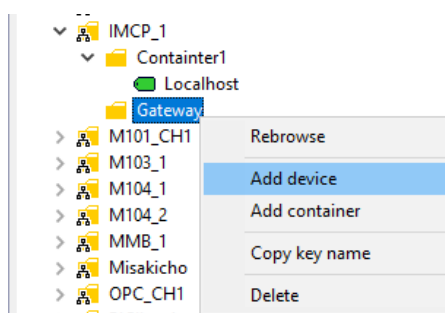


FigureD-125. Add menu Ping container

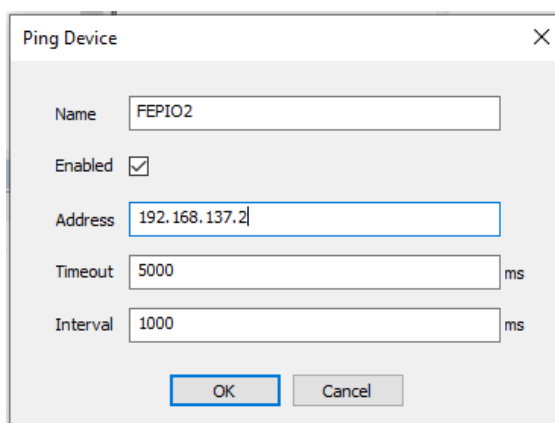


FigureD-126. Container Ping Configuration

- **Add device:** Right Click container, select **Add device**



FigureD-127. Add Ping device menu



FigureD-128. Ping Channel - Device Configuration

Configuration parameters:

- **Name:** Device Name
- **Description:** Equipment Description

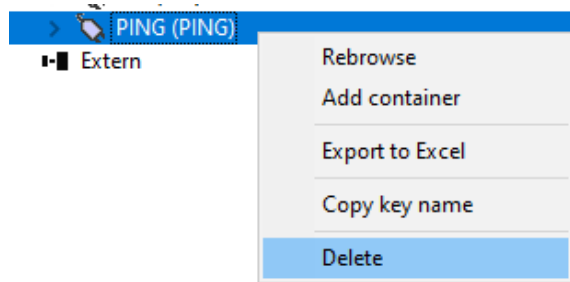
- **Address:** The IP address of the device
- **Timeout:** Controls the time for Ping orders to be executed
- **Interval:** Ping channel cycle

D.3.13.2 Channel Calibration

The FEP Properties *interface* allows users to view and change channel, container, and device information.

D.3.13.3 Delete a channel

Right-click on the channel you want to delete and select **Delete**:



FigureD-129. Ping Channel Removal Menu

D.3.13.4 Export data channels

Similar to the M101 channel.

D.3.14. SEL Channel

The SEL channel supports the SEL Fast Message protocol used to connect to SEL devices.

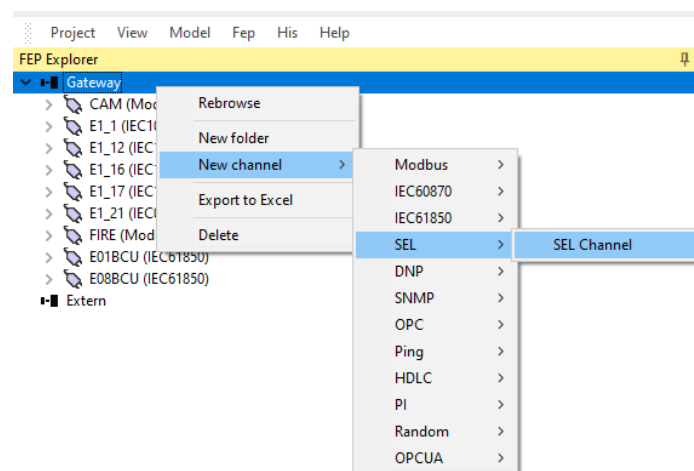
D.3.14.1 Add a new channel

There are two ways to add a new channel: manually and import from a file.

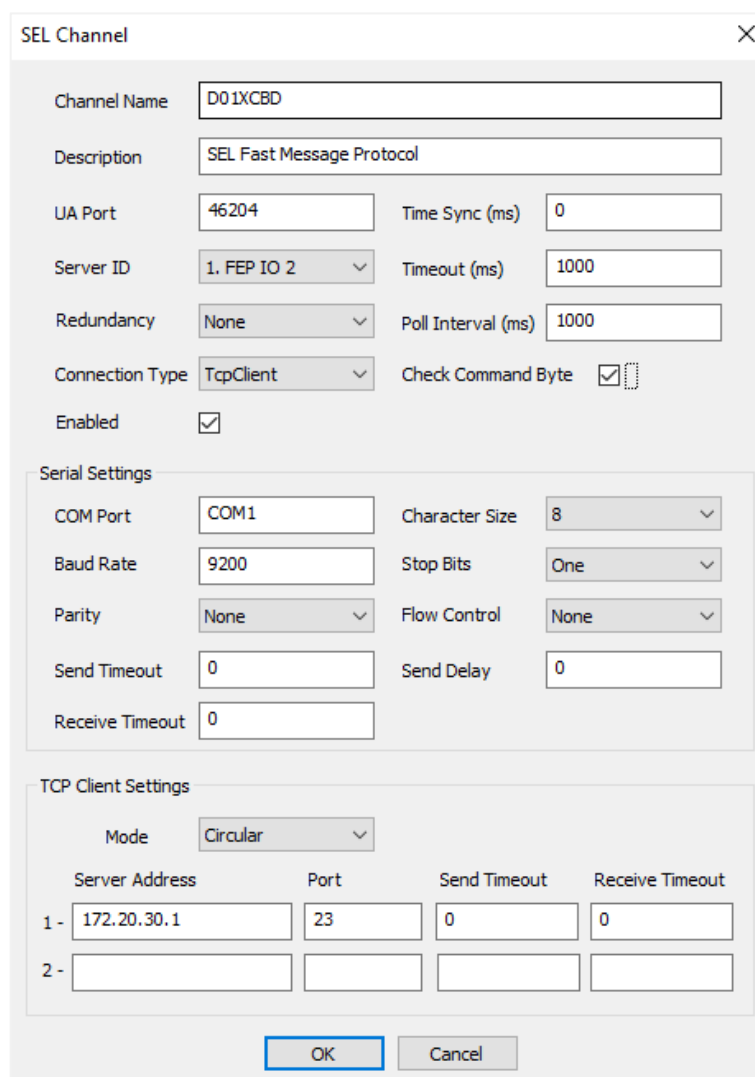
- Import from excel file: Similar to channel M101
- Handicraft

- Add a channel

In the *FEP Explorer* section, Right Click **Gateway**/Folder created >> **New channel** >> **SEL** >> **SEL Channel** to add a new one:



FigureD-130. Add SEL Channel Menu



The dialog box is titled "SEL Channel" and contains the following configuration fields:

- Channel Name:** D01XCBD
- Description:** SEL Fast Message Protocol
- UA Port:** 46204
- Time Sync (ms):** 0
- Server ID:** 1. FEP IO 2 (dropdown)
- Timeout (ms):** 1000
- Redundancy:** None (dropdown)
- Poll Interval (ms):** 1000
- Connection Type:** TcpClient (dropdown)
- Check Command Byte:** ☒
- Enabled:** ☒

Serial Settings:

- COM Port:** COM1
- Character Size:** 8 (dropdown)
- Baud Rate:** 9200
- Stop Bits:** One (dropdown)
- Parity:** None (dropdown)
- Flow Control:** None (dropdown)
- Send Timeout:** 0
- Send Delay:** 0
- Receive Timeout:** 0

TCP Client Settings:

- Mode:** Circular (dropdown)
- Server Address, Port, Send Timeout, Receive Timeout table:**

	Server Address	Port	Send Timeout	Receive Timeout
1 -	172.20.30.1	23	0	0
2 -				

Buttons: OK, Cancel

FigureD-131. SEL Channel Configuration

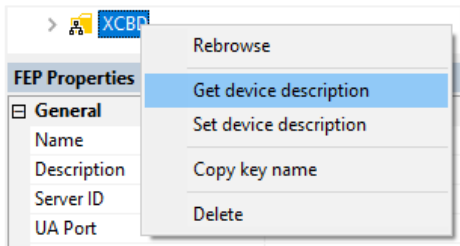
Configuration parameters:

- **Channel Name:** Channel name
- **Description:** Channel description
- **UA Port:** Port return for OPC UA
- **Server ID:** ID of the machine running FEP
- **Enabled:** Enable/Disable Channel
- Depending on the **Connection Type**, configure the parameters for **Serial Settings** or **TCP Client Settings**
- **Check Command Byte:** Select when using a TcpClient connection to ignore Telnet (0xFF) byte commands in the return messages.

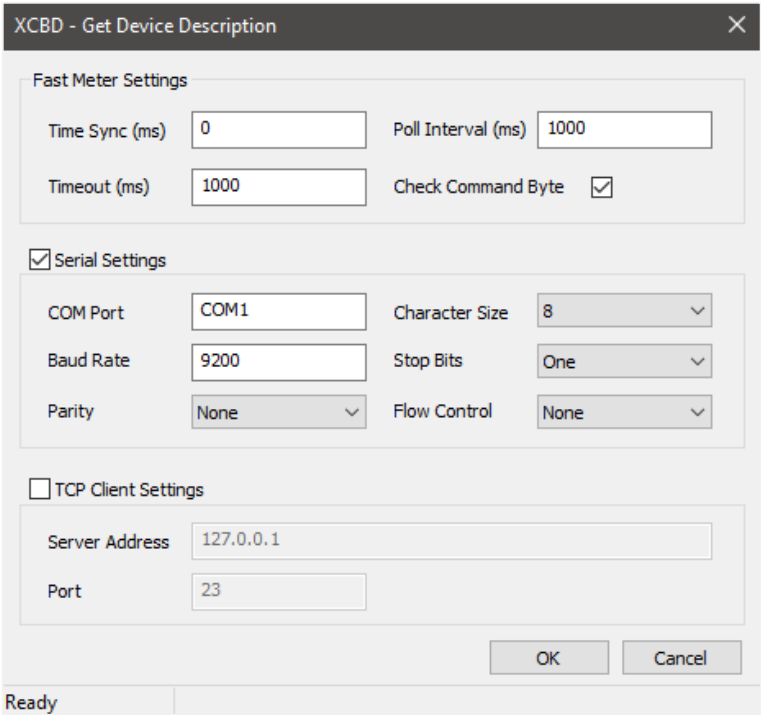
D.3.14.2 Create a data list

Right Click the channel to be created, there are 2 options to create a data list: **Get device description** and **Set device description**.

- **Get device description:** Connect and load the configuration from the device

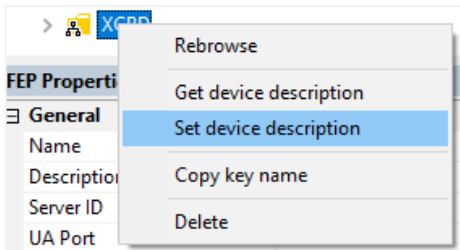


FigureD-132. Get device description menu

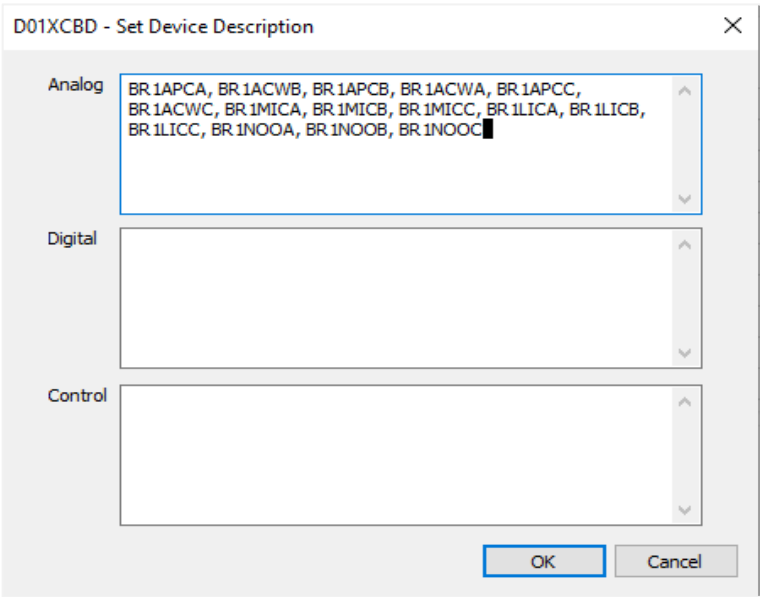


FigureD-133. Device Information

- **Set Devices Description:** create a list based on a given template form



FigureD-134. Menu set device description



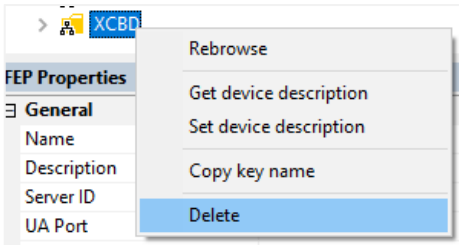
FigureD-135. Configuring Device Description

D.3.14.3 Channel Calibration

The FEP Properties *interface* allows users to view and change channel information.

D.3.14.4 Delete a channel

Right-click on the channel you want to delete and select Delete:



FigureD-136. Delete a SEL channel

D.3.15. PI Client Channel

The PI Client channel is used to connect to the OSI PI database.

D.3.15.1 Add a new channel

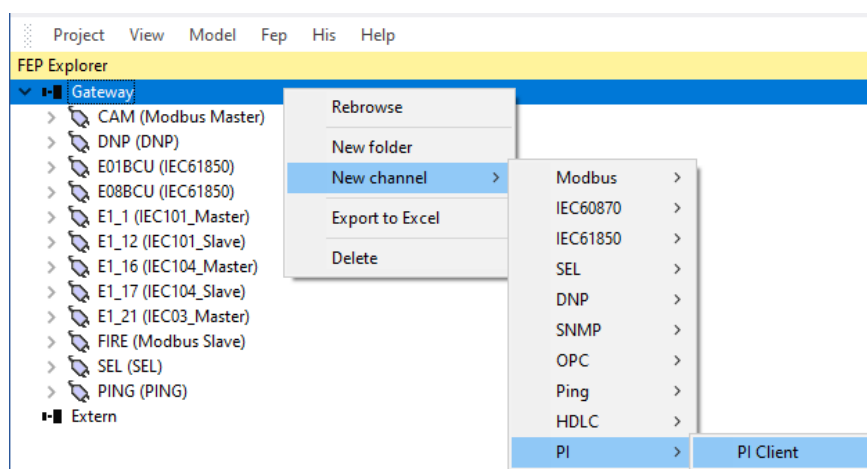
There are 2 ways to add a new channel: Manual and Import from file.

➤ Import from excel file: Similar to channel M101

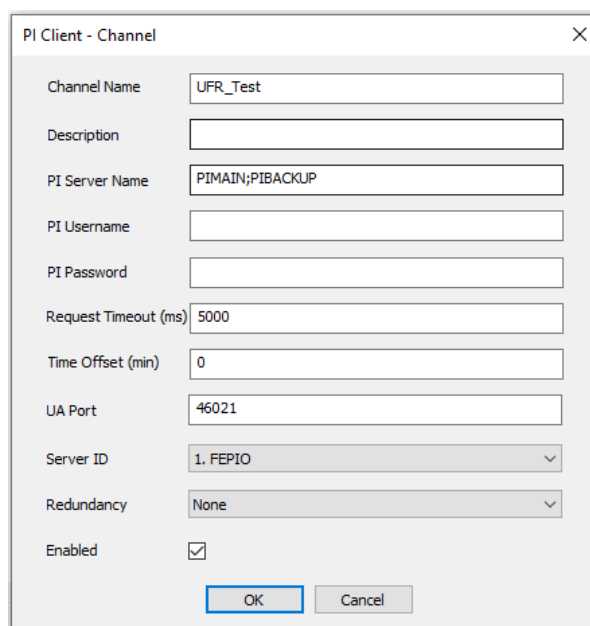
➤ Handicraft:

- Add a channel

In *the FEP Explorer* section, Right Click **Gateway**/Folder has created >> **New channel >> PI >> PI Client** to add new:



FigureD-137. New PI Client channel add menu

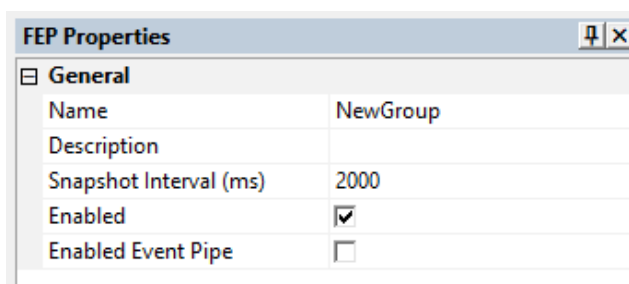


FigureD-138. OPC Client Channel Configuration

Installation parameters:

- **Channel Name:** Channel name

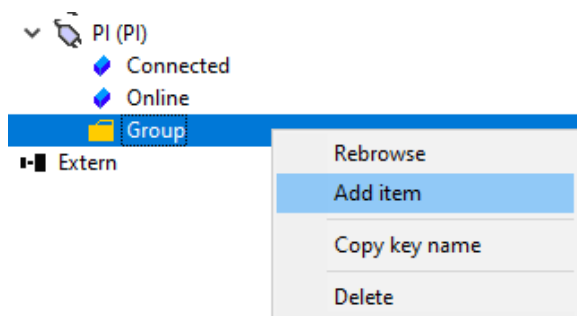
- **Description:** Channel description
 - **UA Port:** Port return for OPC UA
 - **Server ID:** ID of the machine running FEP
 - **Enabled:** Enable/Disable Channel
 - **PI Server Name:** The name of the PI Server declared in the PISDKUtility application. If the server is in the form of Redundancy, it is possible to declare the names of the 2 servers separated by a sign;
 - **Time Offset:** The difference between the time of the computer running FepServer and UTC time, in minutes
 - The remaining parameters are in the form of Reserved
- **Add Group:** Right Click the channel, select **Add group**:



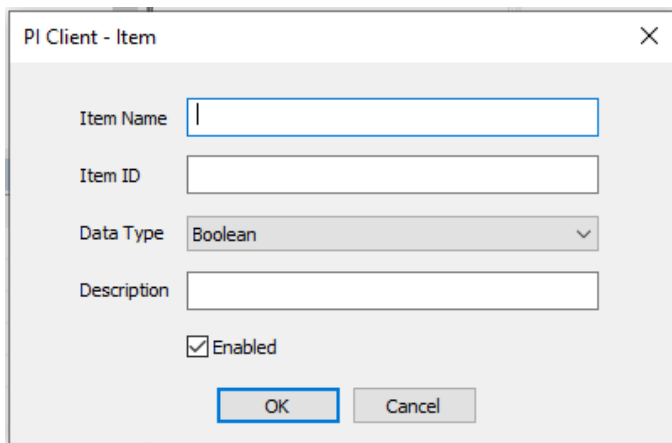
FigureD-139. Configuring PIClient Group

Configuration parameters:

- **Group Name:** Group name
 - **Description:** Group description
 - **Enabled:** Select to activate the group
 - **Snapshot Interval:** Data polling period, if *Snapshot Interval = 0* is selected, the polling feature is not used
 - **Enabled Event Pipe:** The group's data is prioritized to be updated using the event report method. When selected, the group only performs a Get Snapshot once immediately after connecting regardless of the value of the *Snapshot Interval* parameter.
- **Add Item (Point):** Right Click device, select **Add Item**:



FigureD-140. Add item menu



The dialog box titled "PI Client - Item" contains the following fields and controls:

- Item Name:** A text input field.
- Item ID:** A text input field.
- Data Type:** A dropdown menu currently showing "Boolean".
- Description:** A text input field.
- Enabled:** A checked checkbox.
- Buttons:** "OK" and "Cancel" buttons at the bottom.

FigureD-141. Configuring PIClient Items

Configuration parameters:

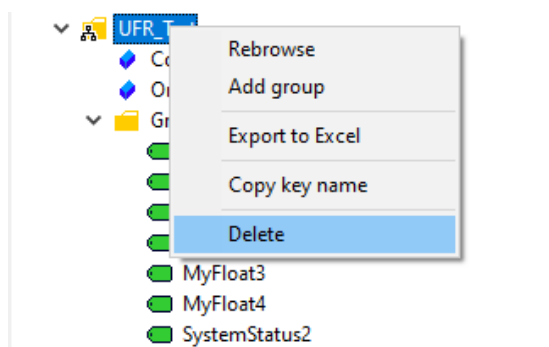
- **Item Name:** The name of the point on the channel's UA Server
- **Item ID:** the name of the point defined in the PI Server connected to
- **Datatype:** data type
- **Description:** description point

D.3.15.2 Channel Calibration

The FEP Properties *interface* allows users to view and change channel, group, and item information.

D.3.15.3 Delete a channel

Right-click on the channel you want to delete and select **Delete**:

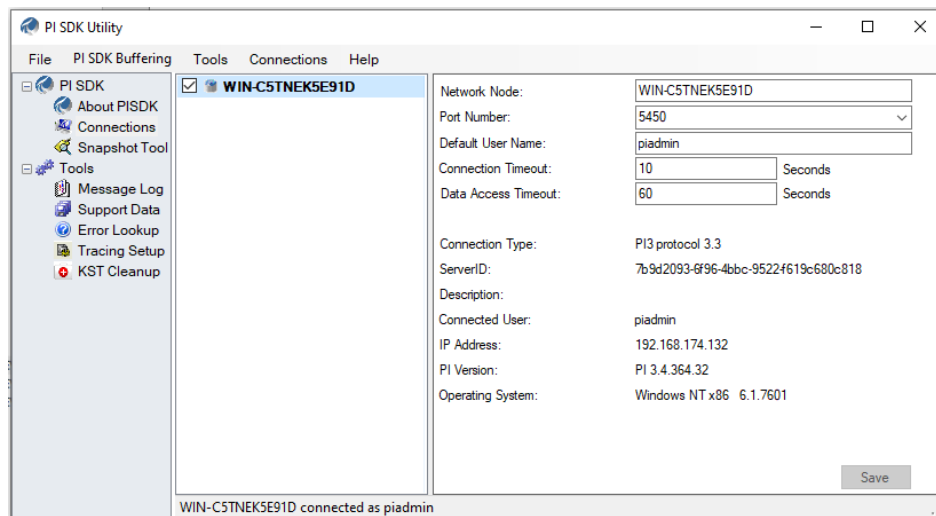


FigureD-142. PI Client channel deletion menu

D.3.15.4 Using a channel

- **Installing PISDKUtility**
 - The PIClient channel does not directly initiate a connection to PIServer, but through the PISDKUtility application.
 - After installing PISDKUtility, it is necessary to start the Pi **Message Subsystem**, **PI Network Manager** and **PIPC Log Server services**.
- **Declare connection to PI Servers**

- Open the PISDKUtility interface in the **Connection menu**, right-click and select AddServer.
- Declare the required parameters and save them.



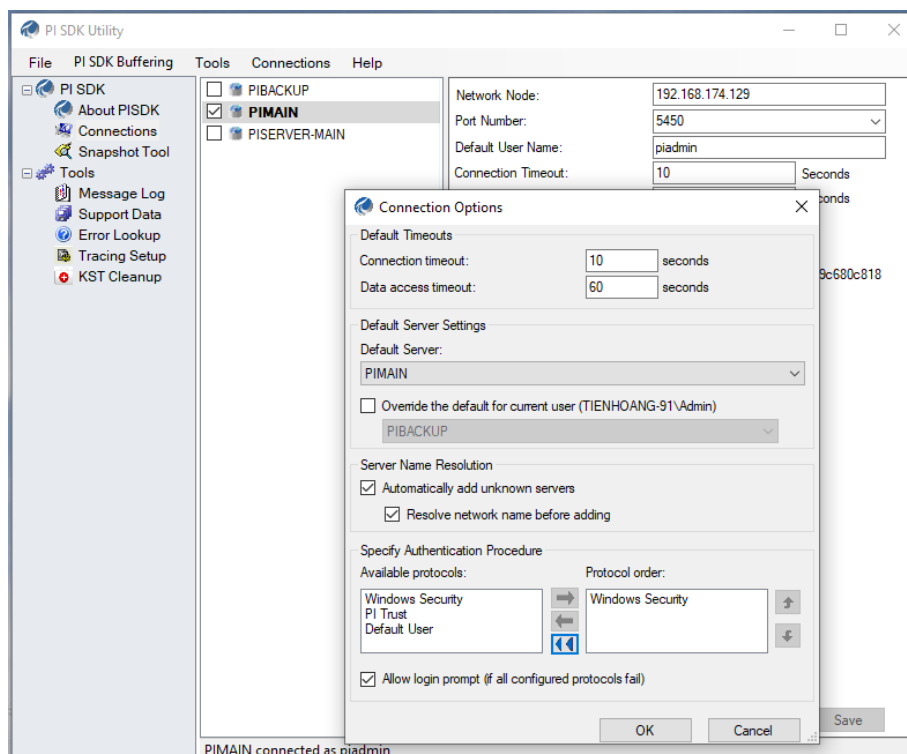
FigureD-143 Declaring a connection to PISDKUtility

- Right-click on the connection name and select **Connect as**, enter the password to access the PI Server to initiate the connection.

- **Declare connection authentication**

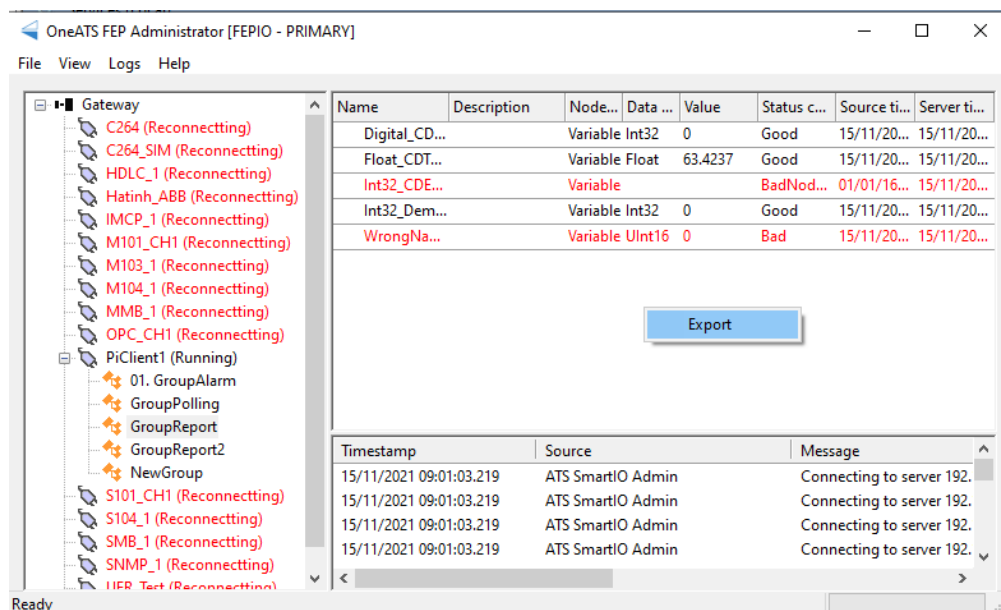
In order for the PIClient channel to have sufficient access and perform read/write features reliably, it is necessary to declare authentication for the connection through PIClient.

On the PISDKUtility menu, select **Connection >> Option**, on the dialog box declaring the authentication method for the PIClient channel.



FigureD-144 Declare the authentication method for the PI Client channel

- **Authenticate with PI Trust**
 - Condition: At the PI Server, declare a PI Trust for the OAFEP application. Channel.PIClient.exe with Fep Server's hostname and IP address
- **Authentication with PI Mapping**
 - Condition: The computer running the PI Server declares a window account and the PI Server must be configured to grant access to that window account.
 - The OAFEP service must be started under the permission of the corresponding window account. To do this, you need to configure Log On for the OAFEP service.



FigureD-145 Configure the account to launch the OAFEP service

- **Function**
 - The PiClient channel supports reading and writing data to the PI Server.
 - These functions may be limited by granting access to specific points in the PI Server.

D.3.15.5 Channel Tracking

- **Track channel status**
 - Connected: Connected = true after successfully establishing a connection between the client and the server.
 - DisconnectedCounter: the number of times the channel has lost connection since launch.

D.3.15.6 Export data channels

Similar to the M101 channel.

D.3.16. DNP Master Channel

D.3.16.1 Add a new channel

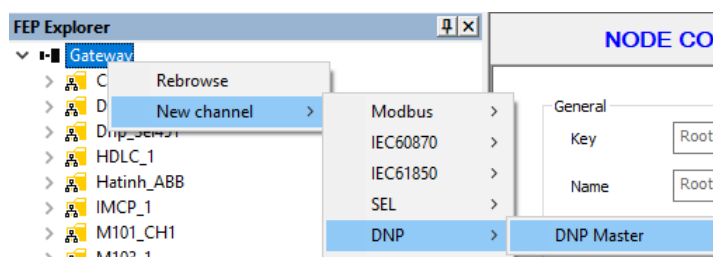
There are two ways to add a new channel: manually and import from a file.

➤ Import from excel file: Similar to channel M101

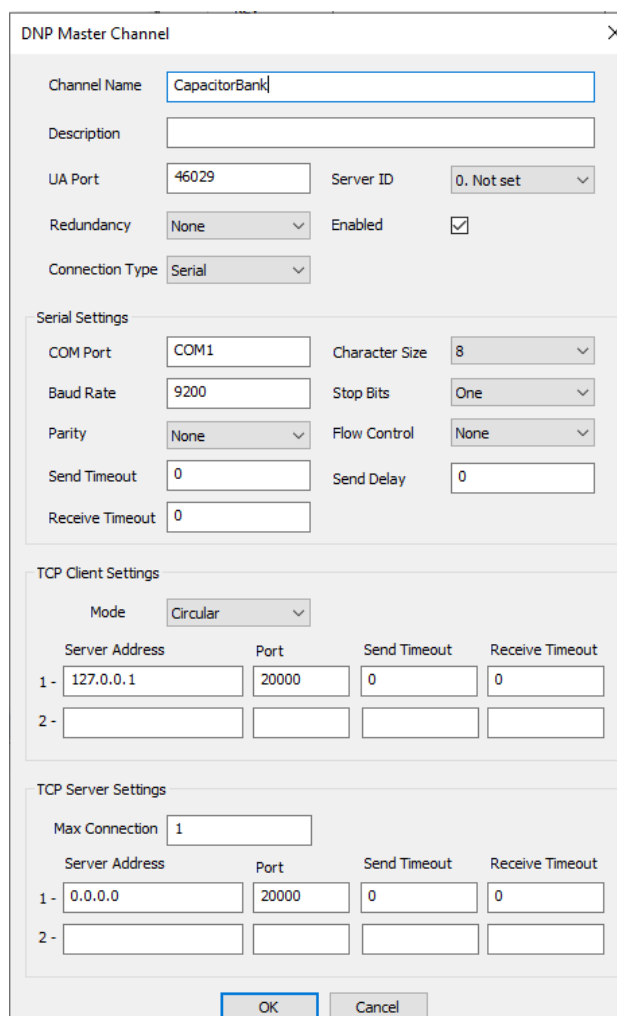
➤ Handicraft

▪ Add a Channel

In the *FEP Explorer* section, Right Click *Gateway*/Folder has created >> *New channel*>> *DNP* >> *DNP Master* to add a new one.



FigureD-146 New DNP Master Channel Addition Menu



The screenshot shows the 'DNP Master Channel' configuration dialog box. It contains the following fields and settings:

- Channel Name:** CapacitorBank
- Description:** (empty)
- UA Port:** 46029
- Server ID:** 0. Not set
- Redundancy:** None
- Enabled:** ☒
- Connection Type:** Serial
- Serial Settings:**
 - COM Port:** COM1
 - Character Size:** 8
 - Baud Rate:** 9200
 - Stop Bits:** One
 - Parity:** None
 - Flow Control:** None
 - Send Timeout:** 0
 - Send Delay:** 0
 - Receive Timeout:** 0
- TCP Client Settings:**
 - Mode:** Circular
 - Server Address, Port, Send Timeout, Receive Timeout:**

	Server Address	Port	Send Timeout	Receive Timeout
1 -	127.0.0.1	20000	0	0
2 -				
- TCP Server Settings:**
 - Max Connection:** 1
 - Server Address, Port, Send Timeout, Receive Timeout:**

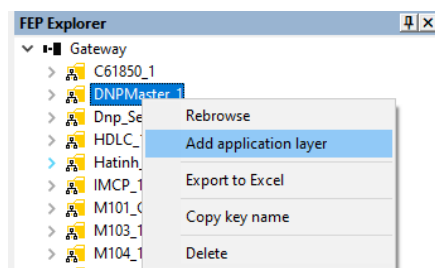
	Server Address	Port	Send Timeout	Receive Timeout
1 -	0.0.0.0	20000	0	0
2 -				

Buttons: OK, Cancel

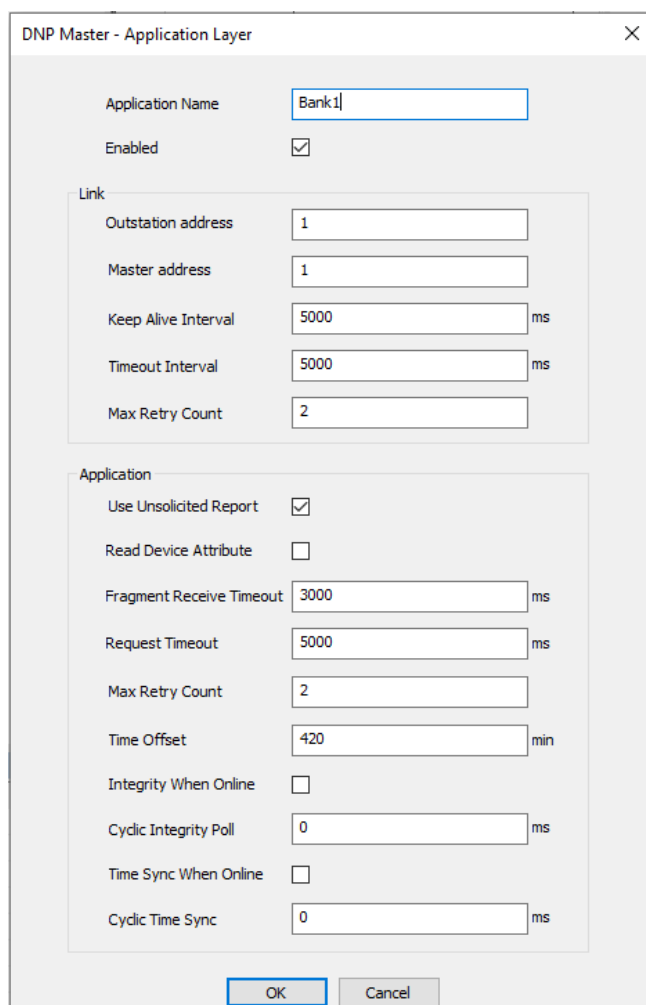
FigureD-147. DNP Master Channel Configuration

Main configuration parameters:

- **Channel Name:** Channel name
 - **Description:** Channel description
 - **UA Port:** Port return for OPC UA
 - **Server ID:** ID of the machine running FEP
 - **Enabled:** Enable/Disable Channel
 - **Connection Type:** Serial/TcpClient
 - **Serial Settings:** the parameter for the Serial connection type
 - **Tcp Client Setting:** the parameter for the TcpClient connection type
- **Add application layer:** Right Click the channel you just created, select **Add application layer**



FigureD-148. Add a new application layer menu



DNP Master - Application Layer

Application Name:

Enabled: ☒

Link

Outstation address:

Master address:

Keep Alive Interval: ms

Timeout Interval: ms

Max Retry Count:

Application

Use Unsolicited Report: ☒

Read Device Attribute: ☐

Fragment Receive Timeout: ms

Request Timeout: ms

Max Retry Count:

Time Offset: min

Integrity When Online: ☐

Cyclic Integrity Poll: ms

Time Sync When Online: ☐

Cyclic Time Sync: ms

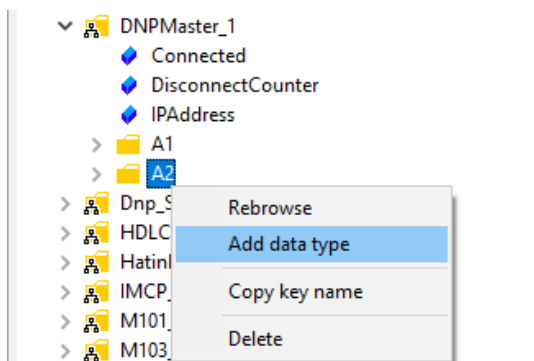
OK Cancel

FigureD-149. Configure the application layer

The DNP Master channel organizes the Link Layer and Application Layer according to DNP standards into the same Application Layer object.

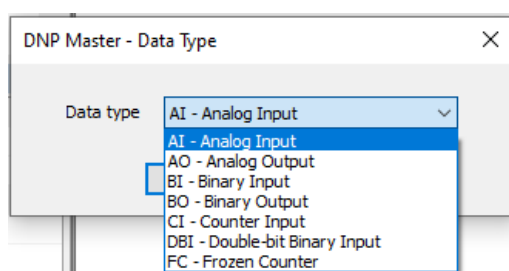
Configuration parameters:

- **Link Settings:**
 - **Outstation Address:** Address of the slave device
 - **Master Address:** Master device address
 - **Keep Alive Interval:** The cycle of sending transmission control bulletins when the channel has no information to exchange
 - **Timeout Interval:** Controls the execution time of a task at LinkLayer
 - **Max Retry Count:** The maximum number of times a task is performed in the LinkLayer
 - **Application Settings:**
 - **Use Unsolicited Report:** Use the unsolicited report as the primary data retrieval method
 - **Read Device Attribute:** Master reads about the variable that represents the slave device state
 - **Fragment Receive Timeout:** A fragment is a unit of data that is transmitted/received at the application layer, which can involve multiple messages at the bottom layer. This parameter controls the time from the first newsletter to the last newsletter of a fragment
 - **Request Timeout:** Controls the execution time of a task at the ApplicationLayer
 - **Time Offset:** The difference between local time and UTC time in minutes
 - **Max Retry Count:** The maximum number of times a task is performed at the ApplicationLayer
 - **Integrity Request Settings:**
 - **Upon Connected:** Send a checksum command as soon as there is a connection
 - **Interval:** Sends a checksum command according to the setting cycle in this parameter
 - **TimeSync Settings:**
 - **Upon Connected:** Send a time sync command as soon as there is a connection
 - **Interval:** Sends the command to synchronize the time according to the setting cycle in this parameter
- **Add DataType:** Right Click the application layer, select **Add data type:**



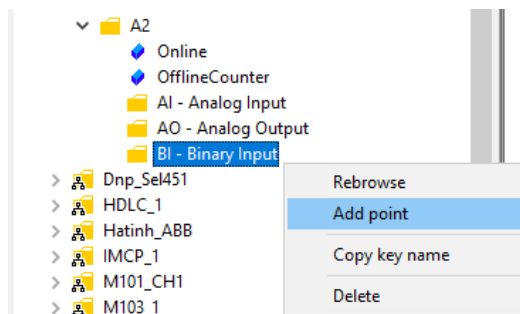
FigureD-150. Add a new data type menu

Select the appropriate data type:



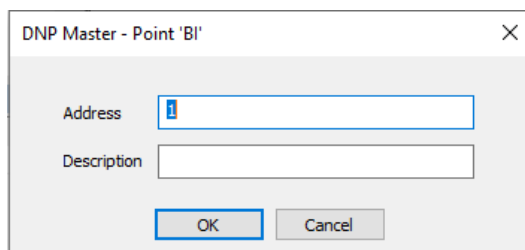
FigureD-151. Select data type

- **Add point:** Right Click Data Type, select **Add point:**



FigureD-152. Menu add new points

- **Warning and measurement points (AI, BI, DBI, FC)**

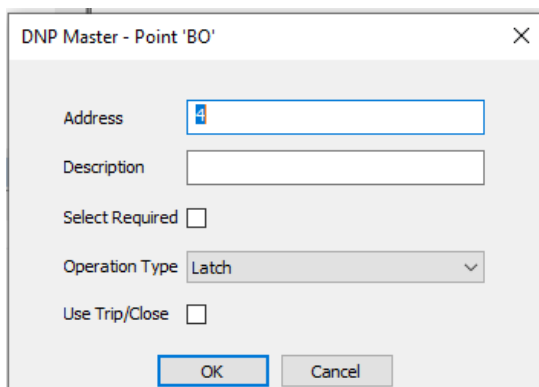


FigureD-153. Configuring Alert Points

Configuration parameters:

- **Address:** Point address
- **Description:** description point

○ **Binary control points**

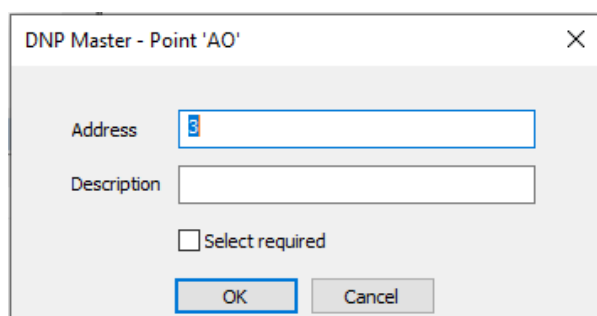


FigureD-154 Configuring Binary Control Points

Configuration parameters:

- **Address:** Point address
- **Description:** description point
- **Select Required:** optional control method of the point
 True: SBO method
 False: Direct Control method
- **Operation Type:** optional output form of the control command
 Latch: hold the output state
 Pulse: Pulsed Output
- **Use Trip/Close:** optional control command output signal meaning
 Disable: The corresponding output of a signal
 (ctlVal = true ~ on; ctlVal = true ~ off)
 Enable: The output corresponds to two signals
 (ctlVal = true ~ close; ctlVal = true ~ trip)

○ **Analog control points**



FigureD-155 Analog Control Point Configuration

Configuration parameters:

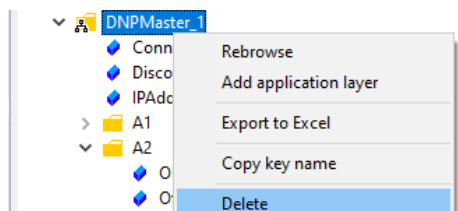
- **Address:** Point address
- **Description:** description point
- **Select Required:** optional control method of the point
 True: SBO method
 False: Direct Control method

D.3.16.2 Channel Calibration

The FEP Properties *interface* allows users to view and change channel information, application layers, and points.

D.3.16.3 Delete a channel

Right-click on the channel you want to delete and select **Delete**:



FigureD-156. Delete a DNP Master channel

D.3.16.4 Using a channel

- **Integrity Request**
 - The Integrity command is used to review all the data of the station, synchronize the master state with the slave.
 - When the master sends an Integrity Request, the slave sends all the data of the points it has to the master.
- **Clock Sync**
 - The Clock Sync command is used to synchronize the slave's time clock with the master.
- **Get Data**
 - The DNP channel uses unsolicited reports as the primary data acquisition method.

D.3.16.5 Channel Tracking

- **Track channel status**
 - Connected: Connected = true after successfully establishing a connection between master and slave.
 - DisconnectedCounter: the number of times the channel has lost connection since launch.
- **Monitor application layer status**
 - Online: Online = true after the application layer in the master and slave shake hands successfully.
 - OfflineCounter: the number of times the channel has lost connection to the adjacent application layer since launch.

D.3.16.6 Export data channels

Similar to the M101 channel.

D.3.17. DNP OutStation Channel

D.3.17.1 Add a new channel

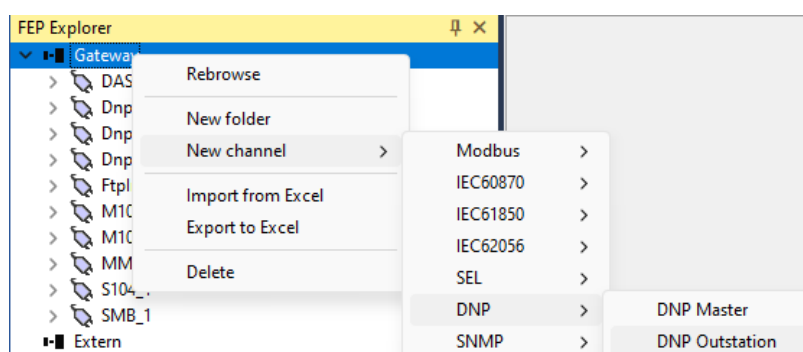
There are two ways to add a new channel: manually and import from a file.

➤ Import from excel file: Similar to channel M101

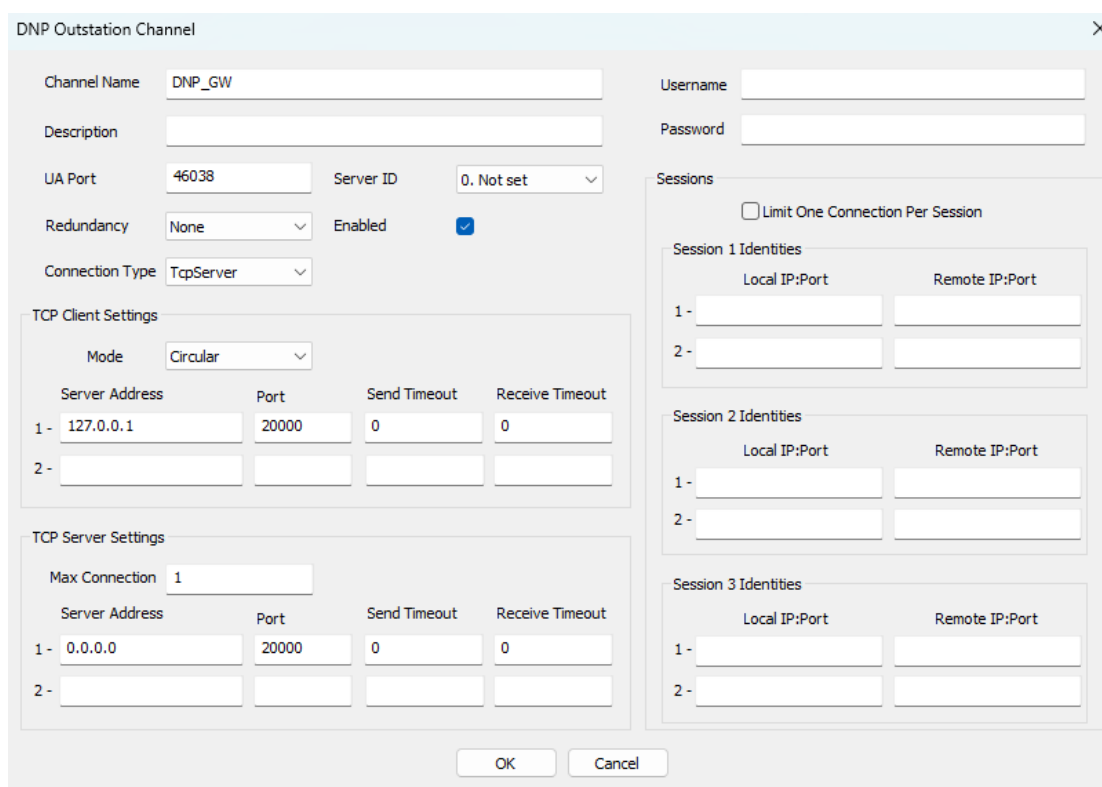
➤ Handicraft

▪ Add a Channel

In *the FEP Explorer* section, Right Click **Gateway**/Folder has created >> **New channel**
>> **DNP** >> **DNP OutStation** to add a new one.



FigureD-157 Add a new DNP OutStation channel menu

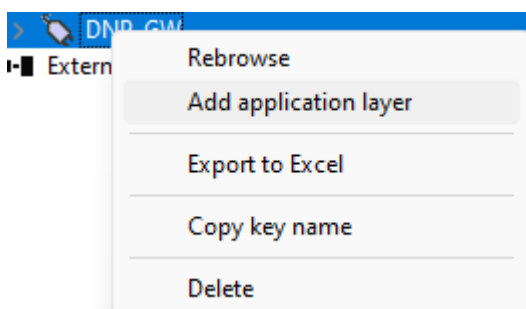


FigureD-158. Configuring the OutStation DNP Channel

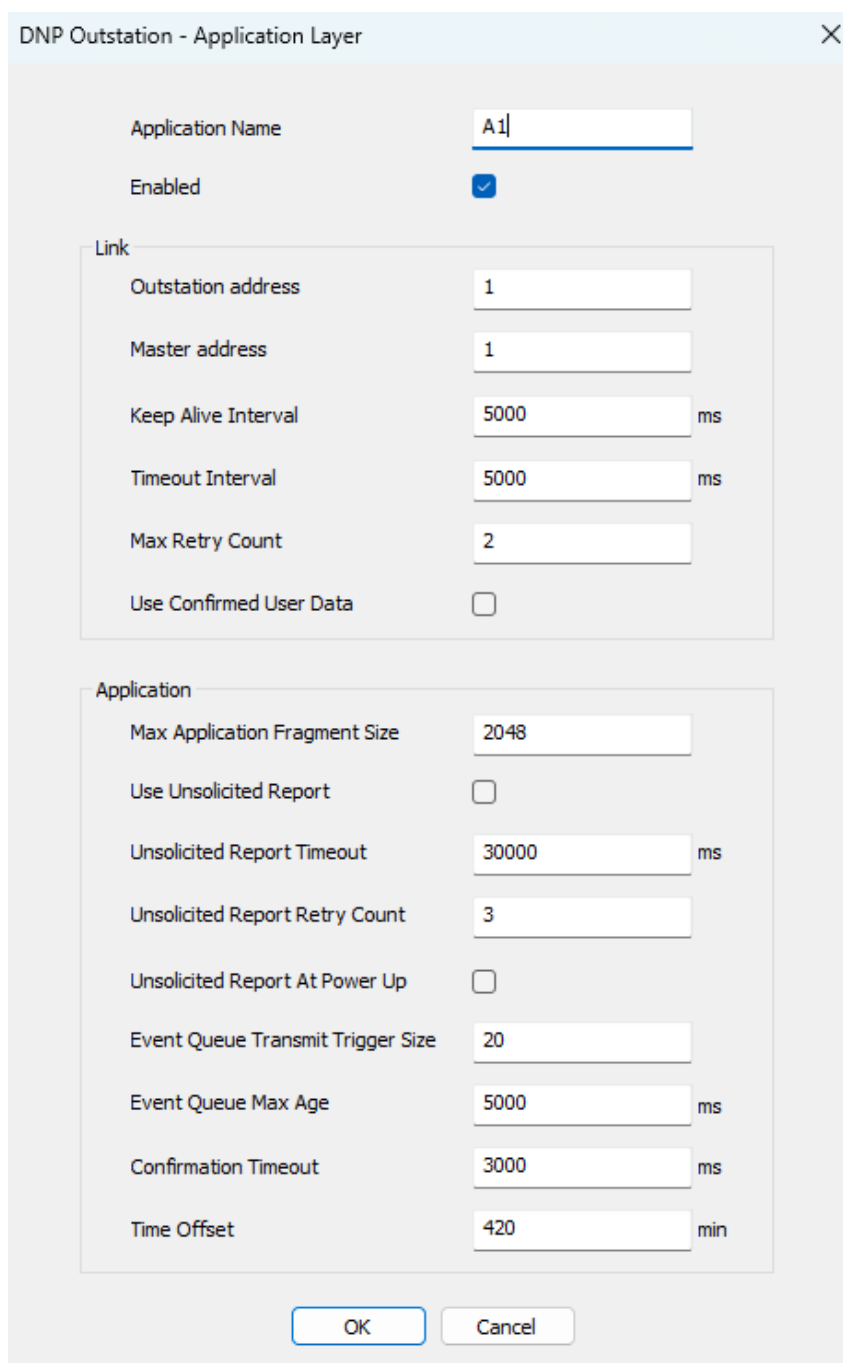
Main configuration parameters:

- **Channel Name:** Channel name

- **Description:** Channel description
- **UA Port:** Port return for OPC UA
- **Server ID:** ID of the machine running FEP
- **Enabled:** Enable/Disable Channel
- **Connection Type:** Serial/TcpClient
- **Username/Password:** The username/password for the channel to log in to the DataServer
- **TCP Server Setting:**
 - o **Max connection:** The maximum number of clients connected to
 - o **Server Address:** 0.0.0.0: The client can connect through all network cards (both physical and Wifi). If a specific IP address is typed, the Client can only connect through this Network Card.
 - o **Port:** 2000 (Default)
- **Sessions:**
 - o **Number of incoming Active Clients,** can add or subtract the number of Sessions
 - o **Session Identities:** includes 4 parameters of local IP, local port, remote IP, remote port; used to distinguish sessions from each other when the DNPOutStation channel needs to serve multiple connected masters.
- **Add application layer:** Right Click the channel you just created, select **Add application layer**



FigureD-159. Add a new application layer menu



DNP Outstation - Application Layer

Application Name: A1

Enabled: ☒

Link

Outstation address: 1

Master address: 1

Keep Alive Interval: 5000 ms

Timeout Interval: 5000 ms

Max Retry Count: 2

Use Confirmed User Data: ☐

Application

Max Application Fragment Size: 2048

Use Unsolicited Report: ☐

Unsolicited Report Timeout: 30000 ms

Unsolicited Report Retry Count: 3

Unsolicited Report At Power Up: ☐

Event Queue Transmit Trigger Size: 20

Event Queue Max Age: 5000 ms

Confirmation Timeout: 3000 ms

Time Offset: 420 min

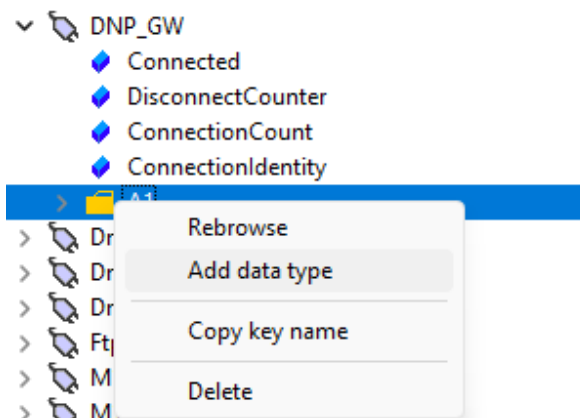
OK Cancel

FigureD-160. Configure the application layer

The OutStation DNP channel organizes the Link Layer and Application Layer according to DNP standards into the same Application Layer object. Configuration parameters:

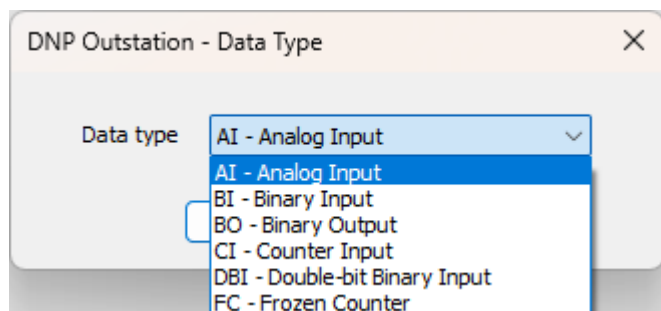
- **Link Settings:**
 - **Outstation Address:** The address of the Outstation device (this channel)
 - **Master Address:** Master device address
 - **Keep Alive Interval:** The cycle of sending transmission control bulletins when the channel has no information to exchange
 - **Timeout Interval:** Controls the execution time of a task at LinkLayer

- **Max Retry Count:** The maximum number of times a task is performed in the LinkLayer
- **Application Settings:**
 - **Max Application Fragment Size:** The limit of the newsletter capacity in the Transportation layer.
 - **Use Unsolicited Report:** Automatically activates unsolicited reports.
 - **Unsolicited Report Timeout (ms):** The time to wait for a newsletter confirmation from DNP Master after sending an unsolicited report.
 - **Unsolicited Report Retry Count:** The maximum number of retries if the unsolicited report has not been confirmed.
 - **Unsolicited Report At Power Up:** Automatically activates the unsolicited report after restarting DNP OutStation.
 - **Event Queue Transmit Trigger Size:** Stretch the cache amount of the event queue
 - **Event Queue Max Age (ms):** The time it takes for event data to change in the event cache.
 - **Confirmation Timeout (ms):** Newsletter validation timeout on ApplicationLayer
 - **Time Offset:** The difference between local time and UTC time in minutes
- **Add DataType:** Right Click the application layer, select **Add data type**:



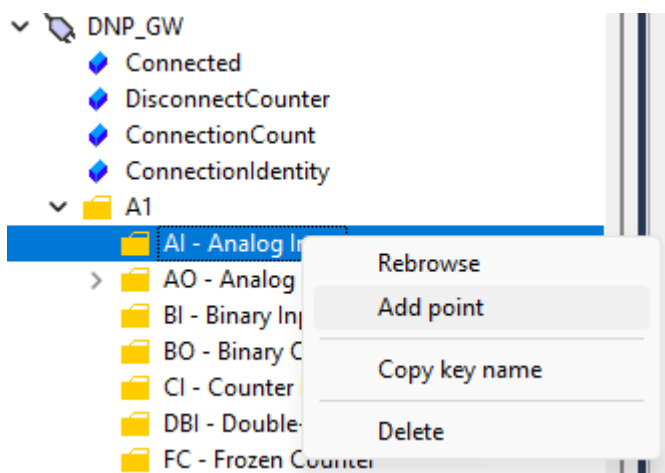
FigureD-161. Add a new data type menu

Select the appropriate data type:



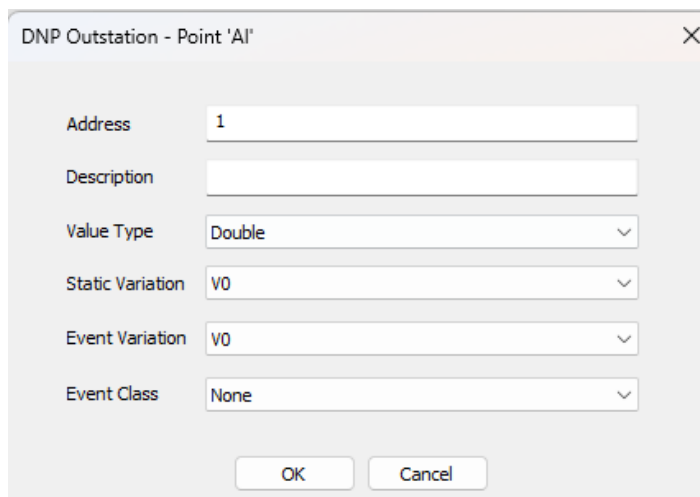
FigureD-162. Select data type

- **Add point:** Right Click Data Type, select **Add point**:



FigureD-163. Menu add new points

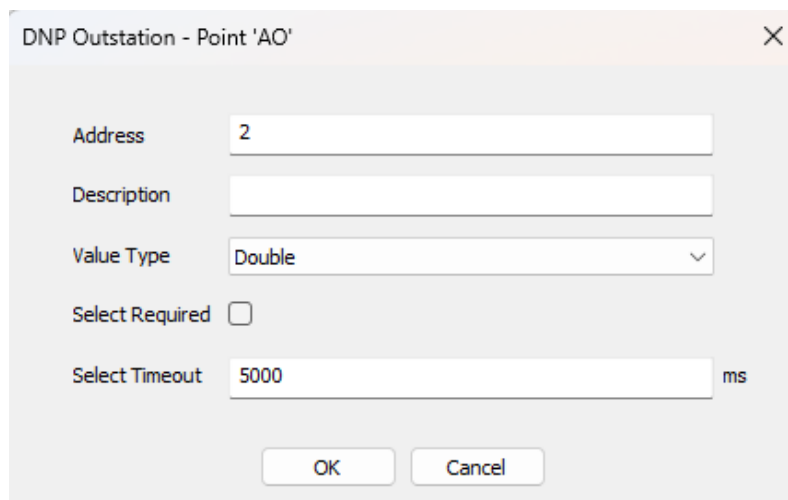
- **Warning and measurement points (AI, BI, DBI, FC)**



FigureD-164. Configuring Alert Points

Configuration parameters:

- **Address:** Point address
 - **Description:** Point description
 - **Value Type:** Displays the data type of the point in the data tree.
 - **Static Variation:** The format of the point value in the non-event newsletter.
 - **Event Variation:** The format of the point value in the event newsletter.
 - **Event Class:** The location of the point in the specific event class.
- **Control points (AO, BO)**



The dialog box titled "DNP Outstation - Point 'AO'" contains the following fields and controls:

- Address:** A text input field containing the value "2".
- Description:** An empty text input field.
- Value Type:** A dropdown menu currently set to "Double".
- Select Required:** An unchecked checkbox.
- Select Timeout:** A text input field containing "5000", followed by a "ms" unit label.
- Buttons:** "OK" and "Cancel" buttons at the bottom right.

FigureD-165 Control Point Configuration

Configuration parameters:

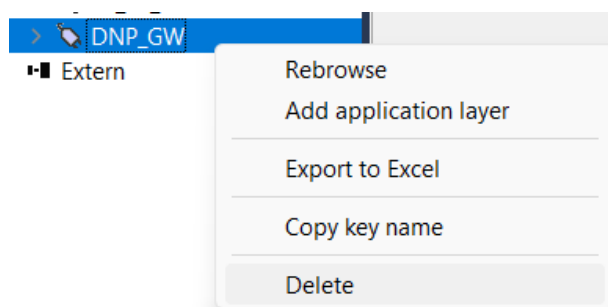
- **Address:** Point address
- **Description:** Point description
- **Value Type:** Displays the data type of the point in the data tree
- **Select Required:** optional control method of the point
 - True:* SBO method
 - False:* Direct Control method
- **Select Timeout:** the time allowed to wait for the control command to be selected

D.3.17.2 Channel Calibration

The FEP Properties *interface* allows users to view and change channel information, application layers, and points.

D.3.17.3 Delete a channel

Right-click on the channel you want to delete and select **Delete**:



FigureD-166. Delete the DNP OutStation channel

D.3.17.4 Using a channel

D.3.17.5 Channel Tracking

- Track channel status

- Connected: Connected = true if the channel logs in successfully on the DataServer.
- DisconnectedCounter: the number of times the channel has lost connection since launch.
- ConnectionCount: The number of clients connected to the channel.
- ConnectionIdentify: The IP address of the client connecting to the channel.

D.3.17.6 Export data channels

Similar to the M101 channel.

D.3.18. HDLC Channel

D.3.18.1 Add a new channel

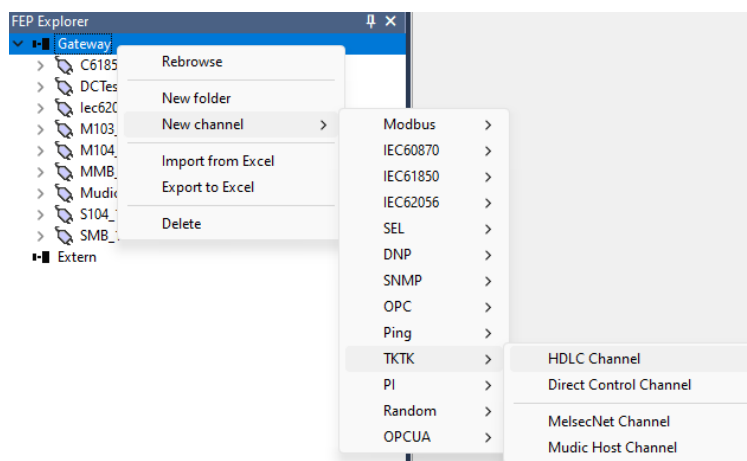
There are two ways to add a new channel: manually and import from a file.

➤ Import from excel file: Similar to channel M101

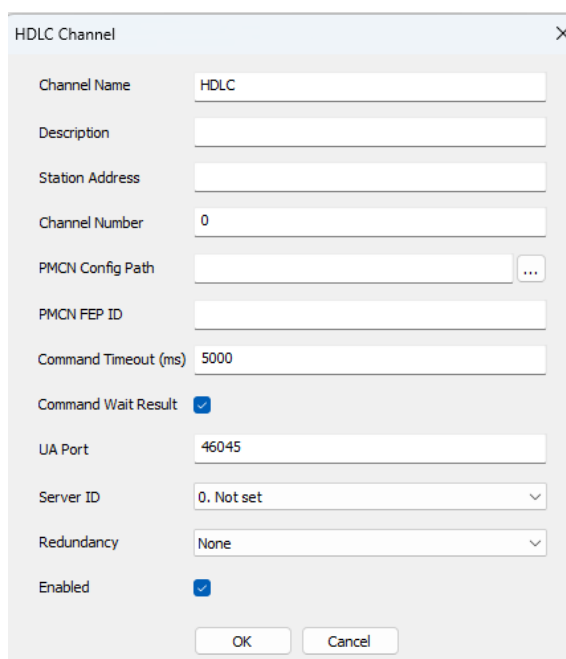
➤ Handicraft

- Add a Channel

In *the FEP Explorer* section, Right Click **Gateway**/Folder has created >> **New channel** >> **TKTK** >> **HDLC Channel** to add a new one.



FigureD-167 Add HDLC Channel Menu

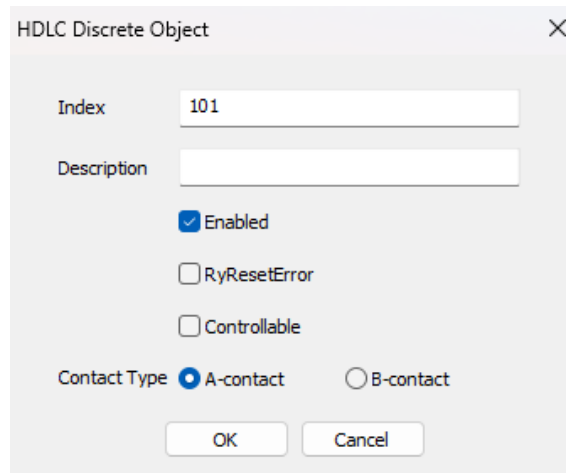


FigureD-168. HDLC Channel Configuration

Configuration parameters:

- **Channel Name:** Channel name
- **Description:** Channel description

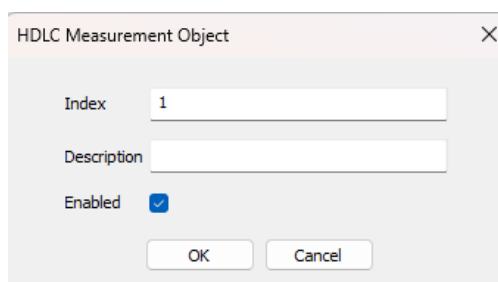
- **Station Address:** The address of the station
- **Channel Number:** The identifier of the channel
- **PMCN Config Path:** The location of the folder containing the PMCN configuration file
- **PMCN FEP ID**
- **Command Timeout(ms):** Controls the execution time of the control command
- **Command Wait Result:** The channel will wait for the result of the order to be executed before running a new command.
- **UA Port:** Port return for OPC UA
- **Server ID:** ID of the machine running FEP
- **Redundancy:** Redundancy Mode
- **Enabled:** Enable/Disable Channel
- **Add Discrete point:** Double Click the channel you just created, Right Click Discrete, select **Add discrete**



FigureD-169. Point Discrete Configuration

Configuration parameters:

- **Enabled:** Enable/Disable point
- **Index:** Index point
- **RyResetError:** Enable/Disable RyResetError
- **Controllable:** Whether the control methods are available or not.
- **ContactType:** Contact Type True(A-contact) / False(B-contact)
- **Add point measurement:** Double Click the channel you just created, Right Click Measurement, select **Add measurement**



The dialog box titled "HDLC Measurement Object" contains the following fields and controls:

- Index:** A text input field containing the value "1".
- Description:** An empty text input field.
- Enabled:** A checkbox that is checked, indicated by a blue checkmark.
- Buttons:** "OK" and "Cancel" buttons at the bottom right.

FigureD-170. Point Measurement Configuration

D.3.18.2 Channel Calibration

The FEP Properties *interface* allows users to view and change channel and point information.

D.3.18.3 Delete a channel

Right-click on the channel you want to delete and select **Delete**.

D.3.18.4 Using a channel

- **Initialization**
 - The FEP module will load the database that performs the init procedure by calling the function equivalent to the TKTK DLL.
- **Data collection**
 - The FEP module will update the data when it receives a data notification or status change from the TKTK DLL.
- **Station Control**

Users can trigger station requests through the FEP module

 - (1) RequestStationStatus: Reads all station statuses
 - (2) RequestDiscrete: Read all Discrete statuses
 - (3) RequestMeasurement: Read all statuses, Measurement values
 - (4) Normalize: Reset Channel
- **Point Discrete Control**

The user can activate the control command for the controllable Discrete point through the SBO control flow.

Users can manually activate the station request through the FEP module

 - (1) Select
 - (2) Execute
 - (3) Cancel

D.3.18.5 Channel Tracking

- **Track channel status**
 - **Connected:** Shows the channel that ran successfully.
 - **Online:** Displays the DLL of the TKTK ready to exchange data.
 - **RequestResults:** Displays the final data request result
 - **ControlResults:** Displays the result of the last discrete order.

D.3.18.6 Export data channels

Similar to the M101 channel.

D.3.19. Direct Control Channel

D.3.19.1 Add a new channel

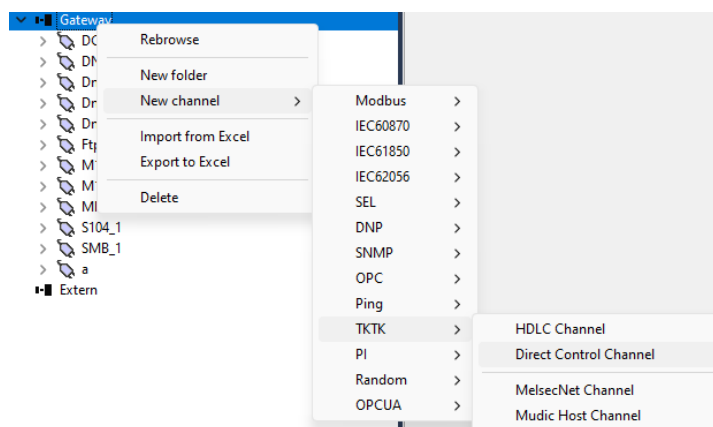
There are two ways to add a new channel: manually and import from a file.

➤ Import from excel file: Similar to channel M101

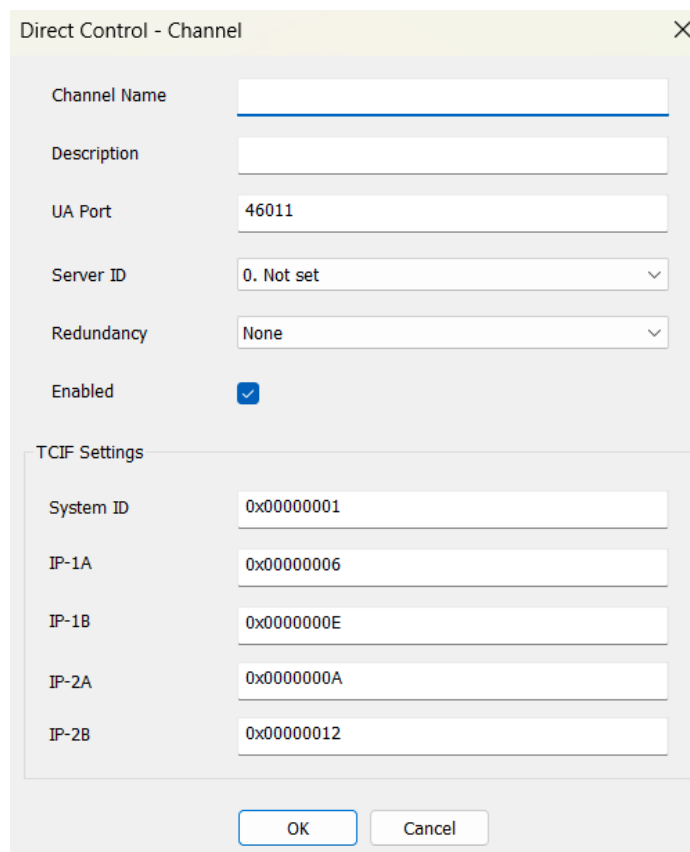
➤ Handicraft

▪ Add a Channel

In *the FEP Explorer* section, Right Click **Gateway**/Folder has created >> **New channel** >> **TKTK** >> **Direct Control Channel** to add a new one.



FigureD-171 Add a new Direct Control Channel menu



The dialog box 'Direct Control - Channel' contains the following fields and options:

- Channel Name: [Text input field]
- Description: [Text input field]
- UA Port: 46011
- Server ID: 0. Not set (dropdown menu)
- Redundancy: None (dropdown menu)
- Enabled: ☒
- TCIF Settings section:
 - System ID: 0x00000001
 - IP-1A: 0x00000006
 - IP-1B: 0x0000000E
 - IP-2A: 0x0000000A
 - IP-2B: 0x00000012

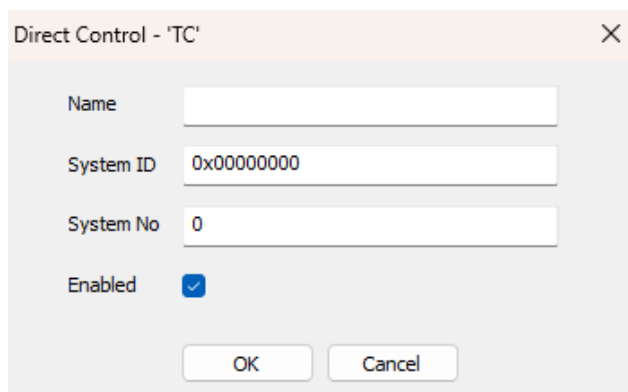
Buttons: OK, Cancel

FigureD-172. Direct Control Channel Configuration

Main configuration parameters:

- **Channel Name:** Channel name
- **Description:** Channel description
- **UA Port:** Port return for OPC UA
- **Server ID:** ID of the machine running FEP
- **Enabled:** Enable/Disable Channel
- **Redundancy:** Redundancy Mode
- **TCIF Setting:**
 - *SystemID:* TCIF System ID
 - *IP-1A:* TCIF IP1 A address
 - *IP-1B:* TCIF IP1 B address
 - *IP-2A:* TCIF IP2 A Address
 - *IP-2B:* TCIF IP2 B address

- **Add TeleCommunication Unit (TC):** Right Click the channel you just created, select **Add TC**

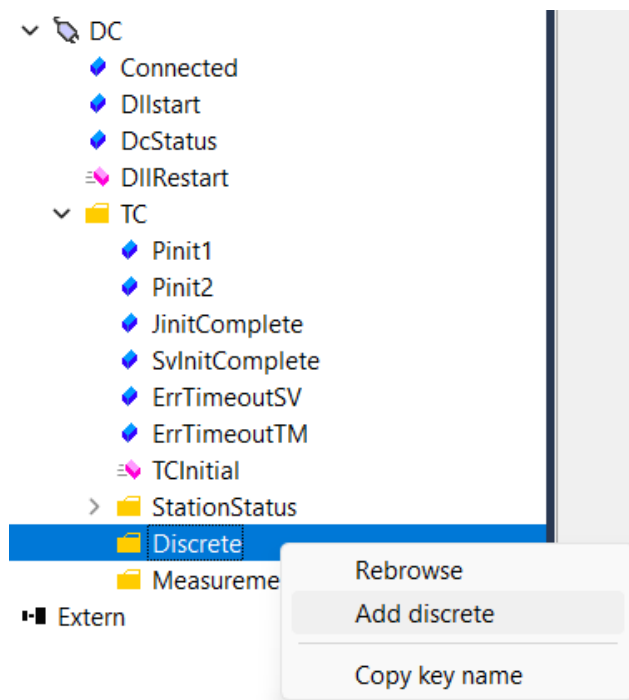


The image shows a dialog box titled "Direct Control - 'TC'". It contains four input fields: "Name" (empty), "System ID" (containing "0x00000000"), "System No" (containing "0"), and "Enabled" (checked). At the bottom are "OK" and "Cancel" buttons.

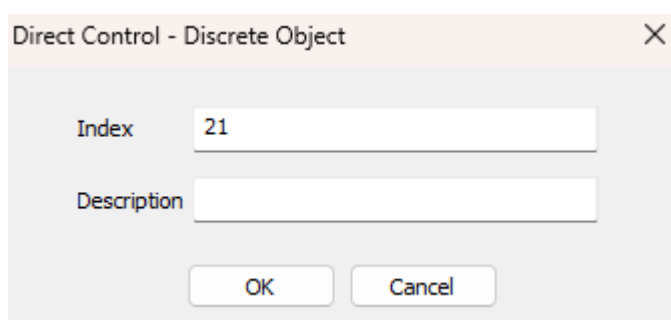
FigureD-173. TeleCommunication Unit Configuration

Configuration parameters:

- **Name:** Name of TC
- **SystemID:** The ID of the TC
- **System No:** Number of TC
- **Enable:** Enable/Disable status
- **Add point Discrete:** Double Click the TC you just created, Right Click **Discrete**, select **Add discrete:**



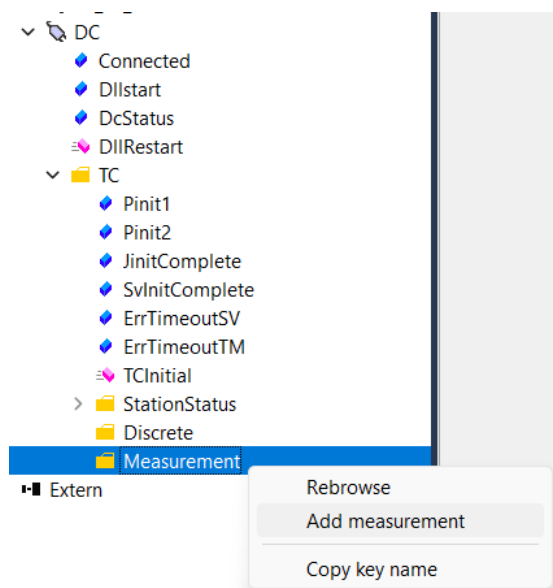
FigureD-174. New menu add point discrete



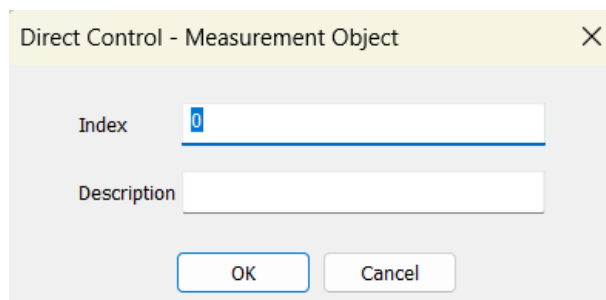
FigureD-175. Point Discrete Configuration

Configuration parameters:

- **Index**: Index point
- **Description**: Point description
- **Add point Measurement**: Double Click TC just created, Right Click **Measurement**, select **Add measurement**:



FigureD-176. Add a new point measurement menu



FigureD-177. Point measurement configuration

Configuration parameters:

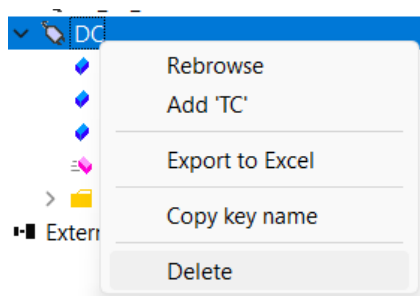
- **Index:** Index point
- **Description:** Point description

D.3.19.2 Channel Calibration

The FEP Properties *interface* allows users to view and change channel information, TCs, and points.

D.3.19.3 Delete a channel

Right-click on the channel you want to delete and select **Delete**:



FigureD-178. Delete a Direct Control channel

D.3.19.4 Using a channel

- **EtcRestart:**
 - The DllRestart command is used to reset all the state of the channel.
 - When the user invokes the DllRestart command, all TCs are also restarted.
- **TCInitial**
 - The TCInitial command is used to reset each TC.

D.3.19.5 Channel Tracking

- **Track channel status**
 - Connected: The channel is connected
 - Dllstart: The DLL was successfully run
 - DcStatus: Active channel with no errors
- **Track app status**
 - Pinit1, Pinit2: Display the result of the Pinit step in TCInitial
 - JinitComplete: Displays the result of the Jinit step in TCInitial
 - SvInitComplete: Displays the result of the SV request step in TCInitial
 - TimeoutErrSV: Displays whether the SV cycle has exceeded the timeout
 - TimeoutErrTM: Displays whether the TM cycle has exceeded the timeout

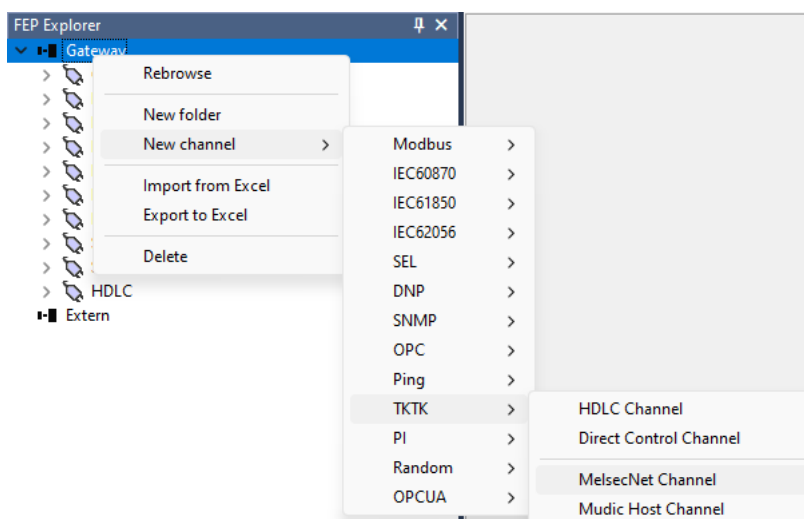
D.3.19.6 Export data channels

Similar to the M101 channel.

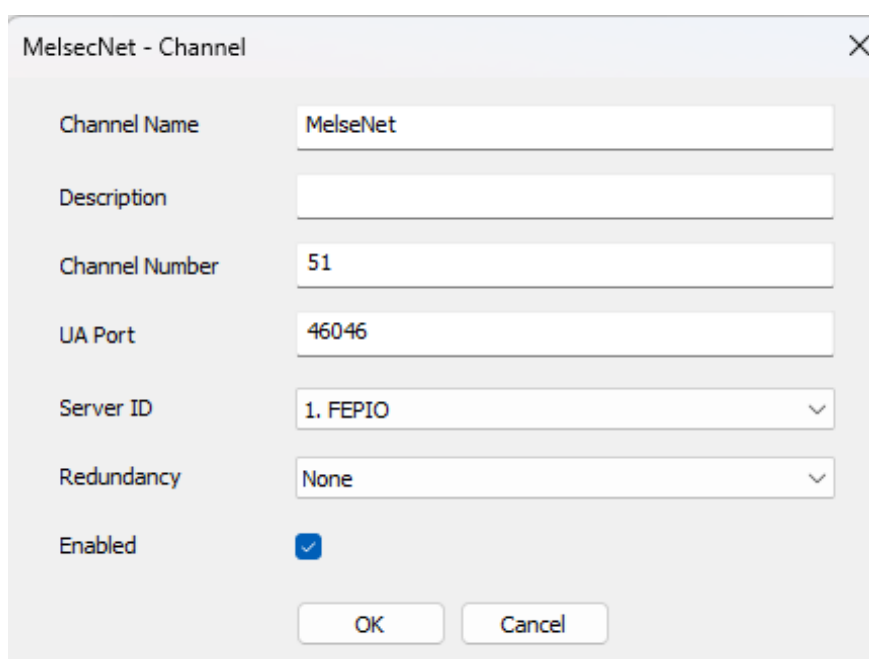
D.3.20. MelsecNet Channel

D.3.20.1 Add a new channel

In the *FEP Explorer* section, Right Click **Gateway**/Folder has created >> **New channel** >> **TKTK** >> **MelsecNet Channel** to add a new one:



FigureD-179 MelsecNet Channel Addition Menu

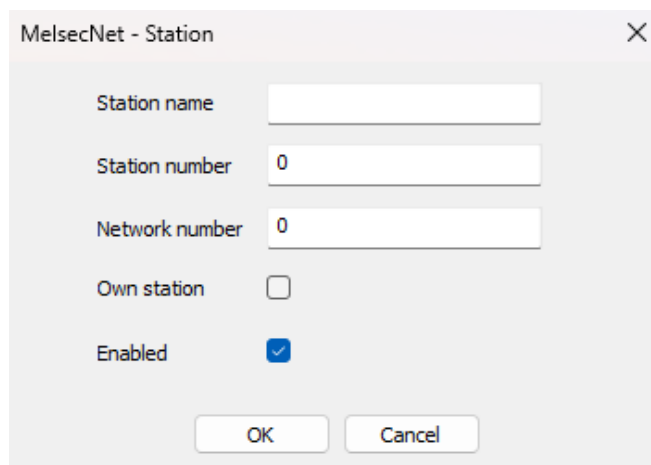


FigureD-180. Configuring the MelsecNet Channel

Configuration parameters:

- **Channel Name:** Channel name
- **Description:** Channel description
- **Channel Number:** Channel identifier
- **UA Port:** Port return for OPC UA
- **Server ID:** ID of the machine running FEP

- **Enabled:** Enable/Disable Channel
- **Redundancy:** Redundancy Mode
- **Add Station:** Right Click the channel you just created, select **Add station**



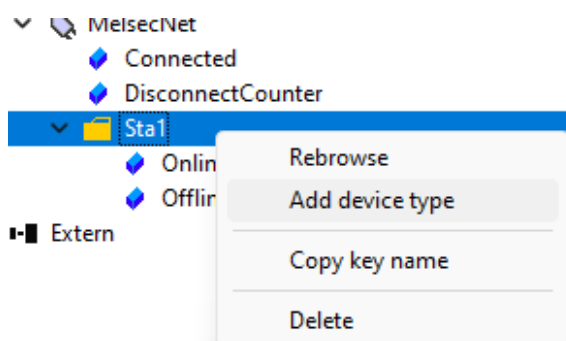
The dialog box titled "MelsecNet - Station" contains the following fields and controls:

- Station name: Text input field.
- Station number: Text input field with the value "0".
- Network number: Text input field with the value "0".
- Own station: Check box, currently unchecked.
- Enabled: Check box, currently checked.
- Buttons: OK and Cancel.

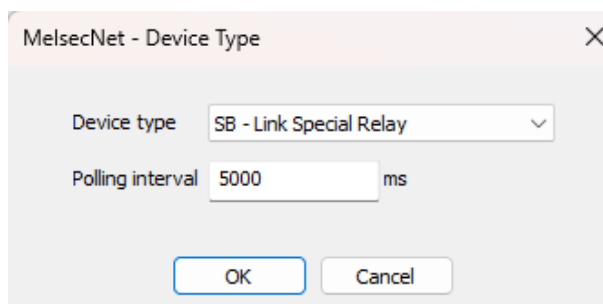
FigureD-181. Configuring More Stations

Configuration parameters:

- **Name:** Station name
- **Station Number:** The name of the station
- **Network Number:** The identifier of the network
- **Own Station:** Configuration data is located in this memory area.
- **Enabled:** The Enable/Disable status of the station
- **Add Device Type:** Right Click Station just created, select **Add device type**, select device type, enter Polling interval and select **OK**.



FigureD-182. Add Device Type Menu



The dialog box titled "MelsecNet - Device Type" contains the following fields and controls:

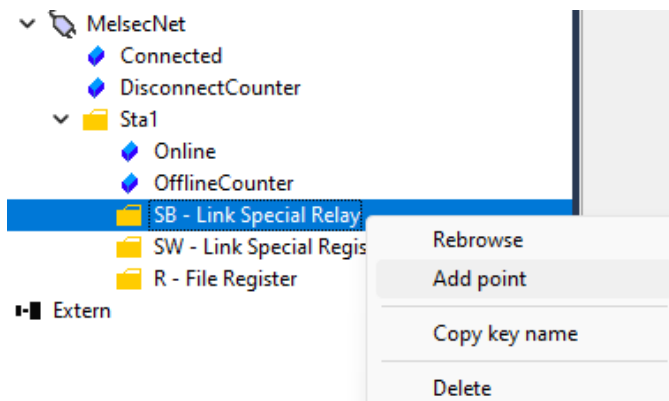
- Device type: Dropdown menu with "SB - Link Special Relay" selected.
- Polling interval: Text input field with the value "5000" and a unit "ms".
- Buttons: OK and Cancel.

FigureD-183 Device Type Configuration

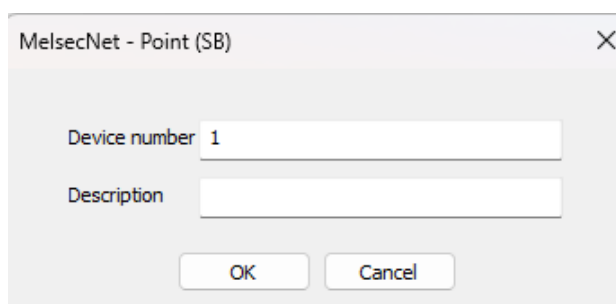
Configuration parameters:

- **Device Type:** Supports the following models:
SB – Link Special Relay
SW – Link Special Register
R – File Register
- **Polling Interval:** The polling request interval

- **Add Relay point:** Right Click SB – Link Special Relay just created, select **Add point**, enter the address and description of the point, click **OK**.

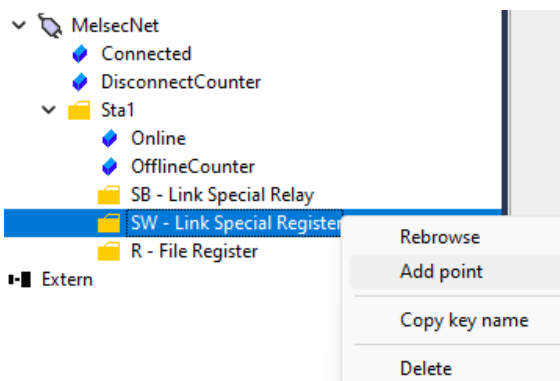


FigureD-184 Add Relay Point Menu

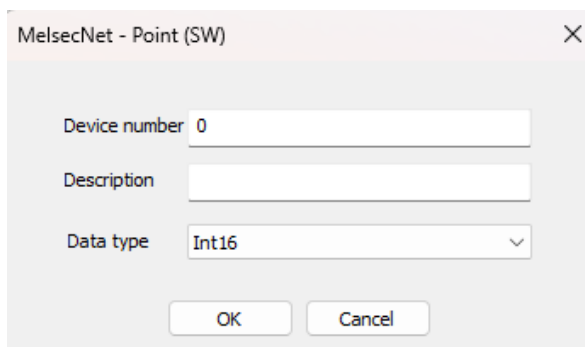


FigureD-185 Relay Point Configuration

- **Add Register Point:** Right Click SW – Link Special Register container or R – File Register just created, select **Add point**, enter the address, description and data type of the point, click **OK**.



FigureD-186 Add Register Point Menu



The image shows a dialog box titled "MelsecNet - Point (SW)". It contains three input fields: "Device number" with the value "0", "Description" which is empty, and "Data type" with a dropdown menu showing "Int16". At the bottom, there are "OK" and "Cancel" buttons.

FigureD-187 Register Point Configuration

D.3.20.2 Channel Calibration

The FEP Properties *interface* allows users to view and change channel, station and point information

D.3.20.3 Delete a channel

Right-click on the channel you want to delete and select **Delete**.

D.3.20.4 Using a channel

- **Conditions of use**
 - Users need to install PCI cart and Drive Melsec Mitsubishi for the FEP server running this channel
- **Data collection**
 - The data configuration will automatically poll the data after the channel is initialized.

D.3.20.5 Channel Tracking

- **Track channel status**
 - Connected: Connected = true after successfully establishing a connection between master and slave.
 - DisconnectedCounter: the number of times the channel has lost connection since launch.
- **Monitor application layer status**
 - Online: Online = true after the first successful data poll.
 - OfflineCounter: The number of times the device went offline.

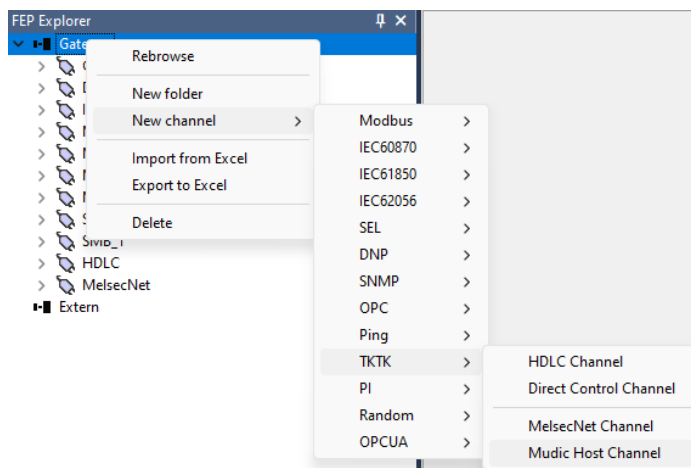
D.3.21. Mudic Host Channel

D.3.21.1 Add a new channel

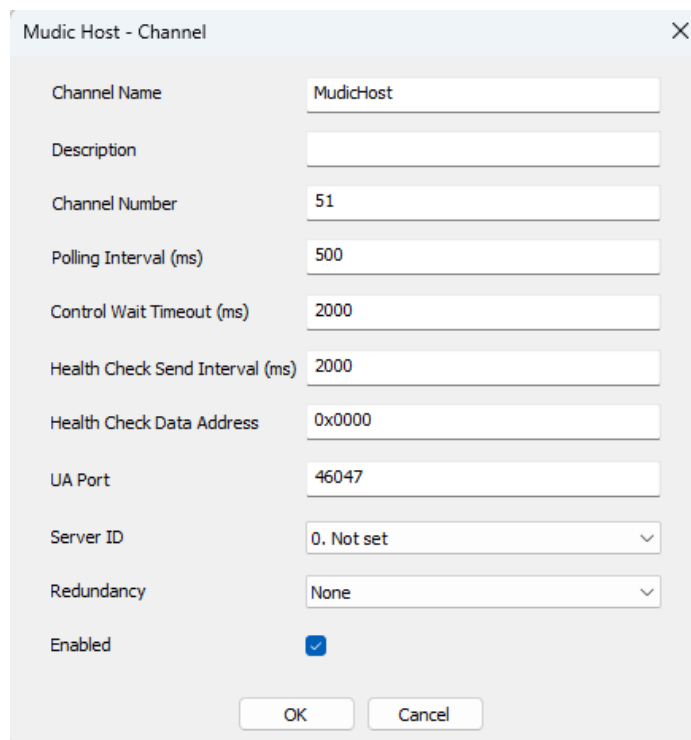
There are two ways to add a new channel: manually and import from a file.

- Import from excel file: Similar to channel M101
- Handicraft
 - Add a Channel

In the *FEP Explorer* section, Right Click **Gateway**/Folder has created >> **New channel** >> **TKTK** >> **Mudic Host Channel** to add a new one.



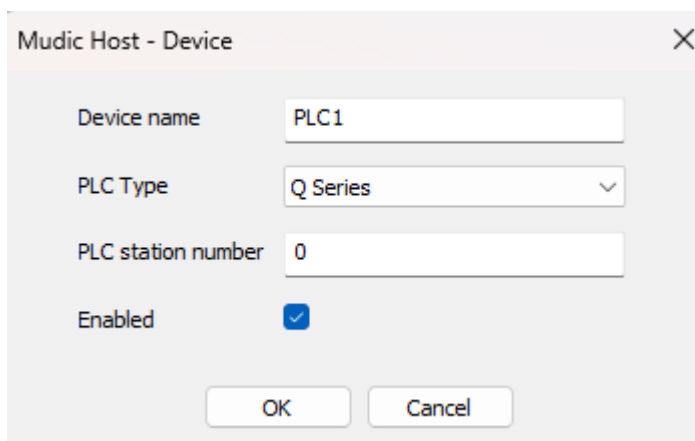
FigureD-188 MudicHost Channel Add Menu



FigureD-189. Configuring the MelsecNet Channel

Configuration Parameters:

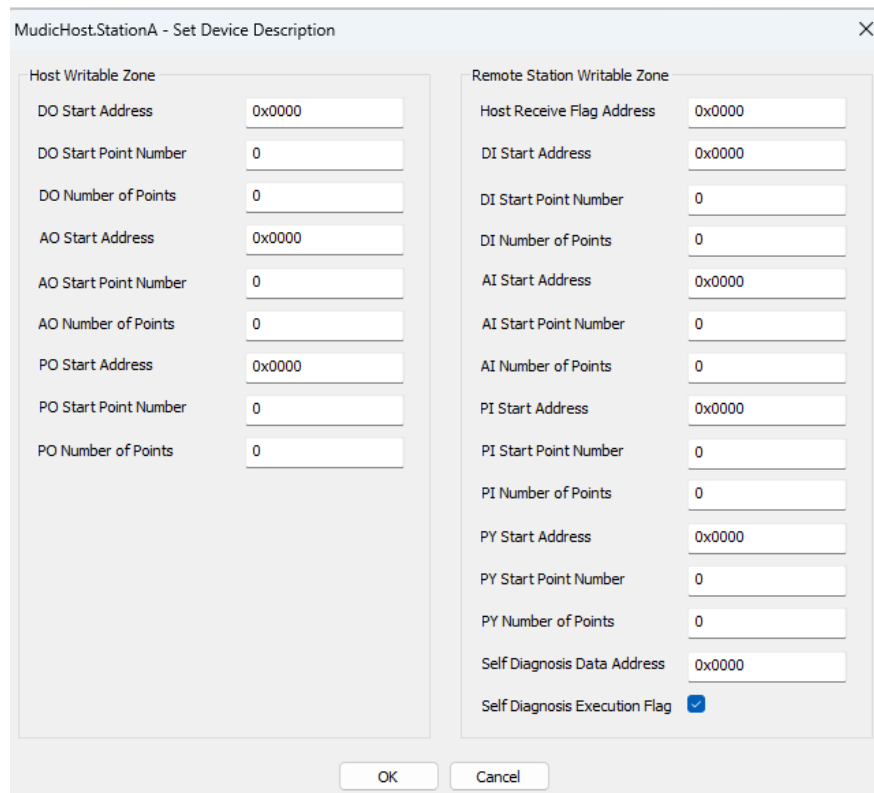
- **Channel Name:** Channel name
 - **Description:** Channel description
 - **Channel Number:** The identifier of the channel
 - **Polling Interval (ms):** Polling cycle
 - **Control Wait Timeout (ms):** Controls the time it takes to execute a control command.
 - **Health Check Send Interval (ms):** HealthCheck Flag send cycle
 - **Health Check Data Address:** The data address for the HealthCheck Flag
 - **UA Port:** Port return for OPC UA
 - **Server ID:** ID of the machine running FEP
 - **Redundancy:** Redundancy Mode
 - **Enabled:** Enable/Disable Channel
- **Add Device:** Right click the channel you just created, select **Add device**.



FigureD-190. Device Configuration

Configuration parameters:

- **Device name:** Device name
 - **PLC Type:** PLC Type: QSeries/iQ-RSeries
 - **PLC Station Number:** Identifier of the device
 - **Enabled:** The Enable/Disable status of the device
- **Add Set Device Description:** Right click the device you just created, select **Set Device Description**, configure the parameters, click **OK**.



FigureD-191. Configure Set Device Description

D.3.21.2 Channel Calibration

The FEP Properties *interface* allows users to view and change channel and device information.

D.3.21.3 Delete a channel

Right-click on the channel you want to delete and select **Delete**

D.3.21.4 Using a channel

- **Conditions of use**
 - Users need to install PCI cart and Drive Melsec Mitsubishi for the FEP server running this channel
- **Data collection**
 - Data configuration will automatically poll data after channel initialization
- **Control Point**
 - Users can activate control commands to other PLCs:
 - DOClose, DOOpen, DOSTop
 - AOSetVal, AOReset, AOExecute
 - POSetVal, POReset, POExecute
 - The control process follows the MudicHost design documentation.

D.3.21.5 Channel Tracking

- **Track channel status**

- Connected: Connected = true after successfully establishing a connection between master and slave.
- **HealthCheckData**: Displays the HealthCheck Flag value
- **ChannelFunctionResults**: Displays the results of the MelsecNet *OpenCommResult* operation
 - ReadSBResult*
 - SBDeviceOnReadFile*
 - ReadSWResult*
 - SWDeviceOnReadFile*
 - WriteHealthCheckDataResult*
- **PcBoardInfos**: Display MelsecNet Communication Status:
 - UnitCommunicationStatus*
 - UnitStationNo*
 - UnitCyclicTransmissionStatus*
 - UnitBatonPassStatus*
 - UnitForwardLoopStatus*
 - UnitReverseLoopStatus.*
- **Monitor application layer status**
 - Online: Online = true after the first successful data poll.
 - ChannelFunctionResults**: Displays the results of MudicHost requests
 - ReadSelfDiagnosisResult* *ReadAIResult* *ReadDIResult* *ReadPIResult*
 - ReadPYResult* *ReadDOReceiveFlagResult* *ReadAOReceiveFlagResult*
 - ReadPOReceiveFlagResult* *WriteDOSendFlagResult* *WriteDODataResult*
 - WriteAOSendFlagResult* *WriteAODataResult* *WritePOSendFlagResult*
 - WritePODataResult*

D.3.21.6 Export data channels

Similar to the M101 channel.

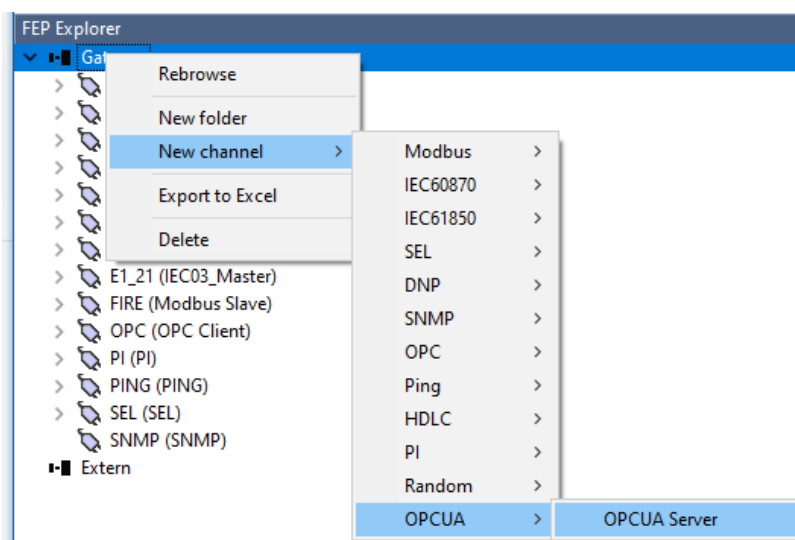
D.3.22. OPCUA Server Channel

D.3.22.1 Add a new channel

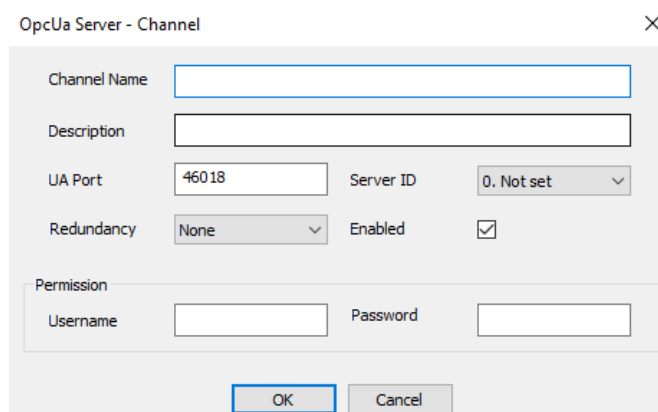
There are 2 ways to add a new channel: Manual and Import from file.

- Import from excel file: Similar to channel M101
- Handicraft:

In the *FEP Explorer* section, Right Click **Gateway**/Folder has created >> **New channel** >> **OPCUA** >> **OPCUA Server** to add a new one:



FigureD-192. Add a new OPCUA Server channel menu



The screenshot shows the 'OpcUa Server - Channel' configuration dialog box. It contains the following fields and controls:

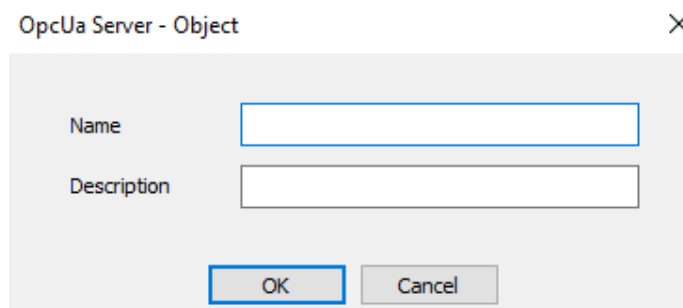
- Channel Name:** A text input field.
- Description:** A text input field.
- UA Port:** A text input field with the value '46018'.
- Server ID:** A dropdown menu with the value '0. Not set'.
- Redundancy:** A dropdown menu with the value 'None'.
- Enabled:** A checkbox that is checked.
- Permission:** A section containing:
 - Username:** A text input field.
 - Password:** A text input field.
- Buttons:** 'OK' and 'Cancel' buttons at the bottom.

FigureD-193. OPCUA Server Channel Configuration

Main parameters:

- **Channel Name:** Channel name
- **Description:** Channel description
- **UA Port:** Port return for OPC UA
- **Server ID:** ID of the machine running FEP
- **Enabled:** Enable/Disable Channel
- **Username/Password:** Account to log in to the Database

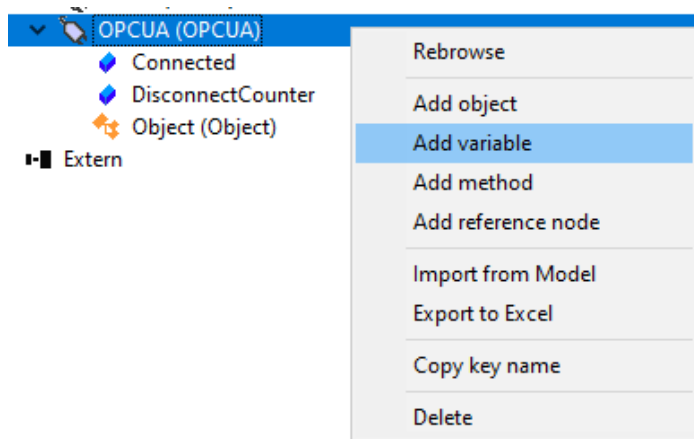
- Add **Object**: Right Click the channel, select **Add object**:



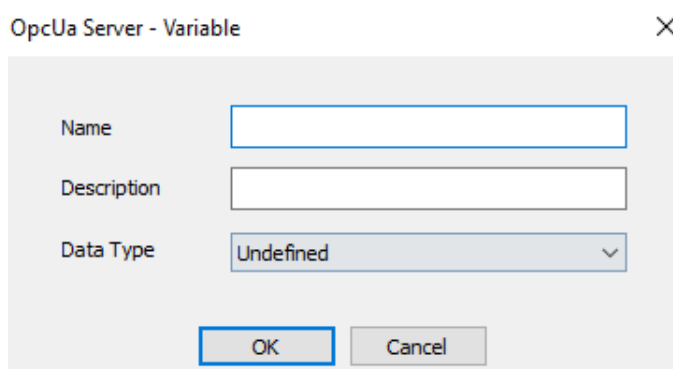
FigureD-194. Add object menu

Configuration parameters:

- **Name**: Object name
- **Description**: Object description
- Add **variable**: Right Click the object or data channel, select **Add variable**:



FigureD-195. Menu Add Variable

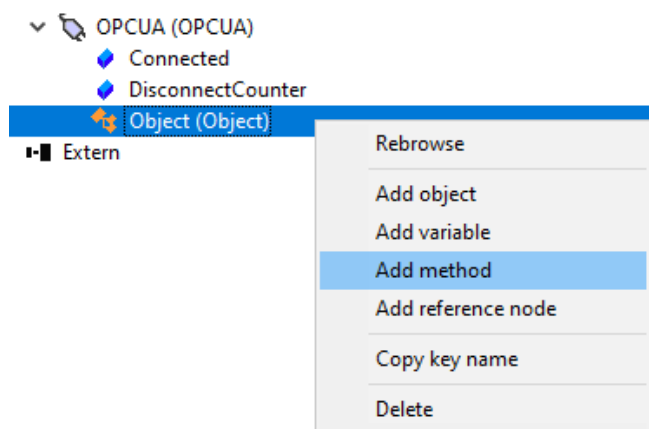


FigureD-196. Configuring OPCUA Server variable

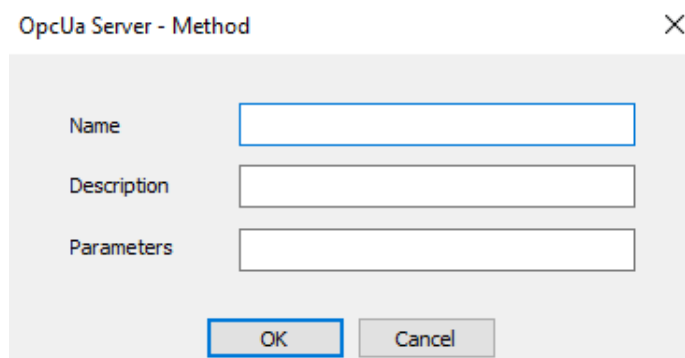
Configuration parameters:

- **Name**: Variable name
- **Description**: variable description
- **Data Type**: data type

- Add **method**: Right Click object or data channel, select **Add method**:



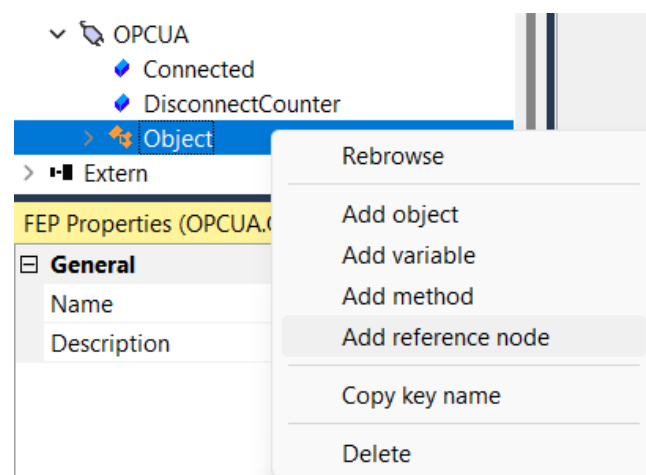
FigureD-197. Add method menu



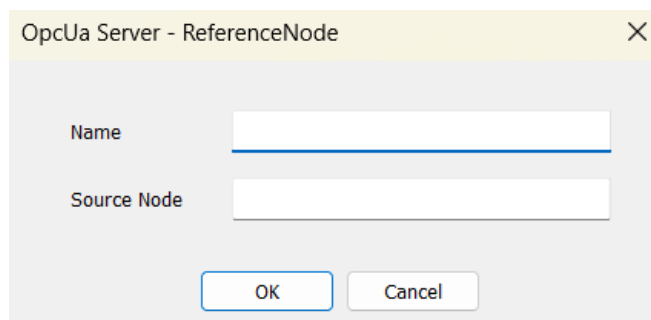
FigureD-198. Configuring the OPCUA Server method

Configuration parameters:

- **Name**: Method name
 - **Description**: description method
 - **Parameters**: the parameter of the method
- Add a **reference node**: Right Click the object or data channel, select **Add reference node**:



FigureD-199. Menu to add reference node



FigureD-200. Configuring the OPCUA Server reference node

Configuration parameters:

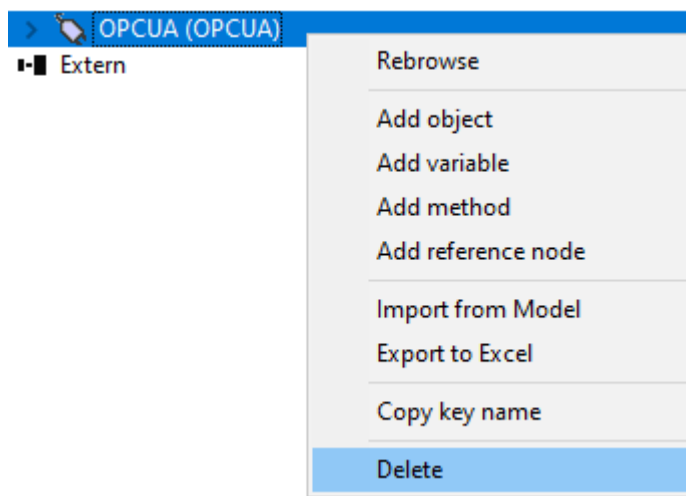
- **Name:** Reference node name
- **Source Node:**

D.3.22.2 Channel Calibration

The FEP Properties *interface* allows users to view and change channel information, objects, variables, methods, and reference nodes.

D.3.22.3 Delete a channel

Right-click on the channel you want to delete and select **Delete**:



FigureD-201. OPCUA Server channel deletion menu

D.3.22.4 Using a channel

D.3.22.5 Channel Tracking

- **Track channel status**
 - Connected: Connected = true after successfully establishing a connection between the client and the server.
 - DisconnectedCounter: the number of times the channel has lost connection since launch.

D.3.22.6 Export data channels

Similar to the M101 channel.

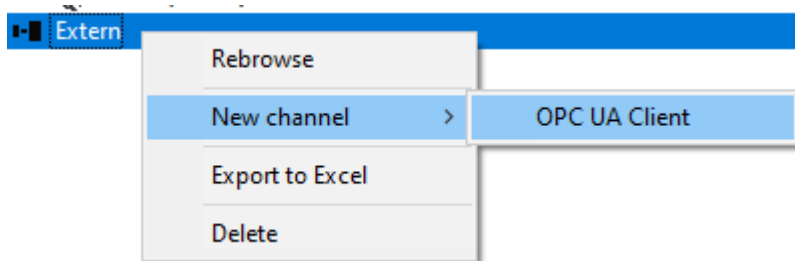
D.3.23. OPCUA Client Channel

D.3.23.1 Add a new channel

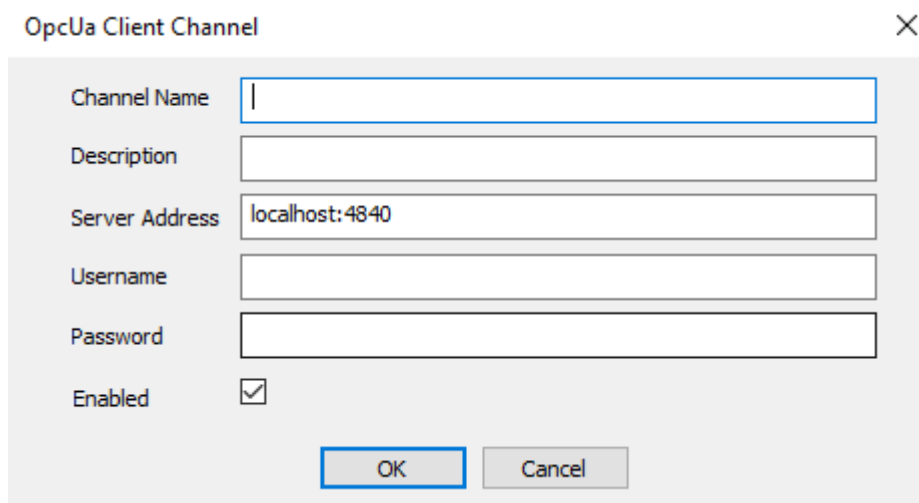
There are 2 ways to add a new channel: Manual and Import from file.

- Import from excel file: Similar to channel M101
- Handicraft:

In the *FEP Explorer section*, Right Click *Extern >> New channel >> OPC UA Client* to add a new one:



FigureD-202. Add a new OPC UA Client channel

A screenshot of the 'OpcUa Client Channel' configuration dialog box. The dialog has a title bar with a close button (X). It contains several input fields: 'Channel Name' (empty), 'Description' (empty), 'Server Address' (containing 'localhost:4840'), 'Username' (empty), and 'Password' (empty). There is a checkbox for 'Enabled' which is checked. At the bottom, there are 'OK' and 'Cancel' buttons.

FigureD-203. OPC UA Client Channel Configuration

Main parameters:

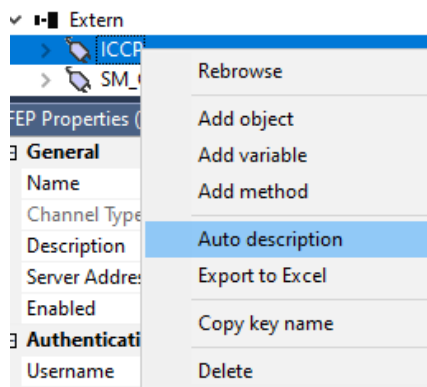
- **Channel Name:** Channel name
- **Description:** Channel description
- **Server Address:** Ip/port of the server
- **Enabled:** Enable/Disable Channel
- **Username/Password:** Account to log in to the Database

D.3.23.2 Add new data

There are 2 ways to add new data: Manual and automatic

D.3.23.2.1 Add new data automatically

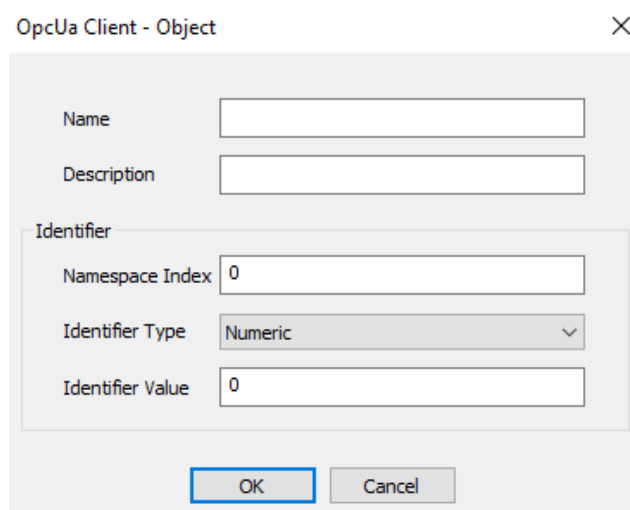
Right Click the channel, select **Auto Description**:



FigureD-204. Add new automatic data on the OPC UA Client channel

D.3.23.2.2 Add new manual data

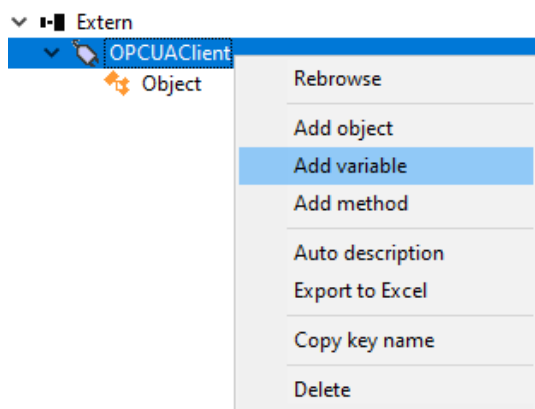
- Add **Object**: Right Click the channel, select **Add object**:



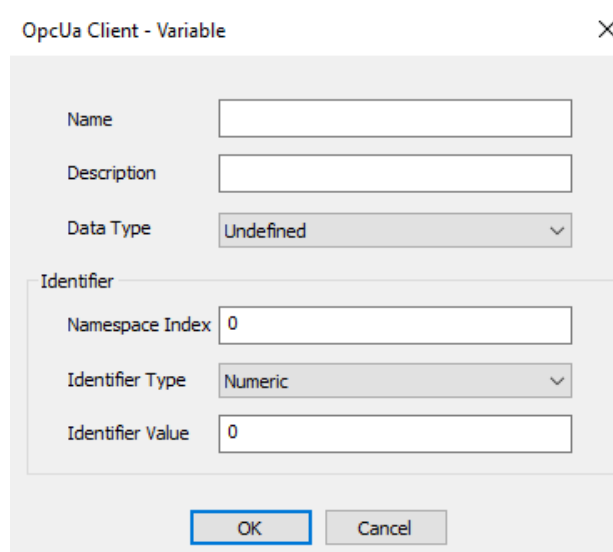
FigureD-205. Add object menu

Configuration parameters:

- **Name**: Object name
 - **Description**: Object description
 - **Namespace Index**
 - **Identifier Type**
 - **Identifiel Value**
- Add **variable**: Right Click the object or data channel, select **Add variable**:



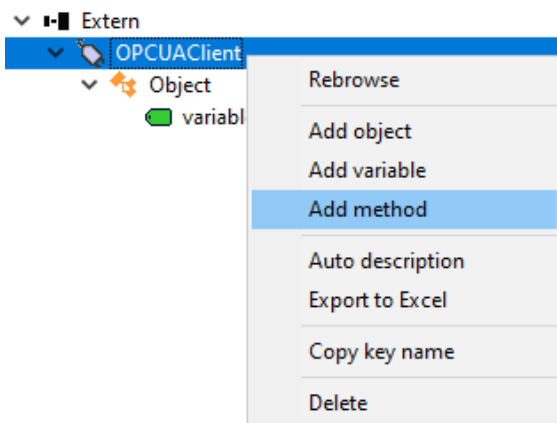
FigureD-206. Menu Add Variable

A screenshot of the 'OpcUa Client - Variable' configuration dialog box. It contains fields for 'Name', 'Description', and 'Data Type' (set to 'Undefined'). Below these is an 'Identifier' section with 'Namespace Index' (0), 'Identifier Type' (Numeric), and 'Identifier Value' (0). 'OK' and 'Cancel' buttons are at the bottom.

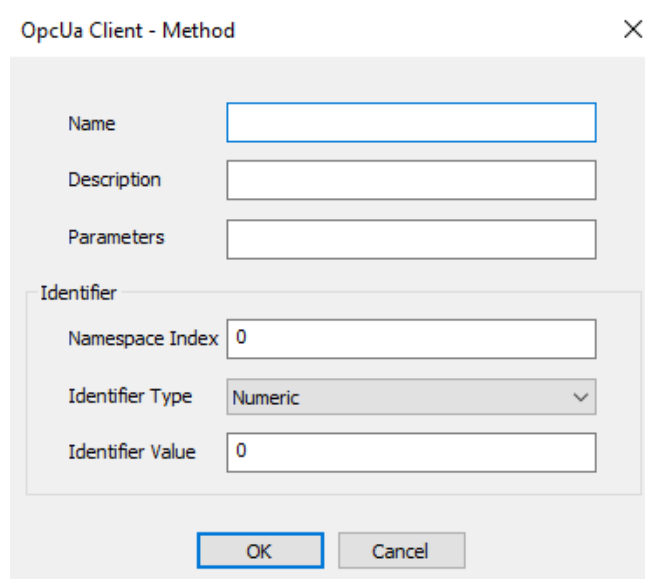
FigureD-207. Configure OPC UA Client variable

Configuration parameters:

- **Name:** Variable name
 - **Description:** variable description
 - **Data Type:** data type
 - **Namespace Index**
 - **Identifier Type**
 - **Identifier Value**
- Add **method**: Right Click object or data channel, select **Add method**:



FigureD-208. Add method menu



FigureD-209. Configuring the OPC UA Client method

Configuration parameters:

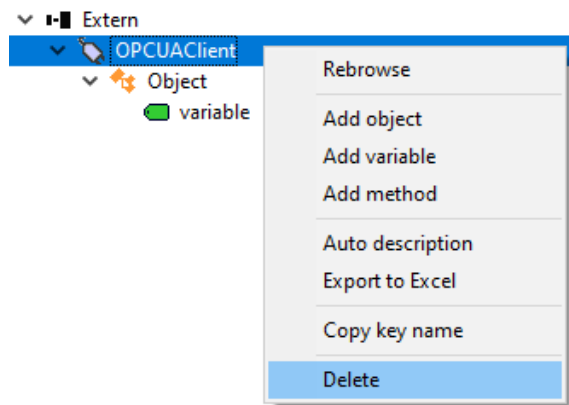
- **Name:** Method name
- **Description:** description method
- **Parameters:** the parameter of the method
- **Namespace Index**
- **Identifier Type**
- **Identifiel Value**

D.3.23.3 Channel Calibration

The FEP Properties *interface* allows users to view and change channel information, objects, variables, and methods.

D.3.23.4 Delete a channel

Right-click on the channel you want to delete and select **Delete**:



FigureD-210. OPC UA Client Channel Deletion Menu

D.3.23.5 Using a channel

D.3.23.6 Channel Tracking

- **Track channel status**
 - Connected: Connected = true after successfully establishing a connection between the client and the server.
 - DisconnectedCounter: the number of times the channel has lost connection since launch.

D.3.23.7 Export data channels

Similar to the M101 channel.

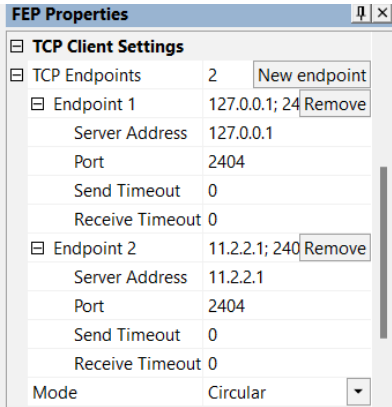
D.3.24. TCP Client Setting Mode

D.3.24.1 TCP Endpoints:

TCP Endpoints are TCP/IP of TCP Server.

2 TCP Servers (IEC104 Server) connected from 1 TCP Client

A list of endpoints to which the TCP Client channel connects.



FigureD-211. TCP Endpoints

Normally, each channel only has 1 Endpoint, a typical case that needs to declare multiple Endpoints is when the channel needs to connect to the redundancy server.

Endpoint configuration parameters include:

- **Server address:** The IP address of the TCP server
- **Port:** The port that works on the TCP server
- **Send Timeout (ms):** If the outgoing message fails after the Send Timeout timeout, the TCP Client will switch to a disconnected state and try to reconnect;

Default = 0, which means that the above configuration is not enabled.

- **Receive Timeout (ms):** If the TCP Client does not receive a new newsletter after the Receive Timeout timeout, the TCP Client will enter a disconnected state and attempt to reconnect.

Default = 0, which means that the above configuration is not enabled.

D.3.24.2 Fashion

How to connect when you have multiple Endpoints.

- **Circular:** TCP Client only maintains a connection to 1 Endpoint, if it loses connection it will try to connect to the next Endpoint. If the last endpoint in the list loses connection, it will go back to trying to connect to the first endpoint.
- **Concurrent:** The TCP Client will maintain a simultaneous connection to all Endpoints.
- **Priority:** The TCP Client only maintains a connection to 1 Endpoint, it will prioritize the connection to the Endpoint configured in **the Priority Endpoint box**. if it loses the connection it will try to connect to the next Endpoint.

Mode	Priority
Priority Endpoint	1
Keep Old Connection	☐

In Priority mode if **Keep Old Connection** is selected, the TCP Client will keep the current connection even if the priority endpoint is in good shape again.

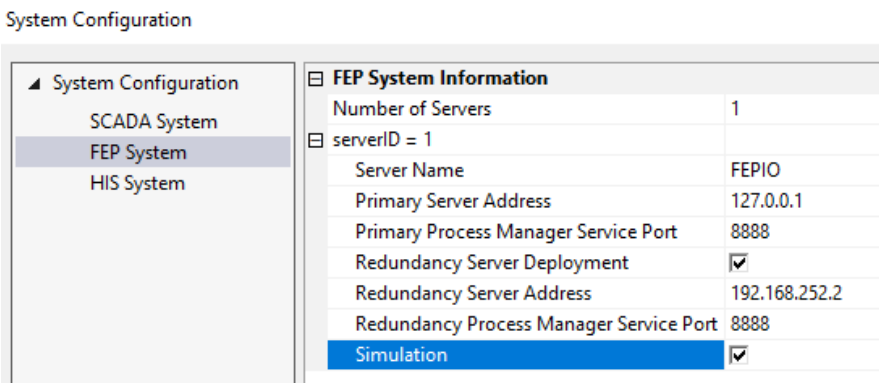
D.4. FEP simulation mode

As a runtime mode, OneATS DataServer will connect to FEP Server Simulation to retrieve data.

D.4.1. Simulation Settings

On the OneATS DataEditor interface, select the following path: **Project/System Configuration**. Click on the **FEP System** section, check **Simulation**.

Click **OK** to save.

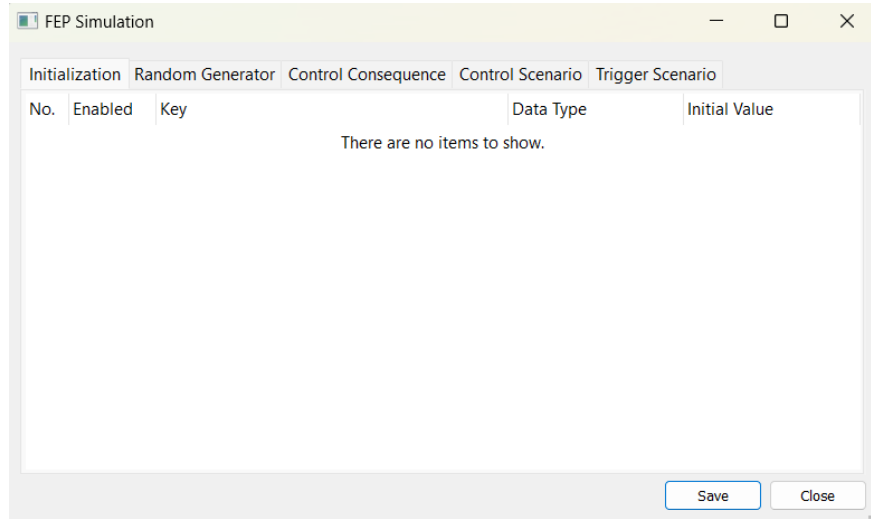


FigureD-212. Setting FEP Simulation Mode

D.4.2. Configure Simulation Data

On the **OneATS DataEditor** interface, select the following path: **FEP/Simulation** to open the **FEP Simulation** window.

Here users can configure many different Simulation modes such as: **Initialization**, **Random Generator**, **Control Consequence**, **Control Scenario**, **Trigger Scenario**.



FigureD-213. FEP Simulation Window Interface

D.4.2.1 Initialization

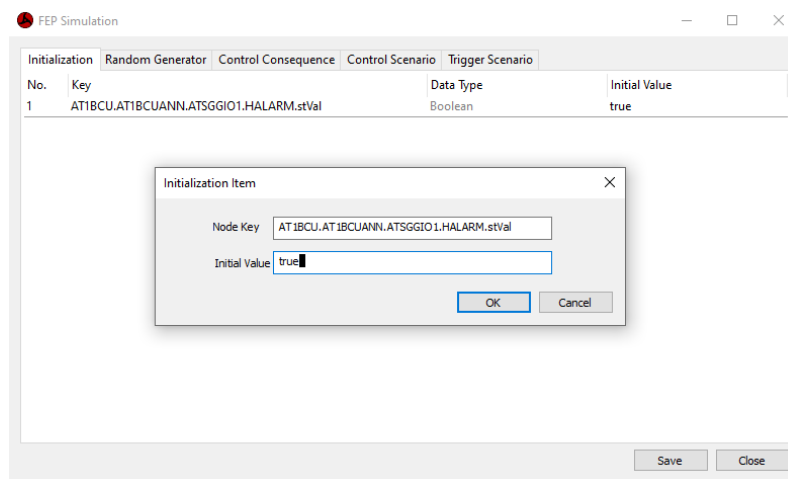
Initialization mode is the value mode of pre-initialized variables.

Open **the FEP Simulation window**, click on the **Initialization tab**.

Right Click on the empty area and select **New**. The **Initialization** window appears.

Fill in the information **Node Key**, **Initial value**. Click **OK**.

Lick **Save** to save and close **the FEP Simulation window**.



FigureD-214. Initialization Mode Data Configuration

D.4.2.2 Random Generator

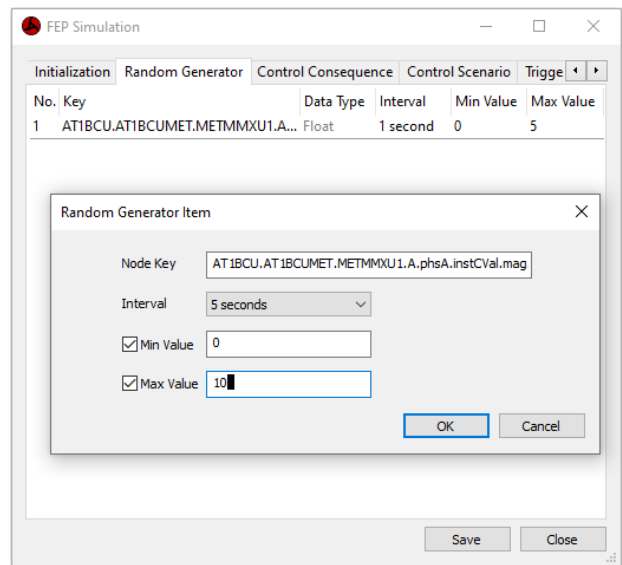
It is a mode in which every **interval of time**, the value of the data is randomly changed within the limit interval [**Min Value**; **Max Value**].

Open **the FEP Simulation window**, click on the **Random Generator tab**.

Right Click on the empty area and select **New**. The **Random Generator Item** window appears.

Fill in the parameters of **Node Key**, **Interval**, **Min Value**, **Max Value**. Click **OK**.

Click **Save** to save and close **the FEP Simulation window**.

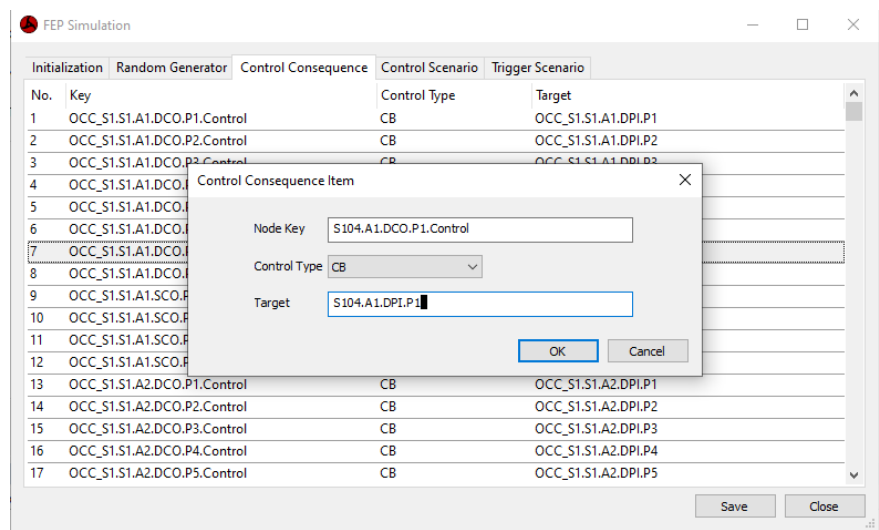


FigureD-215. Random Generator Mode Data Configuration

D.4.2.3 Control Consequence

It is a mode that simulates control in different predefined patterns.
Open the **FEP Simulation** window, click on the **Control Consequence** tab.
Right Click on the empty area and select **New**. The **Control Consequence Item** window appears.
Fill in the **Node Key**, **Target** and select the control type in the **Control Type** section. Click **OK**. Click **Save** to save and close the **FEP Simulation** window.

Node Key: Method is called
Target: variable output



FigureD-216. Configuring Control Consequence Mode Data

D.4.2.4 Control Scenario

Allows the creation of scenarios for control methods.

Open **the FEP Simulation window**, click on the **Control Scenario tab**.

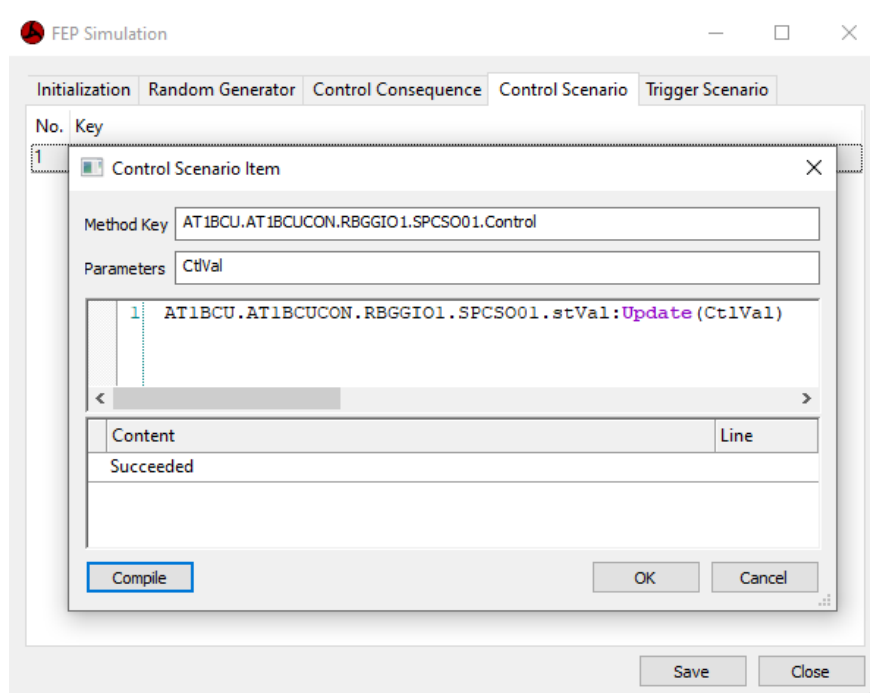
Right Click on the empty area and select **New**. The **Control Scenario Item** window appears. The window interface is similar to **the Method Implementation window** when building a **Method**.

Fill in **the Method Key, Parameters**, the executable when the method is called.

Click **Compile** to compile and check for errors.

Press **OK** when there is a **Succeeded message**.

Click **Save** to save and close **the FEP Simulation window**.



FigureD-217. Control Scenario mode data configuration

D.4.2.5 Trigger Scenario

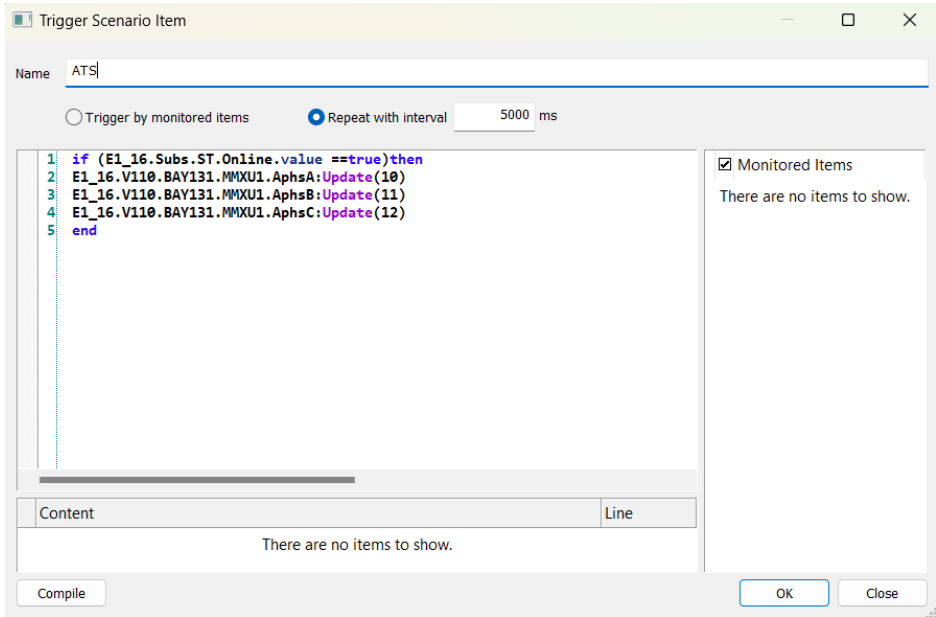
Allows the creation of simulation scenarios for programs.

Open **the FEP Simulation window**, click on the **Trigger Scenario tab**.

Right Click on the empty area and select **New**. The **Trigger Scenario Item window** appears. The window interface is similar to the interface when building a program in a DataModel.

Building trigger scenarios is similar to adding a new program to the DataModel.

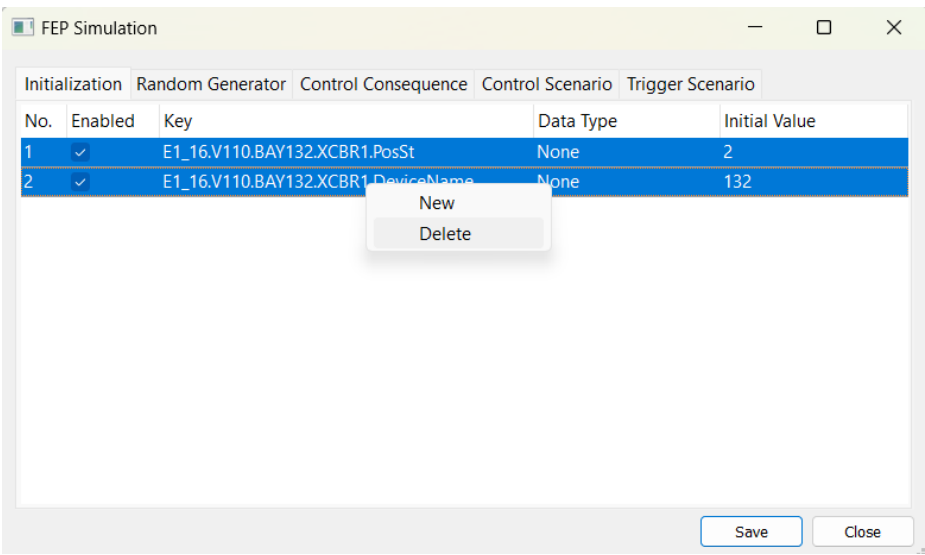
Click **Save** to save and close **the FEP Simulation window**.



FigureD-218. Configuring Trigger Scenario Mode Data

D.4.2.6 Delete simulation data

Open the FEP Simulation window, right-click select the data to be deleted, select Delete.



FigureD-219. Delete simulation data

D.5. Configure runtime deployments

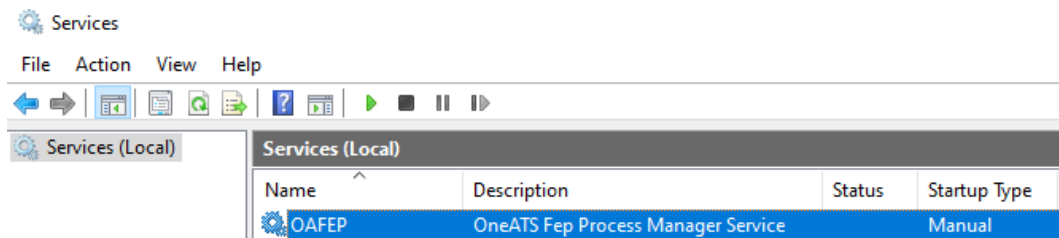
D.5.1. Configure OneATS FEP service

Refer [to D.2.3. App Configuration](#)

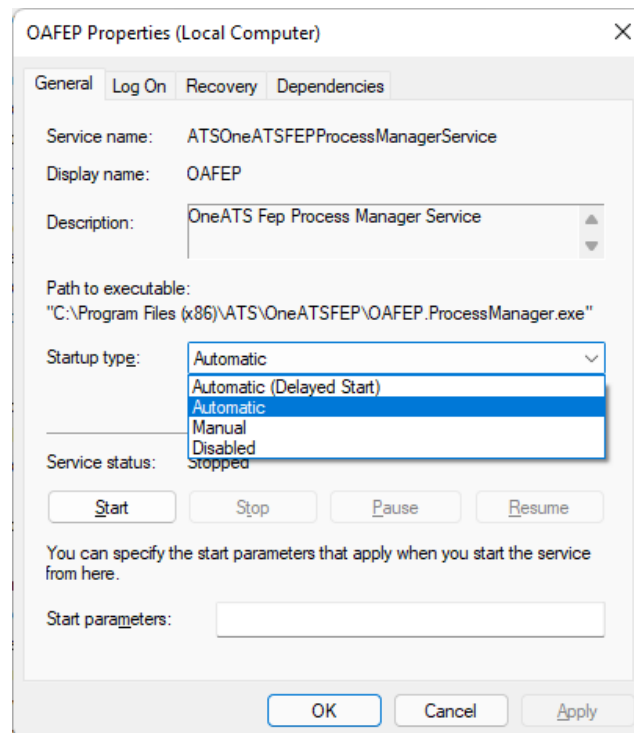
D.5.2. Boot ATS FEP service with Windows

To start OneATS FEP service with Windows:

- Run the Window Services app

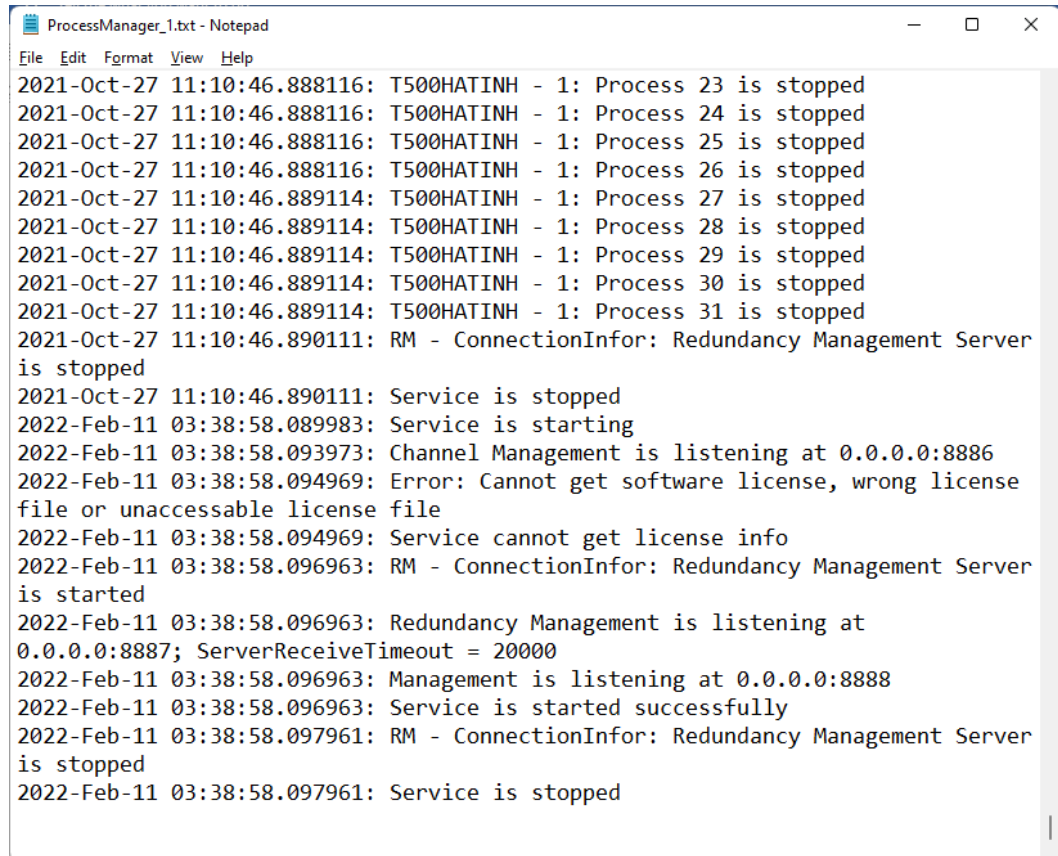


- Setting up OAFEP service : Start-up Type = Automatic



D.6. Monitoring Log - runtime

The logs of the OneATS FEP service operation are stored in the file **ProcessManager_*.txt** in the configured folder (<Log> / <Path>) in the OneATS FEP service configuration file.



```
ProcessManager_1.txt - Notepad
File Edit Format View Help
2021-Oct-27 11:10:46.888116: T500HATINH - 1: Process 23 is stopped
2021-Oct-27 11:10:46.888116: T500HATINH - 1: Process 24 is stopped
2021-Oct-27 11:10:46.888116: T500HATINH - 1: Process 25 is stopped
2021-Oct-27 11:10:46.888116: T500HATINH - 1: Process 26 is stopped
2021-Oct-27 11:10:46.889114: T500HATINH - 1: Process 27 is stopped
2021-Oct-27 11:10:46.889114: T500HATINH - 1: Process 28 is stopped
2021-Oct-27 11:10:46.889114: T500HATINH - 1: Process 29 is stopped
2021-Oct-27 11:10:46.889114: T500HATINH - 1: Process 30 is stopped
2021-Oct-27 11:10:46.889114: T500HATINH - 1: Process 31 is stopped
2021-Oct-27 11:10:46.890111: RM - ConnectionInfor: Redundancy Management Server
is stopped
2021-Oct-27 11:10:46.890111: Service is stopped
2022-Feb-11 03:38:58.089983: Service is starting
2022-Feb-11 03:38:58.093973: Channel Management is listening at 0.0.0.0:8886
2022-Feb-11 03:38:58.094969: Error: Cannot get software license, wrong license
file or unaccessable license file
2022-Feb-11 03:38:58.094969: Service cannot get license info
2022-Feb-11 03:38:58.096963: RM - ConnectionInfor: Redundancy Management Server
is started
2022-Feb-11 03:38:58.096963: Redundancy Management is listening at
0.0.0.0:8887; ServerReceiveTimeout = 20000
2022-Feb-11 03:38:58.096963: Management is listening at 0.0.0.0:8888
2022-Feb-11 03:38:58.096963: Service is started successfully
2022-Feb-11 03:38:58.097961: RM - ConnectionInfor: Redundancy Management Server
is stopped
2022-Feb-11 03:38:58.097961: Service is stopped
```

FigureD-220. OneATS FEP application log file

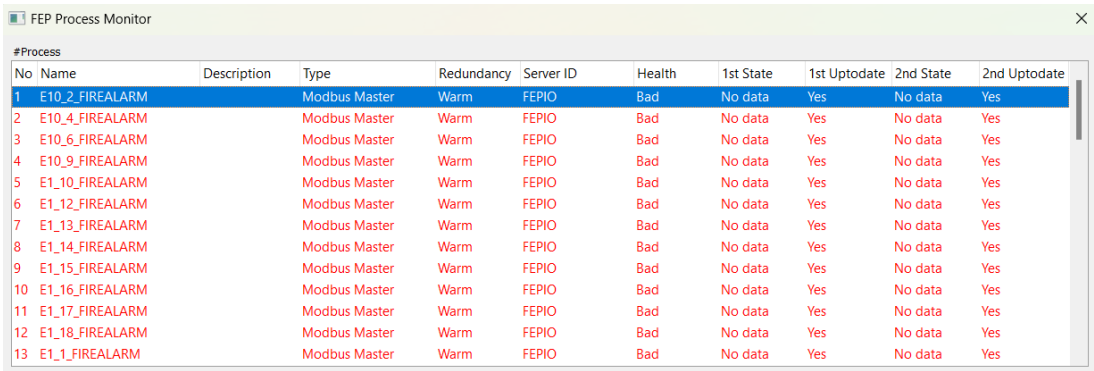
Basic Information:

- Configuration file loading complete & configuration file error if applicable
- License error (if any)
- Connecting to Database / project name
- Redundancy status
- ...

D.7. Online Data Monitoring

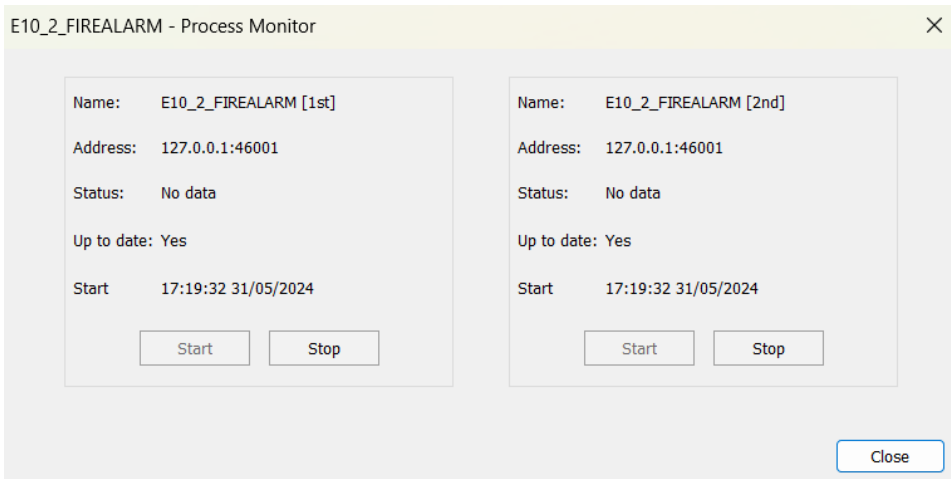
D.7.1. Process Monitor

From the Menu Bar select *Fep >> Process Monitor*. Monitor the status and connection status of the data channel.



No	Name	Description	Type	Redundancy	Server ID	Health	1st State	1st Uptodate	2nd State	2nd Uptodate
1	E10_2_FIREALARM		Modbus Master	Warm	FEPIO	Bad	No data	Yes	No data	Yes
2	E10_4_FIREALARM		Modbus Master	Warm	FEPIO	Bad	No data	Yes	No data	Yes
3	E10_6_FIREALARM		Modbus Master	Warm	FEPIO	Bad	No data	Yes	No data	Yes
4	E10_9_FIREALARM		Modbus Master	Warm	FEPIO	Bad	No data	Yes	No data	Yes
5	E1_10_FIREALARM		Modbus Master	Warm	FEPIO	Bad	No data	Yes	No data	Yes
6	E1_12_FIREALARM		Modbus Master	Warm	FEPIO	Bad	No data	Yes	No data	Yes
7	E1_13_FIREALARM		Modbus Master	Warm	FEPIO	Bad	No data	Yes	No data	Yes
8	E1_14_FIREALARM		Modbus Master	Warm	FEPIO	Bad	No data	Yes	No data	Yes
9	E1_15_FIREALARM		Modbus Master	Warm	FEPIO	Bad	No data	Yes	No data	Yes
10	E1_16_FIREALARM		Modbus Master	Warm	FEPIO	Bad	No data	Yes	No data	Yes
11	E1_17_FIREALARM		Modbus Master	Warm	FEPIO	Bad	No data	Yes	No data	Yes
12	E1_18_FIREALARM		Modbus Master	Warm	FEPIO	Bad	No data	Yes	No data	Yes
13	E1_1_FIREALARM		Modbus Master	Warm	FEPIO	Bad	No data	Yes	No data	Yes

Alternatively, it is possible to perform a Stop/Start data channel by Double Click the data channel and select Stop/Start.



Name: E10_2_FIREALARM [1st]

Address: 127.0.0.1:46001

Status: No data

Up to date: Yes

Start 17:19:32 31/05/2024

Start Stop

Name: E10_2_FIREALARM [2nd]

Address: 127.0.0.1:46001

Status: No data

Up to date: Yes

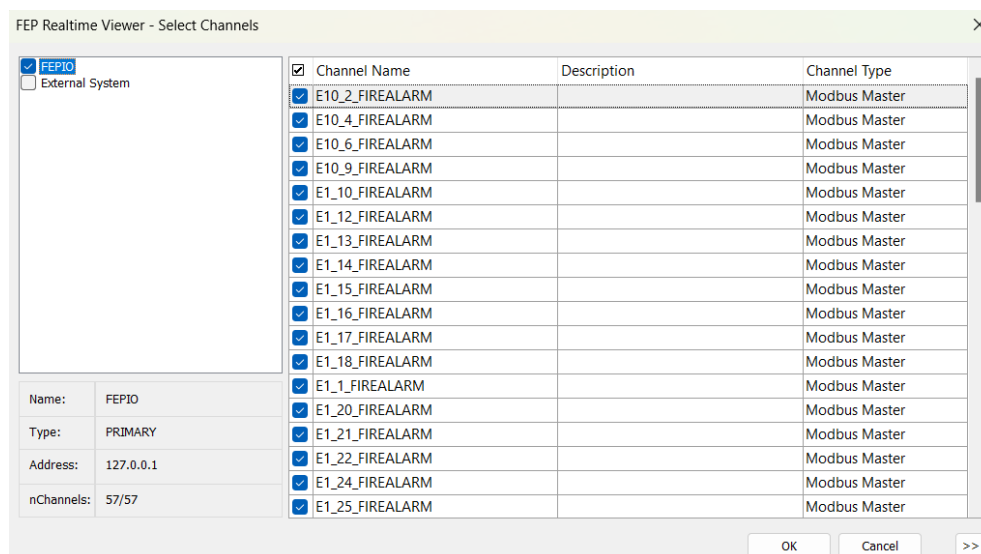
Start 17:19:32 31/05/2024

Start Stop

Close

D.7.2. Realtime Data Monitoring

From the Menu Bar select *Fep >> Realtime Viewer*. Select FEP Server, Data Channel.



D.7.3. Log file

D.7.3.1 General Information

FileLog is a feature of OAFEP that records activity logs to files in .txt format

There are 3 log objects: ProcessManager, Process, and Channel.

The ProcessManager logs the events of the OAFEP service, which is enabled by default.

The Process log records the events of the process window, which is enabled by default.

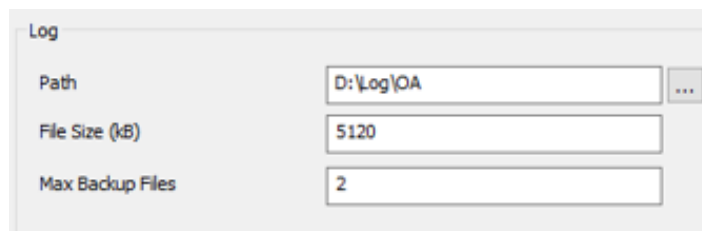
Channel logs record channel activity logs, including 2 parts: channel events and activity logs (update newsletters). Event logging is enabled by default, the activity log needs to be called method via DE to activate.

D.7.3.2 Configure FileLog

Run the **OneMasterTool.exe** app

Select **Configuration**

Select **FEP**



The Path field is the path to the FileLog folder.

File Size (kB) field: the maximum size of a log file.

Max Backup Files field: the number of log files to be backed up when the log file reaches the maximum size.

D.7.3.3 Use

Filelog Name

ProcessManager: ProcessManager_<ordinal number>.txt

Window Process: <Project name> Process_<sequence number>.txt

Channel: <Project name> <Channel name>_<sequence number>.txt

Notes:

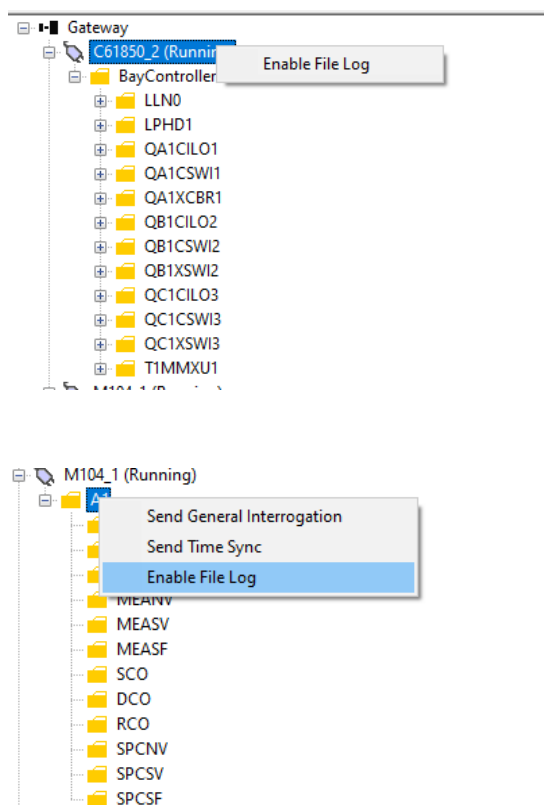
The larger the ordinal number, the older the log.

If a log file .txt overflows Log.Size, the system will generate a new .txt log file, and the data from the old .txt log file will be automatically transferred to the .txt file with a larger order number.

Activity logs of channels.

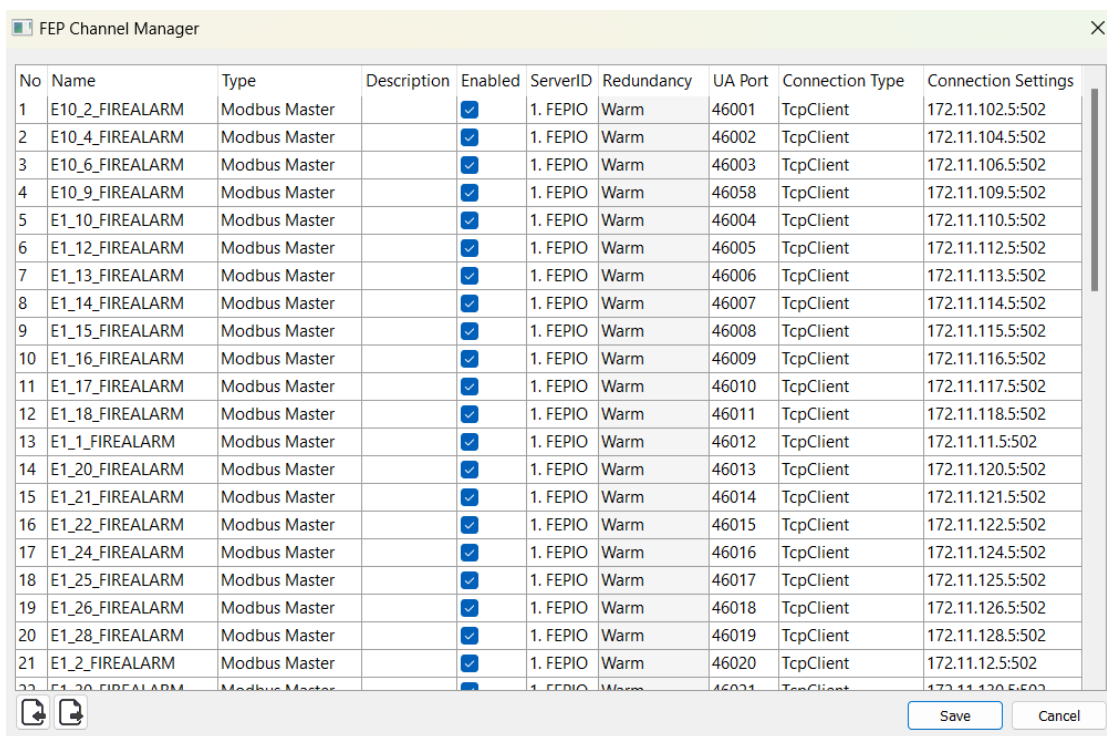
The activity logs of channels are deployed according to the layers corresponding to the protocol, for example, the M104 channel has 2 layers: Channel and Application, the M101 channel has 3 layers: Channel, Link and Application. ChannelLayer logs will record the logs with raw data, and ApplicationLayer logs will analyze the raw logs into clear information.

Activate in DE. FEP. RealTimeViewer: Right-click on Tên_Channel, Tên_Link_Layer, Tên_ApplicationLayer, or Tên_Device and select "EnableFileLog"



D.8. Data channel management

From the Menu Bar, select **Fep >> Channel Manager**



No	Name	Type	Description	Enabled	ServerID	Redundancy	UA Port	Connection Type	Connection Settings
1	E10_2_FIREALARM	Modbus Master		☒	1. FEPIO	Warm	46001	TcpClient	172.11.102.5:502
2	E10_4_FIREALARM	Modbus Master		☒	1. FEPIO	Warm	46002	TcpClient	172.11.104.5:502
3	E10_6_FIREALARM	Modbus Master		☒	1. FEPIO	Warm	46003	TcpClient	172.11.106.5:502
4	E10_9_FIREALARM	Modbus Master		☒	1. FEPIO	Warm	46058	TcpClient	172.11.109.5:502
5	E1_10_FIREALARM	Modbus Master		☒	1. FEPIO	Warm	46004	TcpClient	172.11.110.5:502
6	E1_12_FIREALARM	Modbus Master		☒	1. FEPIO	Warm	46005	TcpClient	172.11.112.5:502
7	E1_13_FIREALARM	Modbus Master		☒	1. FEPIO	Warm	46006	TcpClient	172.11.113.5:502
8	E1_14_FIREALARM	Modbus Master		☒	1. FEPIO	Warm	46007	TcpClient	172.11.114.5:502
9	E1_15_FIREALARM	Modbus Master		☒	1. FEPIO	Warm	46008	TcpClient	172.11.115.5:502
10	E1_16_FIREALARM	Modbus Master		☒	1. FEPIO	Warm	46009	TcpClient	172.11.116.5:502
11	E1_17_FIREALARM	Modbus Master		☒	1. FEPIO	Warm	46010	TcpClient	172.11.117.5:502
12	E1_18_FIREALARM	Modbus Master		☒	1. FEPIO	Warm	46011	TcpClient	172.11.118.5:502
13	E1_1_FIREALARM	Modbus Master		☒	1. FEPIO	Warm	46012	TcpClient	172.11.11.5:502
14	E1_20_FIREALARM	Modbus Master		☒	1. FEPIO	Warm	46013	TcpClient	172.11.120.5:502
15	E1_21_FIREALARM	Modbus Master		☒	1. FEPIO	Warm	46014	TcpClient	172.11.121.5:502
16	E1_22_FIREALARM	Modbus Master		☒	1. FEPIO	Warm	46015	TcpClient	172.11.122.5:502
17	E1_24_FIREALARM	Modbus Master		☒	1. FEPIO	Warm	46016	TcpClient	172.11.124.5:502
18	E1_25_FIREALARM	Modbus Master		☒	1. FEPIO	Warm	46017	TcpClient	172.11.125.5:502
19	E1_26_FIREALARM	Modbus Master		☒	1. FEPIO	Warm	46018	TcpClient	172.11.126.5:502
20	E1_28_FIREALARM	Modbus Master		☒	1. FEPIO	Warm	46019	TcpClient	172.11.128.5:502
21	E1_2_FIREALARM	Modbus Master		☒	1. FEPIO	Warm	46020	TcpClient	172.11.12.5:502
22	E1_20_FIREALARM	Modbus Master		☒	1. FEPIO	Warm	46021	TcpClient	172.11.120.5:502

Change the configuration parameters of a single channel or multiple channels by selecting the data channel, **Right Click** the channel:

Set Enabled
 Set Server ID
 Set Redundancy Mode
 Set UA Port

In which:

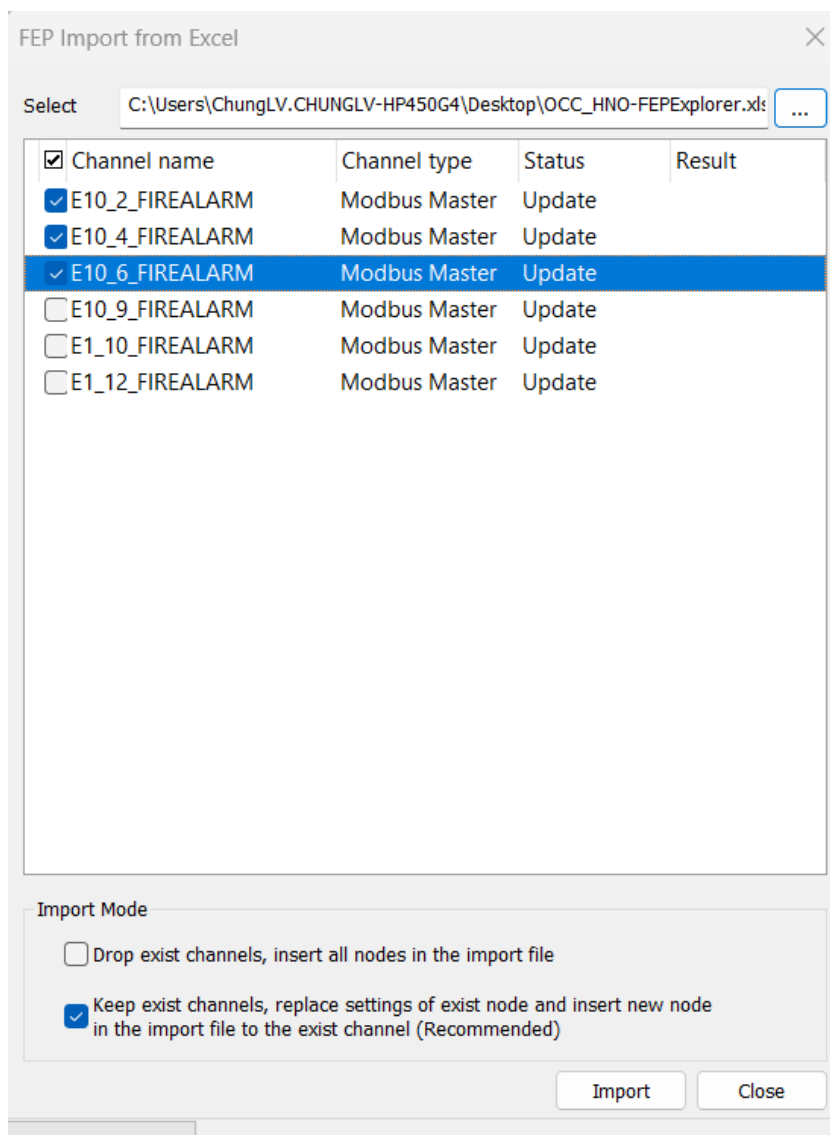
- **Set Enabled:** Enable/Disable data channel operation
- **Set Server ID:** Select Server ID for the data channel
- **Set Redundancy Mode:** Select the Redundancy mode for the data channel
- **Set UA Port:** Configure the output port for the OPC UA. In case of selecting multiple data channels to set UA ports, the UA port value of the data channels will advance by 1 unit

Select **Save** to save the configuration again.

D.9. Import/Export Multiple Data Channels

D.9.1. Import multiple data channels

From the Menu Bar select **Fep >> Import from Excel**. Select the file path, select the channel, and the Import Mode option.



FigureD-221 Import multiple data channels

D.9.2. Export multiple data channels

From the Menu Bar select **Fep >> Export to Excel**. Select the channels, select the location to save the file >> **Ok**.

FEP Export to Excel

☒	Channel name	Description	Channel type
☒	E10_2_FIREALARM		Modbus Master
☒	E10_4_FIREALARM		Modbus Master
☒	E10_6_FIREALARM		Modbus Master
☒	E10_9_FIREALARM		Modbus Master
☒	E1_10_FIREALARM		Modbus Master
☐	E1_12_FIREALARM		Modbus Master
☐	E1_13_FIREALARM		Modbus Master
☐	E1_14_FIREALARM		Modbus Master
☐	E1_15_FIREALARM		Modbus Master
☐	E1_16_FIREALARM		Modbus Master
☐	E1_17_FIREALARM		Modbus Master
☐	E1_18_FIREALARM		Modbus Master
☐	E1_1_FIREALARM		Modbus Master
☐	E1_20_FIREALARM		Modbus Master
☐	E1_21_FIREALARM		Modbus Master
☐	E1_22_FIREALARM		Modbus Master
☐	E1_24_FIREALARM		Modbus Master
☐	E1_25_FIREALARM		Modbus Master
☐	E1_26_FIREALARM		Modbus Master
☐	E1_28_FIREALARM		Modbus Master
☐	E1_2_FIREALARM		Modbus Master

Save To C:\Users\ChungLV.CHUNGLV-HP450G4\Desktop\OCC_HNO-FEPEXplorer.x ...

OK Cancel

FigureD-222 Export multiple data channels



Contact

ATS Building

74 Ba Trieu Str., Hanoi, Vietnam

T. +84-24-3825 1072

F. +84-24-3825 8037

W. www.ats.com.vn

E. contact@ats.com.vn

